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Carboniferous palaeomagnetism of the Werrie Block, northwestern Tamworth Belt, and the New England pole path*

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Palaeomagnetic, rock-magnetic and magnetic-fabric results are presented for Carboniferous and Lower Permian volcanic and volcaniclastic rocks (Merlewood Formation, Currabubula Formation, Werrie Basalt and Boggabri Volcanics), mainly from the Werrie Block of the northwestern Tamworth Belt, southern New England Orogen. Detailed thermal demagnetisation results (91 sites, 998 samples) show two groups of magnetic components with low (<400°C; LT) and high (500–700°C; HT) unblocking temperature ranges. Rock magnetic tests indicate the HT components to reside in magnetite, with single or pseudo-single domain and multidomain characteristics, and in hematite. Detailed demagnetisation up to 700°C demonstrates subtle directional differences between primary components, optimally cleaned and derived from the magnetite–hematite carrier (HT-P2), and pseudo-components, incompletely cleaned and derived from the mainly magnetite carrier (HT-P1). Directional results evidence three phases of magnetic overprinting which arose from: (i) a possible Middle–Late Cenozoic regional weathering event (LT); (ii) fluid movements associated with the Permo-Triassic Hunter–Bowen Orogeny (HT); and (iii) formation of the Bowen–Gunnedah–Sydney Basin system in the latest Carboniferous – Early Permian (HT). Magnetic fabric results show the K_{max} and K_{min} axes of the susceptibility anisotropy ellipsoids to better concentrate in stratigraphic than in geographical (*in situ*) coordinates, suggesting a primary depositional pattern. The prevailing north–south alignment of K_{max} axes changes to east–west towards the top of the Currabubula Formation. The sense of transport could not be established in absence of evidence for imbrication. Well-defined primary magnetisation components (HT-P2) have been determined for 39 of the 69 Currabubula Formation sites and for all sites of the Merlewood Formation (7), Werrie Basalt (9) and Boggabri Volcanics (6). Currabubula Formation site results have been combined into three mean-site results, all with positive fold tests at the 99% confidence level. They show good agreement with correlatable results from the northward adjacent Rocky Creek Block. Integration of the palaeomagnetic poles for the Currabubula Formation and a virtual geomagnetic pole for the Merlewood Formation with earlier determined Visean to Westphalian poles for the Rocky Creek Block outlines a Visean (Tournaisian?) to Stephanian pole path for the northwestern Tamworth Belt. Comparison with a preliminary pole path for the Rouchel and western Gresford Blocks shows no evidence for significant rotational deformation between the northwestern and southwestern Tamworth Belt. Comparison with the Northern Hastings Block indicates counterclockwise rotation of the latter relative to the northwestern Tamworth Belt over about 150°, in agreement with structural estimates.

KEY WORDS: Carboniferous, magnetic fabric, New England Orogen, oroclinal bending, palaeomagnetism, pole path, Tamworth Belt, Werrie Block.

INTRODUCTION

This is the second of a number of detailed palaeomagnetic studies on Carboniferous – Lower Permian successions of the Tamworth Belt, southern New England Orogen, covering the Rocky Creek, Werrie, Rouchel and Gresford Blocks (Figure 1). The main objectives of the studies are: testing oroclinal bending of the Tamworth Belt; defining a Late Palaeozoic pole path for the New England Orogen; comparing this pole path with Australian cratonic pole paths; and contributing to a representative Late Palaeozoic pole path for Australia and eastern Gondwanaland.

The Palaeozoic pole path for the Australian craton has been studied extensively for the past four decades with accumulation of a substantial dataset. The shape of the Late Palaeozoic segment has emerged as a problematic issue (Klootwijk *et al*. 1993; Schmidt *et al*. 1993; Van der Voo 1993; Klootwijk 1996a, 2002, 2003; McKerrow *et al*. 2000; Schmidt & Clark 2000; Tait *et al*. 2000; McElhinny *et al*. 2003; Schmidt & McElhinny 2003) with an important bearing on understanding regional and global tectonics, in particular the origin of the Kanimblan and Alice Springs Orogenies and the evolution of the Gondwana–Laurentia convergence into Pangea.

*Tables 5–7 [indicated by an asterisk (*) in the text and listed at the end of the paper] are Supplementary Papers; copies may be obtained from the Geological Society of Australia's website (http://www.gsa.org.au) or from the National Library of Australia's Pandora archive (http://nla.gov.au/nla.arc-25194).

The Tamworth Belt of the southern New England Orogen may contribute crucial data to the Late Palaeozoic pole path for Australia. Its Carboniferous succession is characterised from the Early Visean onwards by frequent felsic volcanic rocks, mainly ignimbritic, and these have demonstrated considerable potential for defining a detailed, well-dated (U–Pb zircon SHRIMP), pole path for the New England Orogen (Klootwijk 1996a, 2002; Opdyke *et al.* 2000; Geeve *et al.* 2002). Integration with the Australian cratonic pole path is the ultimate aim, but implementation depends on establishing the magnitude and timing of any movements between the New England Orogen and the Australian craton. This requires good control on the time of accretion of New England with the Lachlan–Thompson Orogen and on subsequent displacement and associated evolution of the southern New England oroclines.

The immediate task is defining the Late Palaeozoic pole path for the New England Orogen by integrating detailed studies on representative blocks of the Tamworth Belt. Compartmentalised tectonostratigraphic evolution of the Tamworth Belt has been established from extensive stratigraphic and SHRIMP dating studies (Roberts *et al.* 1991, 1995a, b; Roberts & Geeve 1999; Opdyke *et al.* 2000). Pilot palaeomagnetic studies from the main Tamworth Belt blocks (Rocky Creek, Werrie, Rouchel, Gresford, and Myall Blocks: Klootwijk 1996a) have shown no evidence for significant systematic rotational deformation between the northwestern (Rocky Creek, Werrie) and southwestern (Rouchel, western Gresford) parts of the Tamworth Belt, although the study by Geeve *et al.* (2002) on the southern Tamworth Belt implies a contrary opinion. The stratigraphic coverage of blocks from the northwestern (mainly Upper Carboniferous) and southwestern (mainly Lower Carboniferous) Tamworth Belt is partly overlapping and mainly complementary, and offers good prospects for integration and development of a well-defined, extensive, Carboniferous pole path. The present study advances towards this goal by integrating the present results for the Werrie Block and the earlier determined results for the Rocky Creek Block (Klootwijk 2002, 2003) into a common pole path for the Carboniferous of the northwestern Tamworth Belt.

This study focuses on the Currabubula Formation, of mainly Westphalian age, studied extensively in sections across the eastern and western limbs of the Werrie Syncline and in a basal section of the Kankool Basin (Figures 2–4). Three other formations have been studied in reconnaissance mode: the Visean(–Tournaisian?) Merlewood Formation (Werrie Syncline), the Lower Permian Werrie Basalt (near Wingen), and the Lower Permian Boggabri Volcanics (north of Boggabri). Their reconnaissance results have been included to help constrain the poorly defined Early Carboniferous and Early Permian segments of the New England pole path. The results from the mainly Westphalian Currabubula Formation are integrated with middle to Late Carboniferous results from the Rocky Creek Block, that is the Visean Caroda Formation and the Namurian to Stephanian succession of Clifden Formation, Rocky Creek Conglomerate and Lark Hill Formation. This detailed pole path for the northwestern Tamworth Belt is compared with a preliminary pole path for the southwestern segment of the belt (Rouchel and western Gresford Blocks: Klootwijk 1996a) in order to explore constraints on rotational deformation of the western region of the Tamworth Belt.

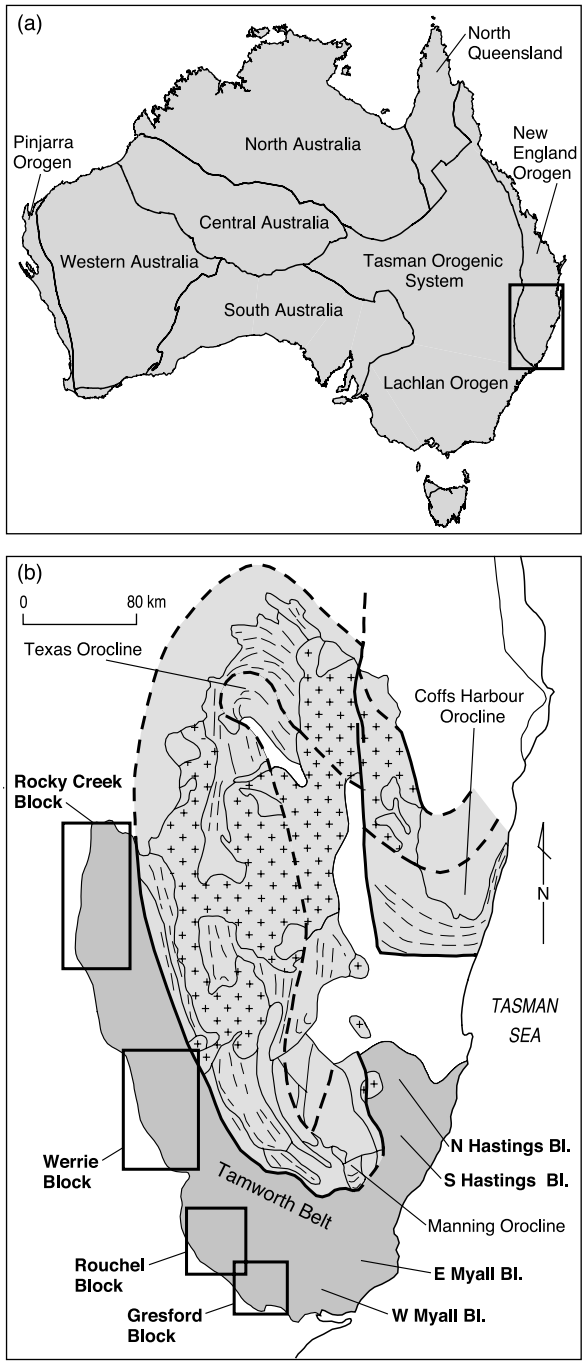


Figure 1 (a) Australian mega-elements, representing continent-scale groups of crustal elements (after Shaw *et al.* 1995, 1996). Inset refers to (b). (b) Schematic overview of the southern New England Orogen (after Korsch & Harrington 1987 figure 2) with outlines of the four studied tectonic units in the Tamworth Belt. Light grey shading, Devono-Carboniferous accretionary wedge rocks and younger granites; dark grey shading, Devono-Carboniferous forearc basin rocks.

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GEOLOGICAL SETTING

Regional framework

The Werrie Block of the northwestern Tamworth Belt is situated between the Rocky Creek Block to the north and the Rouchel and Gresford Blocks to the south (Figure 1b). The Carboniferous stratigraphies of these four blocks are distinctive from the Visean upwards, where they become dominated by felsic volcanic and volcaniclastic rocks. The Werrie Block shares an essentially common Lower Carboniferous succession with the Rocky Creek Block, but their Upper Carboniferous successions differ (Roberts & Geeve 1999). The Werrie Block Carboniferous succession also differs from the Rouchel Block. Roberts and coworkers (Roberts *et al.* 1991; Roberts & Geeve 1999; Opdyke *et al.* 2000) attributed this Late Palaeozoic compartmentalisation of the Tamworth Belt to several fundamental structures, such as the Maules Creek Lineament and the Murrurundi Lineament (Roberts & Geeve 1999 figure 1) which bound the Werrie Block to the north and south respectively. In contrast, McPhie (1987 figure 1) attributed the development of regional stratigraphies to individual volcanic centres, with limited distribution and transport of volcanic and volcaniclastic rocks.

The Carboniferous succession of the Werrie Block forms part of the Tamworth Belt forearc basin, situated between the Hunter–Mooki Thrust system to the west and the Central Complex (Flood & Aitchison 1988), or Tablelands Complex (Korsch 1977), to the east (Figure 2). Facies and petrology point to the former presence of a continental volcanic arc along the western margin of the Tamworth Belt, probably resulting from westwards subduction, and changing from andesitic volcanism in the Devonian to dacitic and finally rhyolitic volcanism in the Carboniferous (Roberts & Engel 1987). This arc has not been located in outcrop, although it has been linked to several surmised volcanic centres in the Hunter region (Roberts *et al.* 1991; Roberts & Geeve 1999). The arc may have been overthrust by the westwards transported Tamworth Belt during the Middle Triassic main phase of the Hunter–Bowen Orogeny, or may have been displaced by dextral strike-slip movement along the Mooki Fault System during the Early Permian (Harrington & Korsch 1985; Korsch & Harrington 1987; Harrington *et al.* 1989; Klootwijk in press a).

Stratigraphic overview

The Tamworth Belt stratigraphy is made up of a lower and upper succession, separated by a middle Carboniferous disconformity that spans a hiatus of variable duration from Late Visean to Early Westphalian (Figure 3) (Opdyke *et al.* 2000). The Rouchel Block hiatus may span less than 10 million years, whereas the Werrie Block hiatus spans about 20 million years. The Carboniferous stratigraphy of the Werrie Block's lower succession starts with the Tulcumba Sandstone of Early–Middle Tournaisian age of limited outcrop in the western limb of the Werrie Syncline (Figure 4a). More dominant in outcrop are the conformably overlying Namoi Formation of Late Tournaisian age, and the possibly disconformably over-

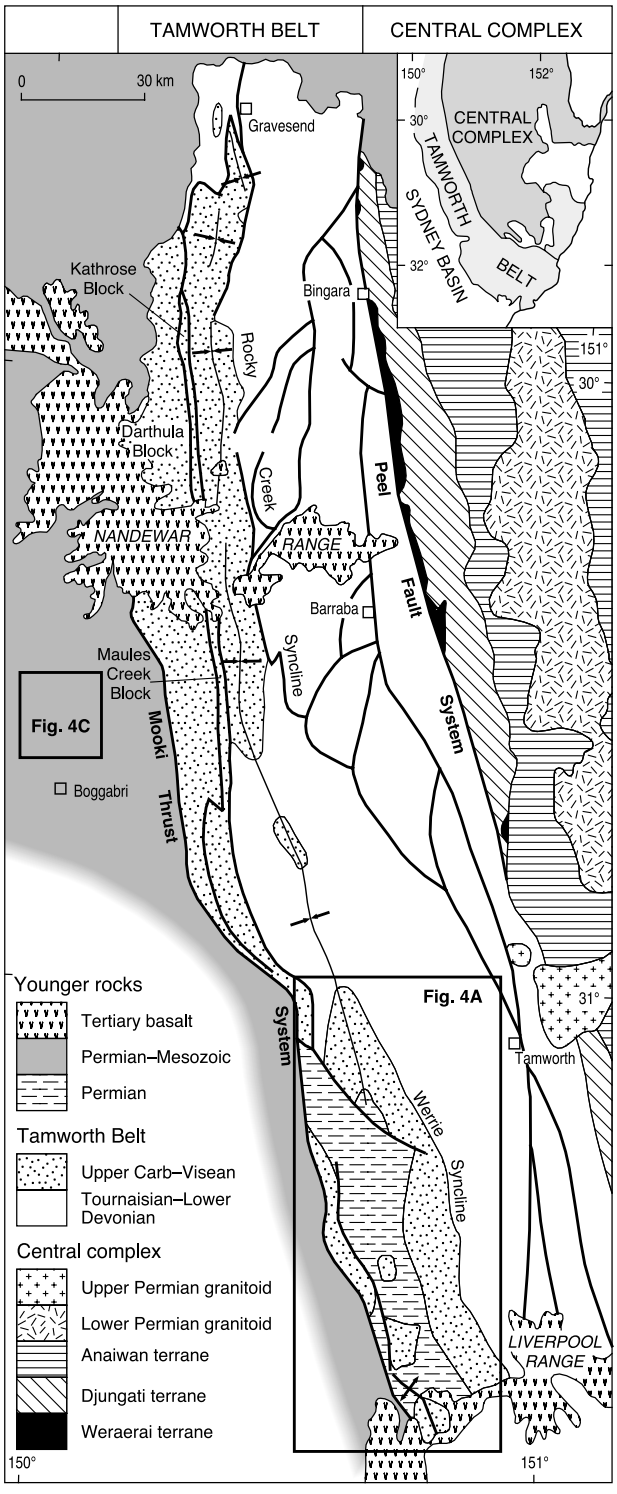
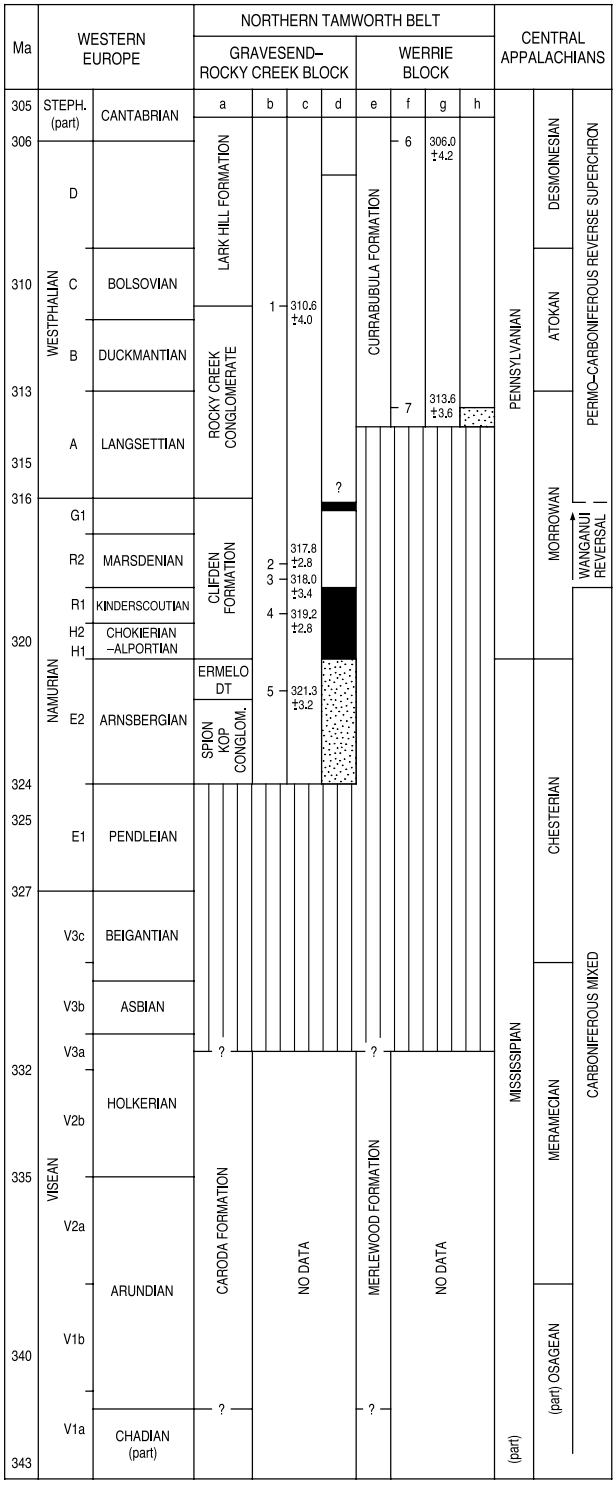


Figure 2 Regional geological map of the northwestern Tamworth Belt (after Opdyke *et al.* 2000 figure 1 and Pogson 1972). The large rectangle indicates the main study area, the Werrie Syncline of the Werrie Block, where the Currabubula and Merlewood Formations have been sampled. The Werrie Basalt has been sampled east of Wingen, to the north of the Liverpool Range (Figure 4b). The Boggabri Volcanics has been sampled north of Boggabri (small rectangle, Figure 4c), to the west of the Werrie Block and across the Mooki Thrust. Terranes of the Central Complex are indicated (after Flood & Aitchison 1988 figure 1).

lying Merlewood Formation of mainly Early to Middle Visean age (Figures 3, 4a). The Merlewood Formation contains volcanolithic sandstone, pyroclastics and three thick pyroxene andesite units (~200 m) at evenly spaced stratigraphic levels. Their mode of emplacement is uncertain (McKelvey & McPhie 1985; Opdyke *et al.* 2000). Some may be sills (Benson 1920; Moore 1974), others have been identified as flows (Carey 1937; Moore 1974; Opdyke *et al.* 2000). A flow origin has been assumed for the lower two units studied here (Figure 4).



The upper succession of the Werrie Block is made up of the Currabubula Formation that dominates the Werrie Syncline in outcrop. The Currabubula Formation lacks clear subdivision, in contrast to the Rocky Creek Block where four formations have been distinguished in the upper succession (Opdyke *et al.* 2000; Klootwijk 2002). The ~2.5 km thick succession is made up of mainly fluvial and fluvioglacial sediments, volcaniclastic rocks, conglomerates, more rarely diamictites and glaciogene laminated mudstone, and is characterised by abundant pyroclastics, silicic ignimbrites of distal origin and air-fall tuffs (McPhie 1983, 1987; McKelvey & McPhie 1985; Opdyke *et al.* 2000). Four ignimbrites have been defined as mappable members throughout the Werrie Syncline (Figures 5, 6) (McPhie 1983; Opdyke *et al.* 2000): stratigraphically they are (from the base) the Cana Creek Tuff, Iventure Ignimbrite, Taggarts Mountain Ignimbrite and Piallaway Trig Ignimbrite Members. These are relatively coarse-grained, crystal-rich, lithic, variably welded ignimbrites. Air-fall tuffs and ignimbrites of more local extent are generally crystal-poor and finer grained with less pumice.

McPhie (1983, 1987 figure 1) suggested correlation of the Taggarts Mountain Ignimbrite with the Plagyan Ignimbrite in the lower Rocky Creek Conglomerate of the Rocky Creek Block, but SHRIMP dating (Opdyke *et al.* 2000) showed the Taggarts Mountain Ignimbrite to be substantially younger than the Plagyan Ignimbrite. No stratigraphic overlap could be established between the main part of the Werrie Block and the Rouchel Block. The Kankool Basin of the southernmost Werrie Block is located north of the Murrurundi Fault Zone, but its volcanic succession is attributed entirely to a more southern eruption centre of the Rouchel Block (McPhie 1984a, 1987). The Kankool succession includes various thin ignimbrites of limited distribution. Three volcanic units are of mappable extent (Figure 4a) (McPhie 1984a figure 3), the uppermost being a more than 100 m-thick flow-banded silicic lava. The main part of the Kankool Basin succession pre-dates the four ignimbrite members of the Currabubula Formation (Figure 5) (McPhie 1983).

The chronostratigraphic position of the Currabubula Formation is constrained by two SHRIMP dates (Figure 3) (Opdyke *et al.* 2000) for volcanics near the base and the top of the formation. The oldest date is from an unnamed lava (313.6 ± 3.6 Ma) that directly overlies a basal conglomerate, formerly described as Coeypolly Conglomerate (McKelvey

Figure 3 Stratigraphic column for the Rocky Creek and Werrie Blocks of the northern Tamworth Belt in comparison with Carboniferous time units for western Europe and eastern North America. Compiled after Roberts *et al.* (1993, 1995a), Jenkins *et al.* (1993), Jones (1996), Roberts *in* Opdyke *et al.* (2000 figure 18), and P. J. Jones (pers. comm. 1999). a, e, studied stratigraphic succession (this paper; Klootwijk 2002); b, f, volcanics with SHRIMP U–Pb zircon age control (c, g: Opdyke *et al.* 2000); d, h, magnetostratigraphic control (Opdyke *et al.* 2000). Legend for b, f: 1, Eulowrie Pyroclastics of the Lark Hill Formation; 2, Eastons Arm Rhyolite; 3, Peri Rhyolite; 4, Wanganui Andesite; 5, Ermelo Pyroclastics; 6, Piallaway Trig Ignimbrite Member; 7, unnamed ignimbrite at the base of the Currabubula Formation. Legend for d, h: white, reverse polarity; black, normal polarity; stippled, undetermined.

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& McPhie 1985). The youngest date is from the Piallaway Trig Ignimbrite Member (306.0 ± 4.2 Ma). These dates constrain the age range of the formation from Westphalian A to just straddling the Westphalian–Stephanian boundary (Figure 3) and correlate the formation with the middle to upper part of the Rocky Creek Formation and the overlying Lark Hill Formation of the Rocky Creek Block. The Currabubula Formation is separated by a hiatus of essentially Stephanian age from the overlying Temi Formation and/or the Werrie Basalt of earliest Permian and Early Permian age respectively.

Early Permian volcanism of the Werrie Block and its adjacent regions differs markedly from the subduction-related volcanism of the Upper Carboniferous Currabubula

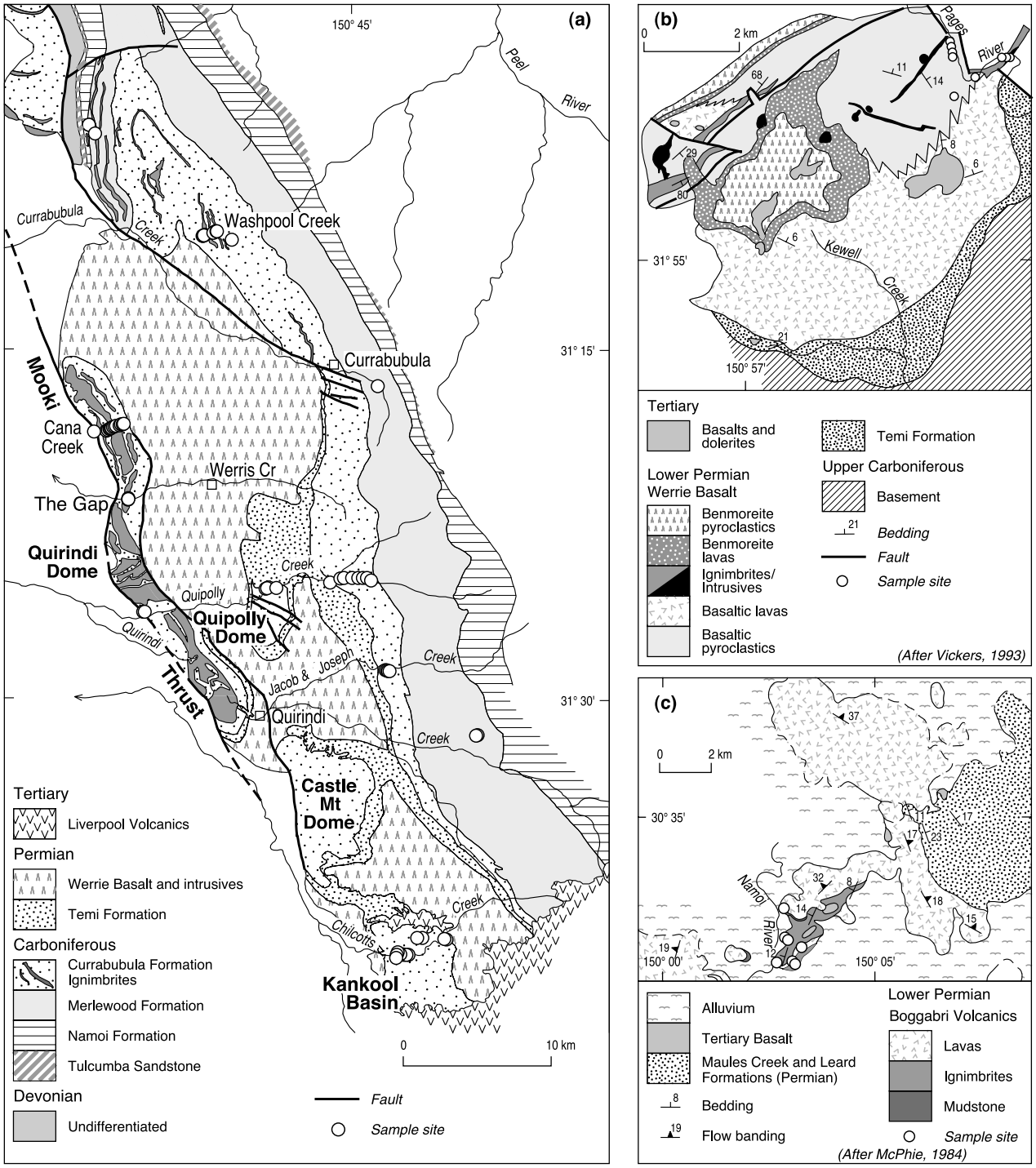


Figure 4 (a) Geological map of Late Palaeozoic and Cenozoic formations of the Werrie Syncline (see Figure 2) with palaeomagnetic sampling sites. Modified from Roberts (*in* Opdyke *et al.* 2000 figure 3); McPhie (1983, 1984a, b); Pogson and Hitchins (1973); Henley *et al.* (1999); R. Barnes (pers. comm. 2000). For simplicity Devonian formations have been omitted or are shown undifferentiated, but Tertiary volcanics are shown. (b) Geological map of the Werrie Basalt east of Wingen (after Vickers 1993 figure 1). (c) Geological map of the Boggabri Volcanics north of Boggabri (after McPhie 1984a figures 5, 6).

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Formation. It is bimodal and may reflect both the earlier subduction process and rifting of the Bowen–Gunnedah–Sydney Basin system (Leitch 1993; Shaw & Flood 1993; Caprarelli 1998; Jenkins & Collins 1998; Scheibner & Basden 1998) during the latest Carboniferous – earliest Permian. Three volcanic centres have been identified for the Werrie Basalt (Figure 4a): a centre near the north-western closure of the Werrie Syncline; the Warrigundi Volcanic Complex in the northern part of the Werrie Syncline (Flood *et al.* 1988); and a more southern centre near Wingen (Manser 1968; Vickers 1993) close to the Liverpool Range (Figure 4b). The latter centre has been studied here, and also by Schmidt (1988), Schmidt and Lackie (1993) and Lackie and Schmidt (1993), for reason of better outcrop. The Boggabri Volcanics crop out on the Boggabri Ridge with a ~600 m succession of alternating flow-banded lavas and ignimbrites (Figure 4c) (McPhie 1984a; Brownlow 1999). The succession is located west of the Mooki Thrust System, but is not necessarily autochthonous. The region is affected by east-dipping thrusts (Tadros 1993; Pratt 1996a; Beckett & Watson 1999) representing frontal splays of a detachment surface that may be located deeper than the Mooki Thrust System, or may be rejoining this thrust further eastwards.

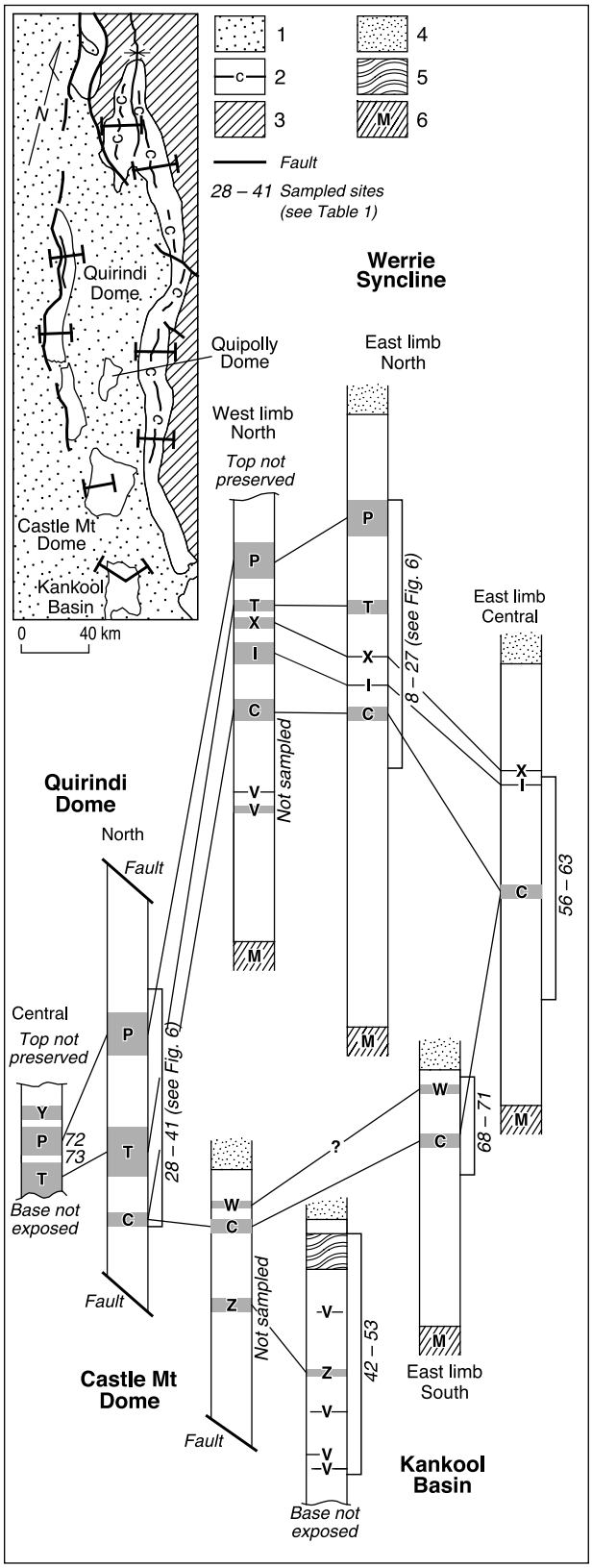


Figure 5 Schematic location of palaeomagnetic sampling sites on sections in the Werrie Syncline described by McPhie (1983 figures 1, 2). The section lines on the inset relate to the stratigraphic columns on the main figure, following approximate relative positions and structure indicators. The numbers to the right of the schematic stratigraphic columns refer to site numbers in Table 1. Legend (inset map): 1, cover consisting of Permian strata, Tertiary volcanics or alluvium; 2, Currabubula Formation and equivalents with Cana Creek Tuff Member indicated; 3, Early Carboniferous and Devonian strata; (stratigraphic sections): 4, Permian strata; 5, lava at The Basins in the Kankool Basin; 6, M, Merlewood Formation; C, Cana Creek Tuff Member; I, Iventure Ignimbrite Member; T, Taggarts Mountain Ignimbrite Member; P, Piallaway Trig Ignimbrite Member; V, W, X, Y, Z, locally mappable ignimbrite units (after McPhie 1983; figures 2, 4).

Tectonic framework and evolution, overprint potential

The Carboniferous succession of the Werrie Block is well exposed in the Werrie Syncline (Figures 2, 4a) (Carey 1934), a gently folded structure with minor, variable, plunge (Moore 1974; Moore & Roberts 1976), a prominent northern closure, and uncomplicated eastern limb. Its western limb is more complex, with splays of the Mooki Thrust System reflecting thin-skinned westwards thrusting of the succession across the Gunnedah Basin during the Middle Triassic (Harrington *et al.* 1989; Glen *et al.* 1993; Korsch *et al.* 1993; Woodward 1995; Pratt 1996a, b, 1998; Beckett & Watson 1999; Greentree & Flood 1999).

Tectonic activity in the New England Orogen spans the Late Palaeozoic and Mesozoic. Eight phases have been identified (Korsch & Harrington 1981; Harrington & Korsch 1985; Harrington *et al.* 1989; Veevers *et al.* 1994; Opdyke *et al.* 2000), four of them dominantly affecting the Tamworth Belt. Secondary magnetisations attributed to these four phases have been established in the Rocky Creek Block (Klootwijk 2002), and the Werrie Block may have been equally affected.

The oldest of the four phases relates to the Late Visean – Early Namurian hiatus of about 20 million years duration between the lower and upper succession (Opdyke

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et al. 2000). Secondary magnetisations caused by intraformational weathering or subsequent fluid flow along this erosional disconformity have been observed in the Rocky Creek Block (Klootwijk 2002) in the top of the lower succession (West Lynne Rhyodacite). None of the Werrie Block samples come from anywhere near the stratigraphic level of this erosional disconformity and such an overprint is not expected in the present study.

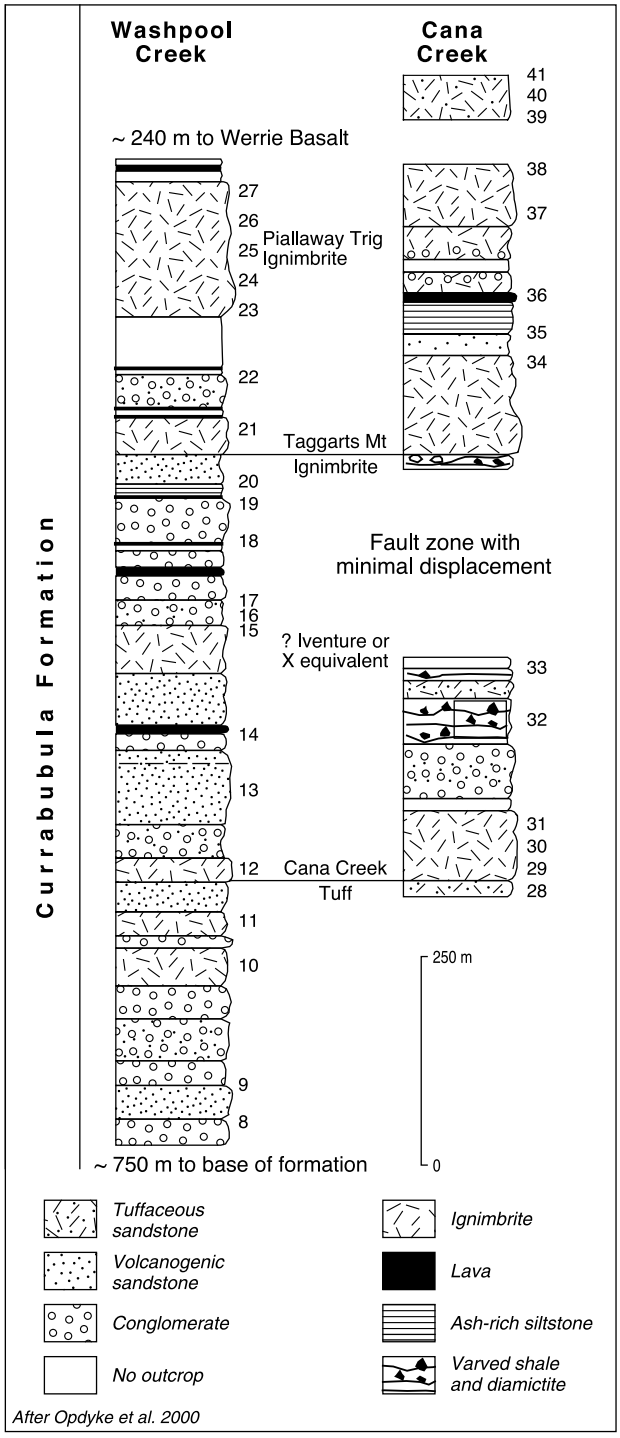


Figure 6 Stratigraphic sections of the Currabubula Formation in the Washpool Creek and the Cana Creek of the Werrie Syncline (after Roberts *in* Opdyke *et al*. 2000 figure 6). Numbers indicate palaeomagnetic sites in Table 1.

The second tectonic phase relates to rifting of the Bowen–Gunnedah–Sydney Basin system (Scheibner & Basden 1998) and subsequent formation of the Texas – Coffs Harbour and Manning Oroclines. Initiation of rifting is generally taken as latest Carboniferous, perhaps as early as 305 Ma (Holcombe *et al*. 1997; Roberts & Geeve 1999; Klootwijk in press a). The time of oroclinal deformation is debated, however, with various opinions (Aubourg *et al*. 1994) ranging from between mid-Namurian and latest Carboniferous (Geeve *et al*. 2002), latest Carboniferous (Murray *et al*. 1987; Lennox & Roberts 1988) and earliest Permian (Fergusson & Leitch 1993) to Late Permian (Collins 1991; Collins *et al*. 1993). Agreement seems emerging for an Early Permian event (Korsch & Harrington 1987; Lennox & Flood 1997; Roberts & Geeve 1999). Overprints attributed to this phase have been observed locally in the Rocky Creek Block (Klootwijk 2002). Geeve (2000a) also observed pre-folding palaeomagnetic overprints in the Rouchel, Gresford and Myall Blocks which he interpreted as of latest Carboniferous to earliest Permian acquisition.

The third tectonic phase relates to the Middle Triassic main phase of the Hunter–Bowen Orogeny, when the Tamworth Belt overthrusted westwards across the Bowen–Gunnedah–Sydney Basin system (Harrington *et al*. 1989; Glen *et al*. 1993; Veevers *et al*. 1994; Woodward 1995; Roberts & Geeve 1999) up to the eastern extension of the Lachlan Orogen (Korsch *et al*. 1993, 1997). Extensive magnetic overprinting attributed to the Hunter–Bowen Orogeny has been observed in the Rocky Creek Block (Klootwijk 2002).

The fourth tectonic phase relates to opening of the Tasman Sea during the Late Cretaceous (*ca* 95–84 Ma: Veevers *et al*. 1991; Symonds *et al*. 1996) and a preceding thermal event along the eastern seaboard (*ca* 105–95 Ma: Moore *et al*. 1986; O'Sullivan *et al*. 1996; Raza *et al*. 1998). Pervasive magnetic overprinting with the characteristic polarity and direction of the Cretaceous Normal Polarity Superchron has been observed widely along the eastern seaboard, and more inland along prevalent structural zones (Klootwijk 1985; Lackie & Schmidt 1996; Klootwijk in press a, b) and locally in the Rocky Creek Block (Klootwijk 2002).

Stratigraphic overlap, pole-path integration and comparison

The Currabubula Formation of the Werrie Block is correlated with the middle and upper parts of the Rocky Creek Conglomerate and with the Lark Hill Formation of the Rocky Creek Block (Figure 3) (Opdyke *et al*. 2000). Both blocks have been thrust westwards across the Bowen–Gunnedah–Sydney Basin system and have a dominant north–south tectonic grain in common. There is thus good potential to combine palaeomagnetic results from both blocks into a representative pole path for the northwestern Tamworth Belt, by adding further detail and wider stratigraphic coverage from the Werrie Block to the Rocky Creek block pole path (Klootwijk 2002 figure 15). Integration of results for the upper part of the Currabubula Formation will extend the Rocky Creek pole path upwards into the Stephanian, and results for the Merlewood Form-

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Table 1 Sampling details.

Site no.	Formation / Member / Flow / Site	Acronym	Lat. (°S)	Long. (°E)	Lithology	Section (cm)	Samples, specimens	Initial intensity (mAm^{-1})	Dip direction (°)	Dip (°)
Merlewood Formation										
1	pyroxene andesite, Belgamba	AQMA	31.2763	150.7700	purple coarse crystalline pyroxene andesite	180	10,11	177.5–340.8	269.2	30.3
2	same, ~50 m below AQMA	AQMB	31.2763	150.7700	same, dark blue-grey	640	11,12	44.0–752.8	238.6	30.0
3	pyroxene andesite, Wallabadah	AQMC	31.5288	150.8418	blue-grey coarse crystalline pyroxene andesite	685	9, 10	155.8–317.4	238.4	28.4
4	same, ~80 m above AQMC	AQMD	31.5288	150.8418	same, dark-blue, finer crystalline	145	10,11	34.6–171.8	238.4	28.4
5	pyroxene andesite, Glen Oak	AQOA	31.0865	150.5618	purple-blue coarse crystalline pyroxene andesite	2 m	10,11	533.6–6990.7	72.3	40.2
6	same, ~50 m S of AQOA	AQOB	31.0865	150.5618	same	~50	10,11	40601–102283	72.3	40.2
7	same, ~80 m above AQOA/B	AQOC	31.0937	150.5660	same, slightly weathered	~50	10,11	700.0–1913.7	87.0	39.7
Currabubula Formation										
8	**Washpool Creek** section, base	AQTE	31. 1703	150.6647	silicic volcaniclastics	~1 m	14,14	3.2–29.8	245.5	31.2
9	up section, 150 m upstream	AQTF	31. 1703	150.6647	silicic fine-grained tuffaceous	~50	12,13	5.6–31.6	245.5	31.2
10	up section	AQTG	31.1689	150.6624	silicic medium-grained tuffaceous	~50	12,13	39.6–103.0	245.5	31.2
11	up section	AQTH	31.1686	150.6617	silicic medium-grained tuffaceous	~2 m	12,12	2.1–29.1	246.3	31.5
12	up section	AQTI	31.1675	150.6608	fine-grained greenish tuff	130	12,13	0.18–6.2	246.3	31.5
13	up section	AQTJ	31.1675	150.6608	fine-grained greenish tuff	85	12,13	0.627–1.01	246.3	31.5
14	up section	AQTK	31.1664	150.6572	splintery fine-grained tuff	~20	12,13	4.8–23.7	239.1	31.4
15	up section	AQTL	31.1642	150.6547	medium to coarse-grained greenish tuffaceous	~150	12,13	191.0–323.0	245.9	34.2
16	up section, 10 m above AQTL	AQTD	31.1642	150.6547	same as AQTL	~10 m	10,11	62.4–301.1	235.0	37.6
17	up section	AQTM	31.1642	150.6547	same as AQTL & AQTD, top of bed	~50	12,13	19.3–745.2	245.6	34.2
18	up section, base Taggarts Mountain Ignimbrite Member	AQTN	31.1644	150.6536	greyish fine-grained tuff	100	12,13	1.5–4.4	246.7	32.7
19	Taggarts Mountain Ignimbrite Member	AQTO	31.1658	150.6528	crystal-rich ash flow	105	13,14	115.5–510.6	248.5	33.0
20	Taggarts Mountain Ignimbrite Member	AQTP	31.1658	150.6528	fine-grained tuff	60	12,13	58.7–667.6	248.5	33.0
21	Taggarts Mountain Ignimbrite Member	AQTC	31.1680	150.6490	crystal-rich ash flow	~5 m	10,11	225.6–843.6	249.5	36.7
22	Taggarts Mountain Ignimbrite Member	AQTQ	31.1669	150.6475	medium–fine-grained tuff	20	12,13	118.8–637.2	244.7	33.9
23	base Piallaway Trig Ignimbrite Member	AQTR	31.1665	150.6452	crystal-rich ash flow	140	12,13	761.2–1730.1	244.7	33.9
24	Piallaway Trig Ignimbrite Member	AQTS	31.1669	150.6448	crystal-rich ash flow	~1 m	12,12	1814.5–3320.5	244.7	33.9
25	Piallaway Trig Ignimbrite Member	AQTT	31.1670	150.6444	crystal-rich ash flow	~1 m	12,13	1963.3–7737.0	244.7	33.9
26	Piallaway Trig Ignimbrite Member	AQTB[a]	31.1670	150.6440	crystal-rich ash flow	~7 m	10,11	543.4–1333.3	244.7	33.9
27	Piallaway Trig Ignimbrite Member	AQTA[a]	31.1670	150.6435	crystal-rich ash flow	~3 m	10,11	144.1–877.5	244.7	33.9
28	**Cana Creek**, Cana Creek Tuff Member, section base	AQCN[b]	31.3097	150.5722	coarse-grained lithic greenish tuff	185	13,14	1.1–2.5	63.8	35.8
29	Cana Creek Tuff Member	AQCA	31.3090	150.5633	lithic greenish tuff	~10 m	10,11	0.14–0.72	63.8	35.8
30	Cana Creek Tuff Member	AQCB	31.3090	150.5633	lithic greenish tuff	~4 m	10,11	0.09–0.61	63.8	35.8
31	Cana Creek Tuff Member	AQCM	31.3081	150.5731	lithic greenish tuff	140	13,14	0.16–8.4	74.7	47.9
32	~120 m above AQCM	AQCL	31.3064	150.5747	silicic varved shale	~7.5 m	15,16	86.3–221.0	74.7	47.9
33	~30 m above AQCL	AQCK	31.3064	150.5747	silicic diamictite	~2 m	12,13	32.4–362.6	74.7	47.9
34	Taggarts Mountain Ignimbrite Member	AQCJ	31.3047	150.5797	crystal-rich ash flow	~2 m	12,13	68.9–205.6	61.2	54.2
35	up section	AQCI	31.3044	150.5808	mottled tuff	160	12,13	45.9–329.4	61.2	54.2

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Table 1 (Cont.)

Site no.	Formation / Member / Flow / Site	Acronym	Lat. (°S)	Long. (°E)	Lithology	Section (cm)	Samples, specimens	Initial intensity (mAm^{-1})	Dip direction (°)	Dip (°)
36	~10 m above AQCI	AQCH	31.3044	150.5811	mottled tuff	~3.5 m	13,14	7.6–60.5	61.2	54.2
37	base Piallaway Trig Ignimbrite Member or equivalent level	AQCG	31.3039	150.5831	purple crystal-rich ash flow	425	12,13	493.0–2251.3	62.3	54.3
38	Piallaway Trig Ignimbrite Member or equivalent level, ~30 m above AQCG	AQCF	31.3039	150.5831	purple crystal-rich ash flow	~50	12,14	595.8–844.4	62.3	54.3
39	up section	AQCE	31.3034	150.5878	medium-grained greenish tuffaceous	220	12,13	100.0–300.2	62.0	52.1
40	~60 m above AQCE	AQCD	31.3031	150.5878	fine to medium-grained greenish tuffaceous	~1 m	12,13	22.7–187.3	62.0	52.1
41	~60 m above AQCD	AQCC	31.3019	150.5872	graded ash bed	~0.5 m	12,13	171.4–498.3	62.0	52.1
42	**Chilcotts Creek, Kankool Basin** basal Currabubula Fm.	AQKE	31.6886	150.7928	reddish coarse-grained sandstone on top of conglomerates	25	12,13	22.4–214.2	253.9	10.0
43	up section	AQKF	31.6881	150.7911	base, flow-banded ignimbrite	~0.5 m	13,14	194.2–952.4	251.6	17.5
44	up section	AQKG	31.6867	150.7858	base, medium-grained tuffaceous	80	12,13	74.5–393.4	247.9	30.1
45	~10 m above AQKG	AQKH	31.6867	150.7858	top, finer grained tuffaceous	~3 m	12,13	27.1–253.4	247.9	30.1
46	up section	AQKI	31.6864	150.7856	pinkish medium-grained tuffaceous	~0.5 m	12,13	9.1–83.1	256.9	20.4
47	up section	AQKJ	31.6886	150.7842	thin-bedded rheomorphic greenish-grey tuff	~0.5 m	13,14	37.2–190.1	268.3	36.1
48	up section	AQKD	31.6885	150.7837	medium-grained tuffaceous	~0.5 m	10,11	7.6–77.6	255.8	41.9
49	~15 m above AQKD	AQKK	31.6897	150.7839	thin-bedded siltstone/mudstone	~1 m	14,15	126.4–693.4	268.3	36.1
50	~5 m above AQKK	AQKL	31.6897	150.7839	thin-bedded, fine-grained tuffaceous	~70	12,13	6.6–105.2	268.3	36.1
51	~15 m above AQKL	AQKC	31.6875	150.7838	thin-bedded, fine-grained tuffaceous	~40	10,11	101.3–639.3	255.8	41.9
52	**Chilcotts Creek Road**, up section	AQKM	31.6753	150.7989	medium to fine-grained tuffaceous	~0.5 m	12,13	36.0–206.6	60.4	4.0
53	up section	AQKN	31.6756	150.8033	medium to fine-grained tuffaceous	~0.5 m	12,13	10.9–135.9	59.1	5.3
54	**The Basins, Chilcotts Creek**	AQKB	31.6766	150.8187	flow-banded silicic lava	~2 m	10,10	29.1–159.8	80.8	26.1
55	~5 m above AQKB	AQKA	31.6766	150.8187	flow-banded silicic lava	~50	10,10	32.1–238.1	80.8	26.1
56	**Quipolly Creek-Dam**	AQQL	31.4170	150.7660	conglomerate	~5 m	9, 10	0.23–600.3	243.7	31.8
57	up section	AQQK	31.4153	150.7637	agglomeratic tuff	7 m	10,11	47.3–224.2	243.7	31.8
58	up section	AQQJ	31.4153	150.7590	green lithic agglomeratic tuff	~5 m	10,10	22.6–627.8	236.6	33.4
59	up section	AQQI	31.4153	150.7560	green lithic agglomeratic tuff	~1 m	10,11	0.95–45.8	236.6	33.4
60	up section	AQQH	31.4153	150.7526	vitric tuff	~0.5 m	10,10	9.3–623.0	239.5	24.6
61	up section	AQQG	31.4153	150.7470	agglomeratic tuff	~7 m	10,11	37.2–592.1	242.1	27.5
62	up section	AQQF	31.4153	150.7442	laminated tuff	~1.5 m	10,11	2.2–79.2	242.1	27.5
63	up section	AQQE	31.4180	150.7358	agglomeratic tuff	~3 m	10,11	58.5–445.0	239.6	22.0
64	up section (West limb)	AQQA	31.4221	150.6983	reddish-pinkish laminated fine-grained tuff	30	10,10	12.6–137.7	59.3	15.3
65	~8 m above AQQA	AQQB	31.4221	150.6983	reddish-pinkish fine-grained tuff	~0.5 m	10,10	53.3–110.8	59.3	15.3
66	~15 m above AQQB	AQQC	31.4221	150.6893	brownish medium-grained tuff	~1.5 m	10,11	175.5–353.1	59.3	15.3
67	~8 m above AQQC	AQQD	31.4221	150.6893	boulders (1–5) within greenish tuffaceous matrix (6–10)	~1 m	1–5, 6 6–10,5	54.0–22262	59.3	15.3
68	**Tara Gap (Jacob and Joseph Creek)**	AQGA	31.4815	150.7785	medium–fine-grained lenses in agglomeratic tuff	~9 m	9, 9	11.2–64.1	244.0	31.2
69	~150 m above AQGA	AQGB	31.4815	150.7775	agglomeratic tuff	~8 m	10,11	1.6–3.4	244.0	31.2
70	~40 m above AQGB	AQGC	31.4815	150.7772	brown-purple vitric tuff	~1.5 m	10,11	51.2–230.9	244.0	31.2

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Table 1 (Cont.)

Site no.	Formation / Member / Flow / Site	Acronym	Lat. (°S)	Long. (°E)	Lithology	Section (cm)	Samples, specimens	Initial intensity (mAm^{-1})	Dip direction (°)	Dip (°)
71	~250 m above AQGC	AQGD	31.4815	150.7758	pinkish sandstone (1–3) and vitric tuff (4–11)	~4 m	1–3, 4 4–11,8	6.5–67.5	244.0	31.2
72	**Lower Quipolly**, Piallaway Trig Ignimbrite Member	AQPB	31.4396	150.6013	yellow-pinkish-reddish ignimbrite	~2 m	10,11	122.1– 1917.7	240.8	24.6
73	~30 m above AQPB	AQPA	31.4396	150.6013	yellow-pinkish-reddish ignimbrite	~1.5 m	10,11	4.2–22.6	240.8	24.6
74	**The Gap**, unnamed basal tuff	AQUA	31.3568	150.5897	grey-purple tuff	~2 m	10,11	50.7–148.2	124.3	26.8
75	up section, main flow	AQUB	31.3570	150.5900	dark blue crystal-rich ash flow	~5 m	10,11	552.6– 1775.2	124.3	26.8
76	up section, silicic siltstone	AQUC	31.3575	150.5892	silicic medium-grained tuffaceous	~1 m	10,11	43.2–304.0	91.7	32.8
Werrie Basalt										
77	Pages River, E of Wingen	AQWA	31.8856	150.9766	heavily jointed fine-grained basalt	~1 m	10,11	390.8–5034.8	254.1	24.1
78	up section	AQWC	31.8852	150.9755	haematised basaltic pyroclastics	~1 m	10,11	527.8–8854.5	254.1	24.1
79	up section	AQWB	31.8852	150.9740	purple-grey crystal-rich ignimbrite?	~10 m	10,11	281.9–707.4	254.1	24.1
80	up section	AQWD	31.8856	150.9713	similar to AQWB	~1 m	10,11	241.4–520.5	117.8	23.1
81	up section?	AQWE	31.8877	150.9671	bluish-purple basaltic pyroclastics	~1 m	10,11	414.8–4496.2	32.0	7.0
82	up section	AQWI	31.8859	150.9637	massive basaltic pyroclastics	~1 m	10,11	28.1–348.8	17.3	9.0
83	up section	AQWH	31.8853	150.9634	massive basaltic pyroclastics	~1 m	10,11	186.5–1096.4	352.9	8.0
84	up section	AQWG	31.8847	150.9631	compact basaltic pyroclastics	~1 m	10,11	184.3–2032.4	353.4	17.0
85	up section	AQWF	31.8804	150.9628	blocky, jointed basaltic pyroclastics	~2 m	10,11	541.8–7216.9	347.3	7.0
Boggabri Volcanics										
86	NE of Boggabri	AQBA	30.6324	150.0524	lithic tuff	~1 m	11,12	30.8–3629.4	310.8	47.3
87	~50 m above AQBA	AQBB	30.6297	150.0556	flow-banded lava	~5 m	10,10	15.3–243.0	323.0	31.3
88	up section	AQBD	30.6341	150.0489	purple-grey flow-banded lava	~1 m	10,11	33.4–991.9	313.2	18.9
89	up section	AQBE	30.6252	150.0514	flow-banded lava?	~1 m	9, 10	99.0–6255.5	307.0	12.0
90	~30 m above AQBE	AQBF	30.6250	150.0500	purplish tuff	~1 m	10,11	266.1–3154.9	313.5	9.9
91	up section	AQBC	30.6189	150.0503	purplish crystal-rich tuff	~2 m	10,11	114.4–5275.1	322.4	26.0

[a]Bedding for sites AQTA and AQTB redetermined in 1997 as per sites AQTQ-AQTT.
[b]Bedding for site AQCN redetermined in 1997 as per sites AQCA and AQCB.

ation may extend the pole path downwards, possibly into the Late Tournaisian. Such an integrated pole path for the northwestern Tamworth Belt can be compared with the preliminary pole path for the Rouchel and western Gresford Blocks of the southwestern Tamworth Belt (Klootwijk 1996a) in order to determine rotational deformation, if any. The Lower Carboniferous stratigraphy is more extensively developed in the southwestern part of the Tamworth Belt, whereas the Upper Carboniferous stratigraphy is better developed in the northwestern part. However, there is ample stratigraphic overlap to facilitate comparison of pole-path trajectories for the two regions. Preliminary results so far have shown no evidence for significant systematic rotation between the two regions (Klootwijk 1996a). If sustained, this will allow direct integration of the extensive and largely complementary pole-path trajectories for the northwestern and the southwestern parts of the Tamworth Belt into an overall Carboniferous pole path for the New England Orogen.

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SAMPLING

Sampling of the Werrie Block succession concentrated on ignimbritic volcanic and volcaniclastic rocks following pilot studies in the Tamworth Belt (Klootwijk 1985) showing that ignimbrites can retain their primary magnetisation despite pervasive regional overprinting. Reconnaissance sampling was carried out in 1987 and 1990 with preliminary palaeomagnetic results listed and discussed in Klootwijk (1996a, b). Follow-up sampling of selected sections was carried out in 1997. The present study focuses on four sections through the Westphalian to earliest Stephanian Currabubula Formation of the Werrie Syncline (69 sites, 778 samples: Table 1). In addition, reconnaissance studies were carried out on the Lower–Middle Visean Merlewood Formation (7 sites, 70 samples) and on the Lower Permian Boggabri Volcanics (6 sites, 60 samples) and Werrie Basalt (9 sites, 90 samples).

All samples were collected with a portable water-cooled drill using stainless steel barrels for reduced magnetic contamination. Drillcores of 2.5 cm diameter and up to 15 cm long were orientated with an integrated magnetic and solar orientation device. The sampling concentrated on well-developed stratigraphic sections across regionally and tectonically simple, representative, structures on both limbs of the Werrie Syncline. Bedding control for the volcanic rocks was routinely determined from intercalated volcaniclastic beds and/or bedding indicators within the ignimbrites.

Sampled sites

The geological setting of the sampled regions is sketched in Figure 4 for the Werrie Syncline, Werrie Basalt and Boggabri Volcanic. The stratigraphic position of the sampled sites in the Werrie Syncline (Figure 4a) is indicated on sections (Figures 5, 6) after McPhie (1983) and Opdyke *et al*. (2000) respectively. Site-specific detail on formation, location, lithology and bedding attitude is summarised in Table 1.

MERLEWOOD FORMATION

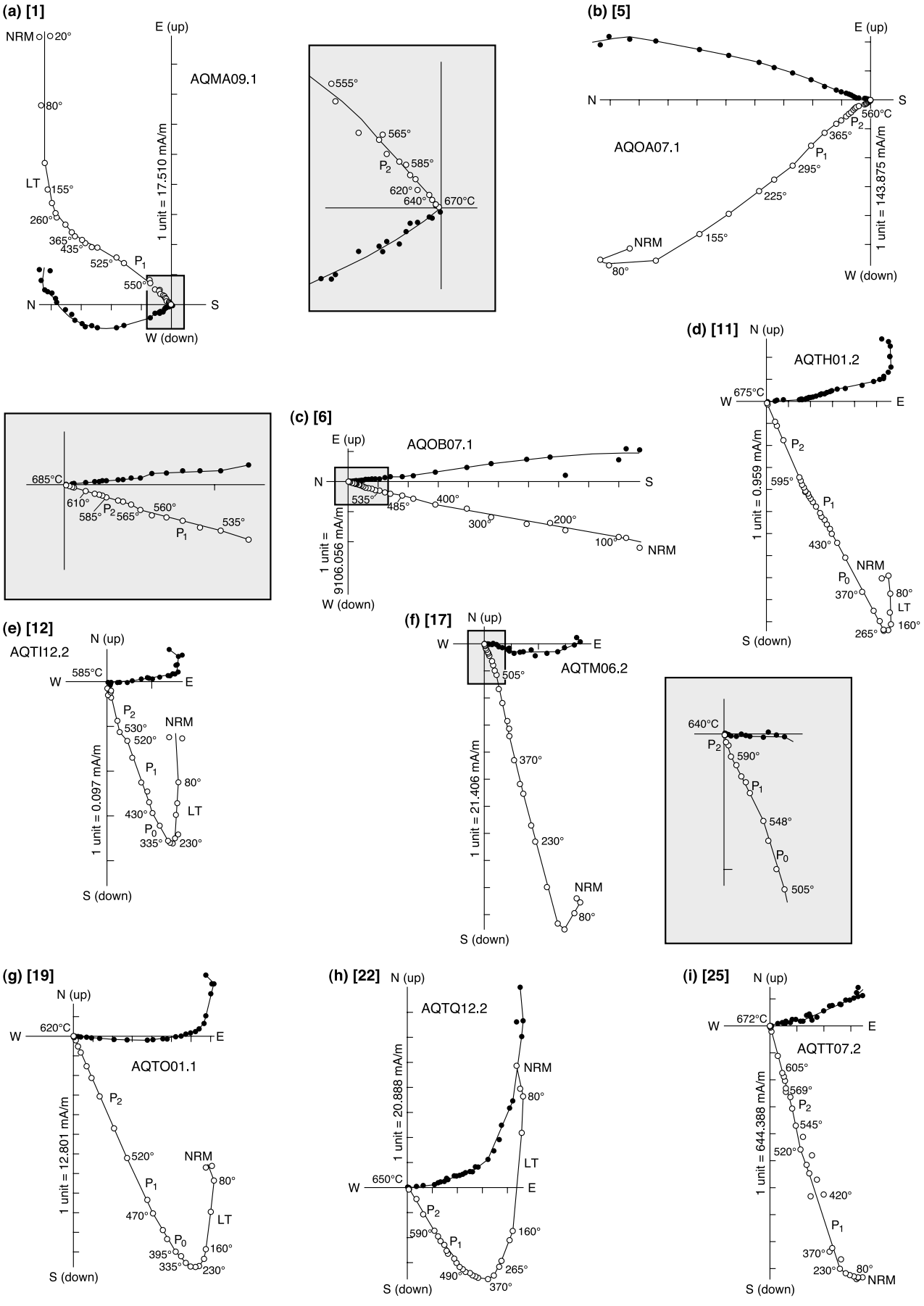
A moderately westwards dipping andesite unit in the middle of the Merlewood Formation was sampled on the eastern limb of the Werrie Syncline, at Belgamba Mountain, east of Currabubula. One site (site 1) was sampled in the purple, weathering-prone, upper part of the prominent volcanic horizon exposed in a creek bed. A second site (2) was sampled in dark-blue crystal-rich volcanics in a railway cutting, about 50 m lower in the stratigraphy. Another two sites were sampled north of Wallabadah in the coarser-grained base (3) and finer-grained upper part (4) of the andesite unit. In the northwestern closure of the Werrie Syncline, south of the Merlewood Homestead, three sites (5–7) were sampled in moderately east-dipping andesites in the lower part of the Merlewood Formation: sites 5 and 6 in the coarse-grained centre of the basal andesite and site 7 in an overlying more weathering-prone coarse-grained andesite.

CURRABUBULA FORMATION

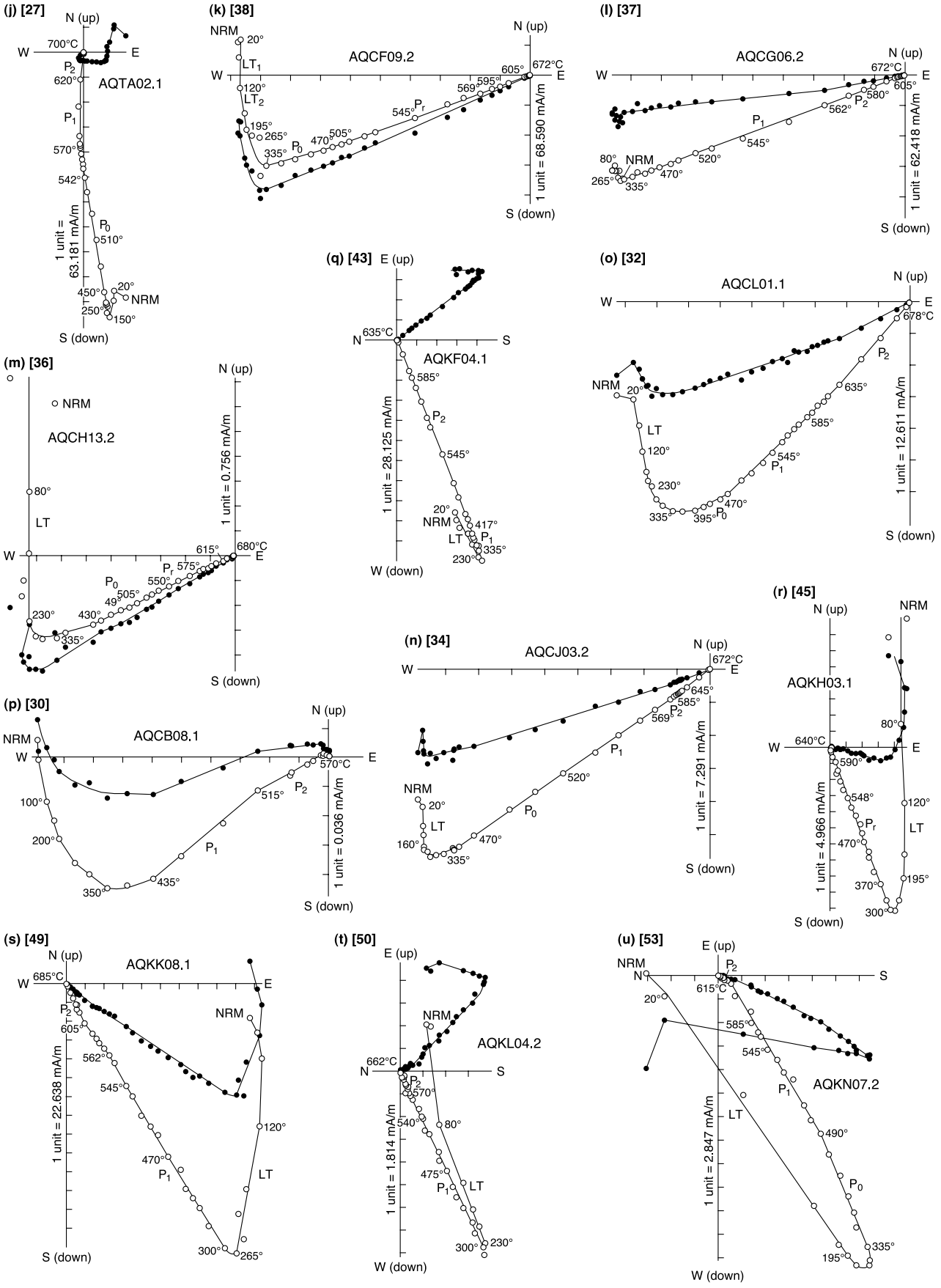
Sampling of the Currabubula Formation focussed on four creek sections across both limbs of the Werrie Syncline (Figure 4a): Washpool Creek on the northeastern limb; Quipolly Creek on the central eastern limb; Cana Creek through the Quirindi Dome on the tectonically reduced western limb; and Chilcotts Creek across the Kankool Basin in the southwestern part of the Werrie Syncline. Limited additional sampling was carried out along Jacob and Joseph Creek through Tara Gap on the eastern limb of the Werrie Syncline, and at The Gap and at the Lower Quipolly Creek through the Quirindi Dome on the western limb. The Werrie Basalt was sampled east of Wingen along the Pages River, and the Boggabri Volcanics in hill sections northeast of Boggabri to the east of the Namoi River.

Sampling of the west-dipping Washpool Creek section (Figures 4–6; Table 1) proceeded stratigraphically upwards from east to west, starting in siliceous volcaniclastic and tuffaceous beds (sites 8–11) directly above the massive conglomeratic horizons in the eastern part of the section, continued through greenish tuff of Cana Creek Tuff Member appearance (12, 13) and greenish tuffaceous (14–17) and tuff horizons (18–20) up to the Taggarts Mountain Ignimbrite Member flow (20), followed by a silicified tuff (22) and five sites through the Piallaway Trig Ignimbrite Member flow (23–27).

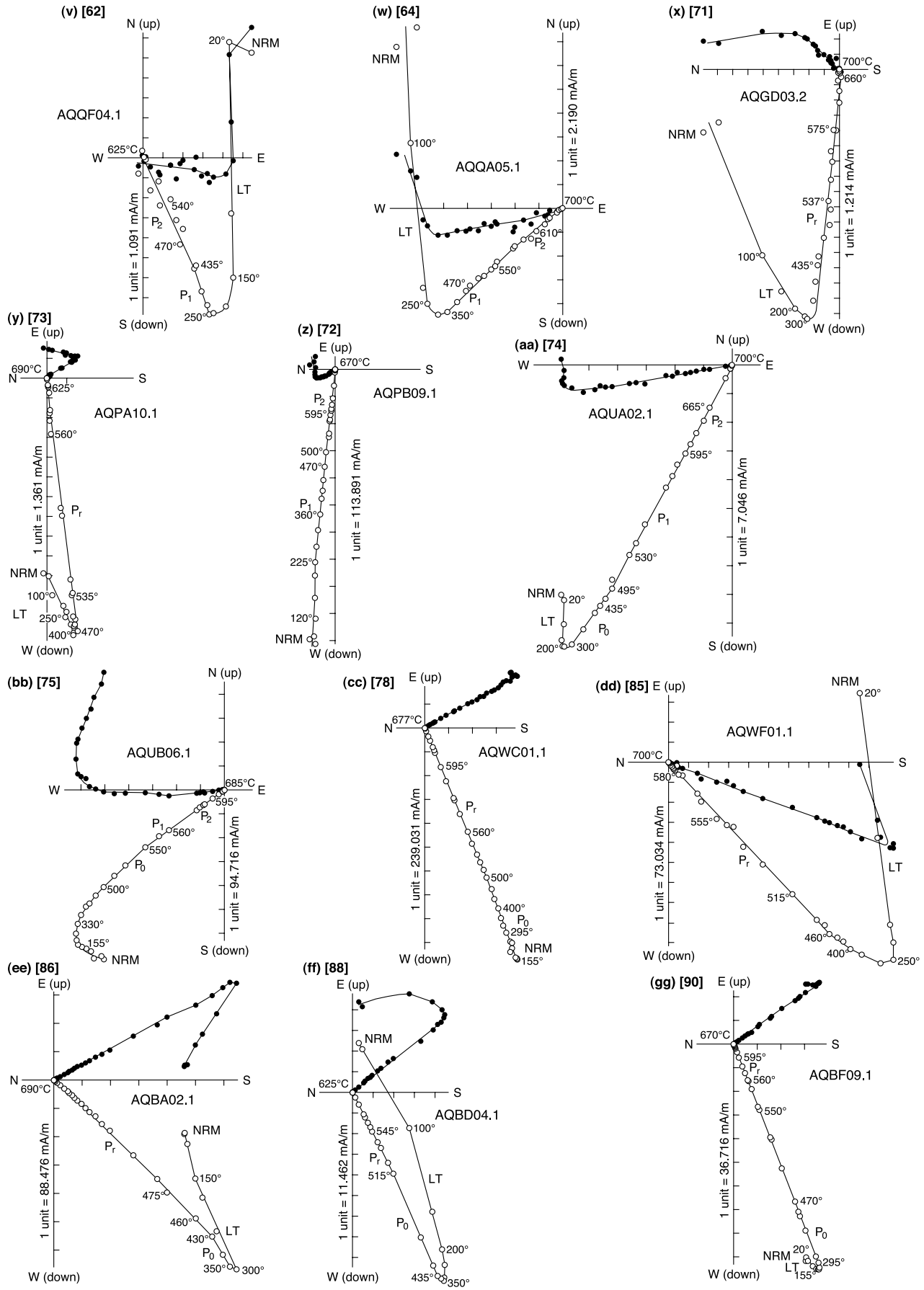
The east-dipping Cana Creek section through the northern part of the Quirindi Dome is the tectonically most affected section sampled, with bounding thrust planes to the east and west of the dome (Figure 4a) and another fault zone midway through the section (Figure 6). Tectonically affected parts of the section with unrepresentative bedding were avoided in the sampling, but fluid flow along these fault zones may have magnetically affected the section well beyond the visibly tectonised areas. Sampling started in the west with four sites through the Cana Creek Tuff Member (28–31) and two sites in overlying silicified varves (32) and silicified diamictites (33). A tectonised part of the section, east of the weir above site 33, was skipped and sampling resumed in a purple ash-flow probably representing the Taggarts Mountain Ignimbrite Member (34). Two



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sites were sampled in a devitrified mottled tuff (35, 36), directly overlying silicified varves, and a further two sites were sampled at the eastern entrance of Cana Creek into Quirindi Dome in an unnamed purple ash-flow (37, 38), apparently stratigraphically below the Piallaway Trig Ignimbrite Member. A further three sites were sampled in the creek bed east of Quirindi Dome, in greenish tuffaceous beds (39, 40) and in an overlying graded ash bed (41).

Stratigraphically upwards sampling of the Chilcotts Creek section of the Kankool Basin started in a sandstone bed (42) directly overlying massive conglomerate horizons in the centre of the basin, followed westwards by an ignimbrite (43) of weathered appearance, an upwards-fining volcaniclastic bed [44 (base)–45 (top)], a pinkish tuffaceous bed (46), greenish-grey medium-grained tuffaceous beds (47, 48), and fine-grained laminated tuffaceous beds (49–51). Bedding attitude of the section increased westwards from minor (42) to moderately steep west-dips (51). Another two sites (52, 53) were sampled along Chilcotts Creek Road in slightly eastwards dipping fine- to medium-grained tuffaceous beds whose stratigraphic position relative to the west-dipping section was not established, but is presumably higher up in the section. A further two sites (54, 55) were sampled in Chilcotts Creek at The Basins, the eastern entry to the Kankool Basin, in flow-banded silicic lavas overlying the succession described above.

The moderately southwest-dipping Quipolly Creek section through the eastern limb of the Werrie Syncline was sampled stratigraphically upwards from a basal conglomerate (56), previously described as the Coeypolly Conglomerate, in mainly agglomeratic (57–59, 61, 63), laminated (62) or vitric (60) tuff and tuffaceous beds, with sites 58 and 59 identified as the characteristically greenish Cana Creek Tuff Member. Another four sites (64–67) were sampled at Quipolly Dome, in a quarry below Quipolly Dam, comprising a fine-grained, partly laminated, slightly west-dipping, tuff succession in the uppermost exposed part of the Currabubula Formation.

Another short section through the eastern limb of the Werrie Syncline was sampled in the Jacob and Joseph Creek at Tara Gap in moderately west-southwest-dipping agglomeratic (68, 69) and vitric (70, 71) tuff beds. A further two short sections were sampled at the Quirindi Dome on the more tectonised western limb of the Werrie Syncline. The Piallaway Trig Ignimbrite Member (72, 73) was sampled here in moderately west-southwest-dipping beds along the Lower Quipolly Creek. At the The Gap crossing of the Quirindi Dome one site each was sampled in a pinkish-reddish unwelded tuff (74), an overlying thick dark-blue crystal-rich ash-flow (75), and a silicified siltstone (76) high up in the section. The unnamed tuff (74) and crystal flow (75) are of local extension with moderate west-southwest-dip and their stratigraphic position is apparently above the Cana Creek Ignimbrite Member and probably also above the Taggarts Mountain Ignimbrite Member (McPhie 1983).

Figure 7 Zijderveld (1967, 1975) plots of representative specimens during thermal demagnetisation. The plots are shown in geographical (*in situ*) coordinates for ease of identification of post-folding overprints. The circles indicate successive positions of the end-points of the resultant magnetisation vector during progressive demagnetisation. Open circles indicate projections on the vertical east–west or vertical north–south plane and dots indicate projections on the horizontal plane. Numbers denote successive peak temperatures in °C of the heating cycles. LT indicates the low temperature overprint. $P2/P_r$ indicate the primary component as per Table 2 or overprint components as per Table 6*. $P1/P_0$ indicate incompletely cleaned (pseudo)-components as per Table 5* (see discussion in the text). A shaded box indicates enlargement. a–c, Merlewood Formation; d–bb, Currabubula Formation; cc, dd, Werrie Basalt; ee, gg, Boggabri Volcanics. Numbers in parentheses following the headings a–gg relate to site numbers in Table 1.

WERRIE BASALT, BOGGABRI VOLCANICS

The Werrie Basalt and Boggabri Volcanics were sampled in reconnaissance mode only. Nine Werrie Basalt sites were sampled in basaltic lavas (77, 78, 81–85) and porphyritic ignimbrites (79, 80) along the Pages River (Figure 4b). Six Boggabri Volcanics sites were sampled in ignimbrites and felsic lavas (86–91) along profiles (Figure 4c) described by McPhie (1984a) north of Boggabri and east of the Namoi River.

MAGNETIC TREATMENT, ANALYSIS AND INTERPRETATION METHODS

Sample preparation, demagnetisation, rock magnetic treatment, magnetic fabric determination, and analysis of results closely followed the procedures detailed and discussed for the Rocky Creek Block study (Klootwijk 2002 pp. 385–389; Klootwijk 2003; Schmidt & McElhinny 2003) to which the reader is referred. Specific for the present study are that: thermal demagnetisation was continued up to 705°C (up to 37 steps), as required for the magnetic intensity to drop below the practical measurement limit of the ScT magnetometer (2×10^{-3} mAm^{-1}); isothermal remanent magnetisation (IRM) was acquired with a PYE electromagnet (up to 500 mT) and a Newport electromagnet (up to 2000 mT); alternating field (AF) demagnetisation was generally limited to a single specimen per site, as a first indication of magnetisation content and behaviour; and thermal demagnetisation, providing better separation of multicomponent magnetisations than AF-demagnetisation, was routinely carried out on at least one specimen per sample.

PALAEOMAGNETIC RESULTS

Component structure

Progressive thermal demagnetisation of the four formations generally showed two thermally distinct groups of magnetic components (Figure 7) that unblocked at low temperatures (LT-components) and high to very high temperatures (HT-components).

LT component

Less than half the sites showed consistent unblocking below 300–400°C and below 30 mT of a single LT component

Table 2 Primary, high temperature (HT), components in stratigraphic coordinates.

Site no.[a]	Comp.	Decl. (°)	Incl. (°)	KD[b]	α95[c] (°)	Sp.[d]	Polarity	Lat. (°S)	Long. (°E)	Palaeolat. (°S)	DP[e] (°)	DM[e] (°)
1	P2	4.8	–44.2	133.4	4.0	11	N	83.2	10.0	25.9	3.1	5.0
2	P2	12.6	–22.5	31.5	16.6	4	N	67.2	4.2	11.7	9.3	17.6
3	P	7.6	–23.9	17.1	12.8	9	N	69.7	352.6	12.5	7.2	13.5
4	P	344.8	–22.5	55.5	6.5	10	N	65.7	292.2	11.7	3.7	6.9
5	P2	24.9	0.8	19.8	15.4	6	N	50.6	12.2	0.4 N	7.7	15.4
6	P2	186.2	15.8	29.4	8.6	11	R	66.3	346.0	8.1	4.5	8.8
7	P2	151.1	4.9	16.0	17.3	6	R	50.4	281.2	2.5	8.7	17.3
(8	*P2*[a]	*218.9*	*62.6*	*9.1*	*26.9*	*5*	*R*	*57.0*	*94.6*	*44.0*	*32.8*	*42.0)*
(9	*P2*[a]	*184.0*	*79.1*	*17.8*	*14.7*	*7*	*R*	*52.2*	*148.3*	*68.9*	*26.5*	*27.9)*
10	P2	237.5	79.7	1231.6	1.2	13	R	40.1	128.5	70.0	2.1	2.3
11	P2	225.3	75.9	836.6	1.4	13	R	47.1	122.7	63.3	2.4	2.6
12	P2	240.2	76.7	167.9	3.4	12	R	40.5	121.4	64.6	5.8	6.2
13	P2	238.3	69.6	131.4	3.6	13	R	43.1	106.5	53.3	5.3	6.2
(15	*P2*[a]	*264.5*	*80.7*	*10.1*	*18.3*	*8*	*R*	*31.2*	*129.4*	*71.9*	*33.9*	*35.2)*
(16	*P2*[a]	*168.5*	*76.1*	*12.7*	*15.0*	*9*	*R*	*56.6*	*159.9*	*63.7*	*25.6*	*27.7)*
17	P2	225.9	68.2	28.3	8.3	12	R	50.9	105.3	51.3	11.8	14.0
(18	*P2*[f]	*240.1*	*81.3*	*86.4*	*4.5*	*13*	*R*	*38.3*	*131.7*	*72.9*	*8.4*	*8.7)*
(19	*P2*[f]	*207.2*	*76.4*	*802.6*	*1.4*	*15*	*R*	*52.9*	*131.4*	*64.2*	*2.3*	*2.5)*
(21	*P*[f]	*241.1*	*67.7*	*90.7*	*4.8*	*11*	*R*	*41.5*	*102.7*	*50.6*	*6.7*	*8.1)*
(22	*P2*[ag]	*182.2*	*80.8*	*24.1*	*8.6*	*13*	*R*	*49.0*	*149.6*	*72.1*	*16.0*	*16.6)*
(23	*P2*[f]	*212.5*	*82.7*	*181.3*	*3.1*	*13*	*R*	*42.9*	*140.1*	*75.6*	*5.9*	*6.0)*
24	P2	244.6	57.0	654.4	1.6	13	R	37.4	86.5	37.6	1.7	2.4
25	P2	241.8	62.6	453.6	2.0	13	R	40.6	94.0	43.9	2.4	3.1
26	P	235.2	61.5	300.0	2.6	11	R	45.2	91.6	42.6	3.1	4.1
27	PH	239.4	54.9	279.7	3.3	8	R	40.9	82.5	35.4	3.3	4.7
29	P2	224.1	67.5	29.8	9.0	10	R	52.4	104.0	50.4	12.5	15.0
30	P2	217.1	72.4	20.4	11.0	10	R	53.5	117.5	57.5	17.2	19.4
35	P2	259.5	68.3	35.5	8.2	10	R	30.2	105.4	51.4	11.7	13.8
36	P2	218.3	68.6	19.4	10.1	12	R	55.3	108.4	51.9	14.5	17.1
37	P2	282.7	67.7	248.1	2.8	12	R	16.4	110.5	50.7	3.9	4.6
38	P2	260.9	72.3	332.8	2.2	14	R	30.7	112.4	57.5	3.4	3.9
39	P2	242.5	68.5	8.4	16.0	12	R	40.7	104.2	51.8	22.8	27.0
40	P2	230.1	70.9	10.8	14.5	11	R	47.7	110.1	55.2	22.0	25.3
41	P2	259.3	75.6	28.2	8.8	11	R	32.3	118.5	62.9	14.8	16.1
43	P2	177.9	68.6	491.6	1.8	14	R	69.8	154.5	51.9	2.6	3.0
44	P2	206.8	71.9	14.5	11.8	12	R	58.9	122.3	56.8	18.3	20.8
45	P2	192.0	68.0	125.9	3.9	12	R	68.7	130.0	51.1	5.5	6.5
46	P2	228.6	69.4	269.9	2.5	13	R	49.3	107.0	53.0	3.7	4.3
(47	*P2*[h]	*210.6*	*47.5*	*40.0*	*6.4*	*14*	*R*	*63.4*	*62.1*	*28.6*	*5.4*	*8.3)*
48	P2	210.1	61.3	125.2	4.1	11	R	63.9	93.5	42.4	4.8	6.3
49	P2	206.9	71.8	113.9	3.7	14	R	58.9	122.1	56.7	5.8	6.6
50	P2	223.4	68.8	34.6	7.5	12	R	52.6	106.9	52.2	10.7	12.7
51	P2	241.4	67.7	68.9	5.5	11	R	41.7	102.6	50.6	7.7	9.3
52	P2	190.7	69.6	217.2	2.8	13	R	67.0	134.3	53.3	4.1	4.8
53	P2	211.2	67.8	32.4	7.7	12	R	60.1	109.6	50.7	10.8	13.0
(54	*P2*[h]	*103.2*	*80.5*	*483.5*	*2.2*	*10*	*R*	*34.0*	*172.7*	*71.5*	*4.1*	*4.2)*
(55	*P2*[h]	*134.9*	*77.3*	*290.6*	*2.8*	*10*	*R*	*46.5*	*175.8*	*65.8*	*5.0*	*5.3)*
58	P2	173.8	67.2	19.4	15.6	6	R	70.9	163.0	50.0	21.6	26.0
59	P2	232.6	78.4	9.7	17.4	9	R	42.8	126.6	67.7	31.1	32.9
(61	*P2*[i]	*305.3*	*81.9*	*14.1*	*18.5*	*6*	*R*	*21.5*	*136.8*	*74.1*	*34.9*	*35.9)*
62	P2	223.5	69.1	27.8	11.7	7	R	52.2	107.7	52.6	16.9	19.8
64	P2	253.1	59.6	78.9	5.8	9	R	31.8	91.7	40.5	6.6	8.8
65	P2	239.5	61.8	29.1	11.4	7	R	42.3	92.3	43.0	13.6	17.6
66	P2	245.0	61.8	20.2	11.0	10	R	38.3	93.1	43.0	13.2	17.1
71	P2	247.5	64.4	136.9	3.7	12	R	37.1	97.5	46.2	4.8	6.0
72	P2	230.7	65.4	66.6	5.6	11	R	48.6	98.5	47.6	7.4	9.2
73	P2	226.5	69.3	544.2	2.0	11	R	50.4	107.2	52.9	2.9	3.3
74	P2	197.0	68.2	166.8	3.6	11	R	66.4	123.4	51.4	5.0	6.0
75	PH	238.5	58.1	111.8	4.6	10	R	42.3	86.5	38.8	5.0	6.8
77	P2	157.9	77.8	139.0	3.9	11	R	52.9	165.3	66.6	6.9	7.3
78	P2	196.2	63.2	94.4	4.7	11	R	72.0	111.1	44.7	5.9	7.4
79	P2	207.2	70.2	70.9	6.2	9	R	60.4	118.3	54.3	9.2	10.6

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Table 2 (Cont.)

Site no.[a]	Comp.	Decl. (°)	Incl. (°)	KD[b]	α95[c] (°)	Sp.[d]	Polarity	Lat. (°S)	Long. (°E)	Palaeolat. (°S)	DP[e] (°)	DM[e] (°)
80	P2[j]	135.0	61.9	554.6	2.1	10	R	53.1	210.2	43.1	2.5	3.2
81	P2	184.2	76.4	319.4	2.6	11	R	57.5	147.5	64.3	4.4	4.7
82	P2[j]	242.8	78.8	423.2	2.2	11	R	39.4	125.9	68.4	4.0	4.2
83	P2[i]	254.9	71.2	250.0	2.9	11	R	34.1	110.0	55.8	4.4	5.1
84	P2	211.4	66.2	45.5	7.7	9	R	61.1	105.4	48.6	10.4	12.6
85	P2	213.3	53.4	425.5	2.3	10	R	62.1	74.3	34.0	2.3	3.3
86	PH	205.5	74.7	1078.8	3.8	3	R	55.0	129.0	61.4	6.2	6.8
87	P2	203.9	82.0	263.9	2.8	11	R	44.7	141.2	74.3	5.3	5.5
88	P	151.5	73.8	41.3	7.2	11	R	55.2	174.9	59.8	11.7	13.0
89	PH	159.3	79.3	203.1	5.4	5	R	49.5	161.2	69.3	9.8	10.2
90	PH	160.8	72.4	441.4	2.6	8	R	59.9	170.6	57.7	4.2	4.7
91	PH	195.1	80.2	33.5	9.7	8	R	48.9	142.6	70.9	17.9	18.6

[a]Result of ambiguous primary or suspected secondary origin, not included in Tables 3 and 4 *(in italics and between parentheses)*, see footnotes f-j and discussion in text. [b]KD, best estimate of the precision parameter κ (Fisher 1953). [c]α95, half-angle of the circle of 95% confidence around the site-mean direction. [d]Sp, number of specimens included in calculation of the site-mean direction. The total number of studied specimens for this site is listed in Table 1. [e]DP, DM, half-angles of the oval of 95% confidence around the pole position. [f]Unclear origin, possibly affected by Permian or Permo-Triassic overprint. [g]Suspected Permo-Triassic overprint? [h]Anomalous direction, possibly due to rheomorphism. [i]Suspected Permian overprint? [j]Outlier.

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[D = 3.7°, I = –65.4°, *n* = 34 (sites), k = 96.2, α_{95} = 2.5°]. Most other sites showed lesser predominance of this single LT component [mean direction across all 91 studied sites: D = 2.1°, I = –65.0°, *n* = 574 (specimens), k = 42.6, α_{95} = 0.9°]. Instead a substantial number of specimens in these other sites showed successive unblocking of two sub-components (LT1, LT2) below 250°C [LT1: (mean-sites) D = 2.8°, I = –64.8°, *n* = 42, k = 190.6, α_{95} = 1.6°; (mean-specimens) D = 3.8°, I = –64.4°, *n* = 175, k = 78.0, α_{95} = 1.2°] and below 400°C [LT2: (mean-sites) D = 3.3°, I = –63.8°, *n* = 42, k = 136.8, α_{95} = 1.9°; (mean-specimens) D = 4.5°, I = –62.1°, *n* = 175, k = 69.2, α_{95} = 1.3°]. Whilst there is only a marginal statistical difference between their respective mean directions at site and/or specimen level (sites: angular distance = 1.0°, linear sum of α_{95} = 3.5°, square root sum of α_{95} = 2.5°; specimens: angular distance = 2.4°, linear sum of α_{95} = 2.5°, square root sum of α_{95} = 1.8°), the two sub-components are well distinguishable within individual specimens with their angular distance exceeding the square root sum of their respective 'madangles' (Kirschvink 1980) for 82% of the specimens (*n* = 175; angular distance: mean = 9.3°, σ = 5.3°; square root sum of 'mad-angles': mean = 5.3°, σ = 4.9°). The mean-sites inclination for the LT2 sub-component is 1.0° (mean-specimens 2.4°) lower than for the LT1 sub-component and 1.6° (mean-specimens 2.5°) lower than for the LT component. If significant, this could be indicative of minor incomplete separation between the upwards pointing LT (sub-) components and the downwards pointing HT components.

HT components

The HT components generally break down between 200 and 700°C and between 15 and 160 mT, are exclusively of reverse polarity (apart from the Merlewood Formation), and show south steep to very steep downwards directions. Generally two HT components can be determined: an apparently incompletely cleaned (pseudo-) component with an upper magnetite unblocking range around 500–550°C (HT-P1: Table 5*), and another component with an uppermost magnetite–hematite unblocking range of 300–700°C, but mainly around 600°C (HT-P2: Table 2). The P2 components predominantly show slightly lower declinations and slightly higher inclinations than the P1 components. In a few sites this shift towards slightly lower declinations is recognisable already at lower temperatures, with unblocking of another apparently even less-completely cleaned (pseudo-) component (HT-P0) between 200 and 600°C and mainly between 350 and 450°C.

Formation and section highlights

MERLEWOOD FORMATION

Sites 1–3 and less so site 4 from the middle-level andesite (Figure 7a) show well-defined LT components, but these are not observed in the sites from the lower-level andesites (5–7), other than in several site 5 samples. HT components are poorly defined in sites 2,3, 5 and 7, but better defined in sites 1, 4 and 6, showing low to moderately low inclinations of mainly normal polarity (Figure 7a, b). Reverse polarities occur only in sites 6 and 7 (Figure 7c) in the lower, but not lowest, part of the formation. Lightning apparently affected a part only of the lowermost site (site 5, samples 1–5) on an exposed ridge.

CURRABUBULA FORMATION

Washpool Creek (*sites 8–27: Figure 7d–j)* LT components are prevalent and well-determined throughout the section, with exception of site 17. HT components are well-determined in the volcanics, but are poorly defined in several volcaniclastic horizons (8, 9, 15, 16). Site 14, a fine-grained tuff of slightly weathered appearance, shows more complex Z-shaped Zijderveld (1967, 1975) plots with a very high temperature component of normal polarity and odd direction.

Cana Creek (28–41: Figure 7k–p) Well-determined LT components are found in all sites except 30 and 37. HT components are well-determined throughout most of the

section, apart from the uppermost volcaniclastic sites 39–41 and the devitrified porphyritic tuff at site 36.

Chilcotts Creek (42–55: Figure 7q–u) Well-determined LT components are present in all sites, apart from a basal coarse-grained sandstone (42) and a thinly bedded reworked tuff (51). Well-defined HT components could be determined from all sites, with the medium-grained volcaniclastics (44) at the base of a massive horizon (44, 45) showing a larger than normal spread in directions. One tuff (47) and two lava (54, 55) sites show anomalous mean directions, which may be related to rheomorphic effects.

Quipolly Creek (56–67: Figure 7v, w) Dominant and well-determined LT components are present in all sites. The consistency of the HT components varies throughout the section. The conglomerate at the base of the section (56) and the boulder bed at the top (67) show well-determined, widely spread, HT components. Whilst the other Quipolly Creek sites generally show well-determined HT components, many sites show considerably spread directions (57–59, 61–63, 66) with better concentration in a few sites only (60, 64, 65).

Jacob and Joseph Creek (68–71: Figure 7x) These sites, apart from 70, show well-determined LT components that

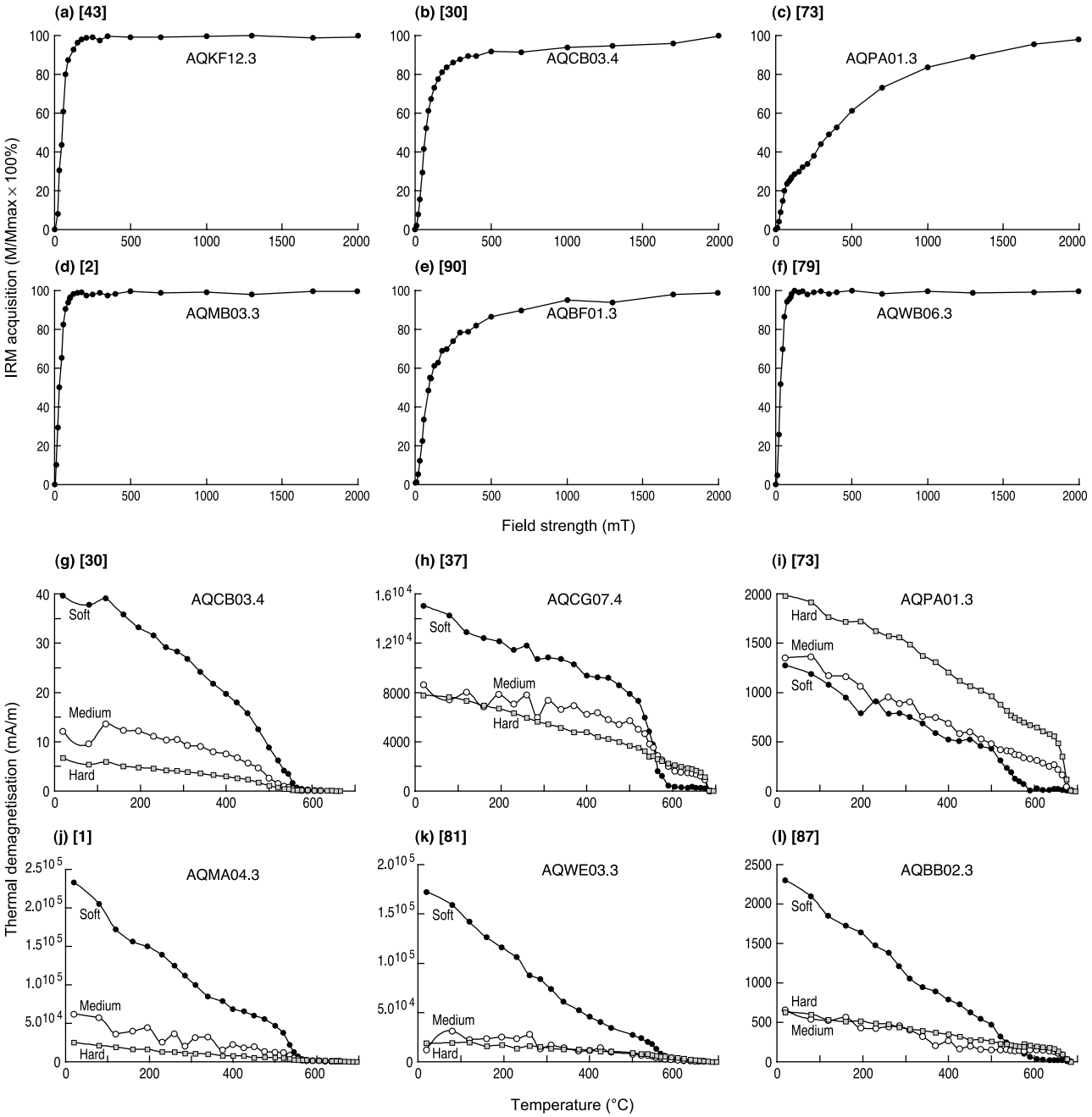


Figure 8 (a–f) Examples of IRM acquisition in constant fields up to 2000 mT. (g–l) Examples of thermal demagnetisation of IRM three-axis magnetisations following Lowrie (1990). Grey-shaded squares, hard IRM acquired in a 2000 mT constant field; open circles, medium IRM acquired at 400 mT; filled circles, soft IRM acquired at 120 mT. Numbers in parentheses following the headings a–l relate to site numbers in Table 1.

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dominate the remanence. HT components generally show noisy demagnetisation trajectories and considerable spread in directions, with site 71 a well-determined exception.

Lower Quipolly (72, 73: Figure 7y, z) LT components are either absent (72), or of limited intensity (73). HT components are well-determined in the upper site (73) of the Piallaway Trig Ignimbrite Member, and fairly so in site 72, showing considerably more dispersed directions.

The Gap (74–76: Figure 7aa, bb) LT components are dominant in the silicified siltstone (76), non-existent in the ash-flow (75), and of limited intensity in the tuff (74). HT components are well-determined in all sites, with site 76 showing complex Z-shaped Zijderveld plots.

WERRIE BASALT

LT components are present with variable intensities: either dominant and well-determined (78–80, 82, 84, 85: Figure 7dd), or subordinate with considerable directional spread (77, 81) or odd direction (83). HT components are generally well-determined (Figure 7cc, dd), although Zijderveld plots for sites 79 and 84 are noisier with a wider directional spread.

BOGGABRI VOLCANICS

An LT component of variable intensity and considerable directional spread is present in most sites (86–88, 91: Figure 7ee, ff), but is of limited or insignificant intensity in sites 89 and 90 (Figure 7gg). All sites show well to very well-determined HT components, with site 91 a notable exception showing considerably spread directions.

ROCK MAGNETIC RESULTS

Lowrie–Fuller test

All sites show higher resistance to AF-demagnetisation up to 160 mT for weak-field ARMs (0.1 mT bias field, 250 mT AF peak-field) than for IRMs (250 mT), conventionally regarded as an indicator for single domain/pseudo-single domain (SD/PSD-) magnetite. However, as discussed below, low-temperature susceptibility cycling indicates the presence of multidomain magnetite in many volcaniclastic and various volcanic rocks. These observations are similar to those of the Rocky Creek Block (Klootwijk 2002) and add to doubts about the general applicability of the Lowrie–Fuller test.

IRM acquisition

The acquisition curves show the dominant presence in the volcanic and volcaniclastic rocks of a magnetite carrier saturated in fields of 200–300 mT (Figure 8a, d, f). Less prevalent, and mainly in volcaniclastic rocks, is a hematite phase not yet saturated at 2000 mT (Figure 8b, c, e) together with a magnetite phase saturated between 300 and 500 mT. This magnetite–hematite combination is characteristically present in the Cana Creek and Quipolly Creek – Lower Quipolly sections of the Currabubula Formation (e.g. sites 28, 30, 32–34, 36, 37, 59, 62, 64, 65, 72, 73) and in the Boggabri Volcanics (86–91), with a more sporadic presence in other Currabubula sections and the Werrie Basalt (e.g. 8, 9, 11, 14, 48, 69, 74, 81).

Three-axis Lowrie (1990) test

Distributed unblocking temperatures below 500°C characterise the sample collection, with a few volcanic and volcaniclastic rocks showing the low unblocking temperatures of goethite (e.g. sites 7, 10, 78, 80). Several volcanic rocks show a distinct magnetite unblocking temperature (e.g. Merlewood Formation, 12, 29, 55, Werrie Basalt: Figure 8g), but most of the volcanic and volcaniclastic rocks show unblocking of a magnetite–hematite phase with slightly higher Curie temperatures up to around 620°C (e.g. 20, 25, 38, 47, 51, 64, 74, Boggabri Volcanics: Figure 8k). A minority of volcanic and volcaniclastic rocks show in addition a distinct hematite phase with Curie temperatures up to 675°C (e.g. 9, 28, 34, 36, 62, 64, 69, 72, Boggabri Volcanics: Figure 8h, i, l). Some of the volcaniclastic rocks show distributed unblocking temperature ranges up to 300°C (e.g. 35, 41, 46, 47, 76, 86), but without clear evidence for presence of maghemite or iron sulfides.

Low-field, low- and high-temperature susceptibility cycling

Low-temperature cycling shows multidomain (MD-) magnetite in various volcanic (e.g. 34, 37: Figure 9b) and volcaniclastic sites (e.g. 32, 33, 40, 47, 49–53, 57, 61, 63, 68, 70, 76: Figure 9a), but not in others (Figure 9c, d, g), with characteristic presence in volcanic rocks of several formations [Merlewood andesites (Figure 9e); Piallaway Trig Ignimbrite Member, 34, 37, Werrie Basalt, Boggabri Volcanics (Figure 9f)]. High-temperature cycling shows the general presence of a magnetite phase with Curie temperatures above 500°C, often accompanied by a hematite phase with Curie temperatures up to 700°C (Figure 9h–o). The characteristic (lepidocrocite)–maghemite–magnetite conversion takes place in many volcanic (e.g. 1, 5, 7, 35, 36, 73, 81–85, Washpool Creek, Boggabri Volcanics: Figure 9i, k, n, o) and volcaniclastic sites (e.g. 8–10, 39, 42, 46, 47, 49, 50, 54, 51, 71, 76, Washpool Creek, Quipolly Creek: Figure 9j). Observations of a sharp Hopkinson peak in several volcanic sites shows evidence for single-domain (SD-) magnetite (e.g. 2, 7, 21, 34, 55, 82, 86–88, 91: Figure 9h). Universal reduction in susceptibility after heating to 700°C indicates ineffectiveness of the argon environment to completely prevent magnetite–hematite oxidation.

MAGNETIC FABRIC ANALYSIS

Measurement results

The volcanic rocks of the Currabubula Formation show a low to moderate anisotropy of magnetic susceptibility (AMS) with anisotropy values (P) generally below 5%, but up to 15% in sites of more banded appearance (e.g. 35, 47, 64–67). Volcanics of the Merlewood Formation, Werrie

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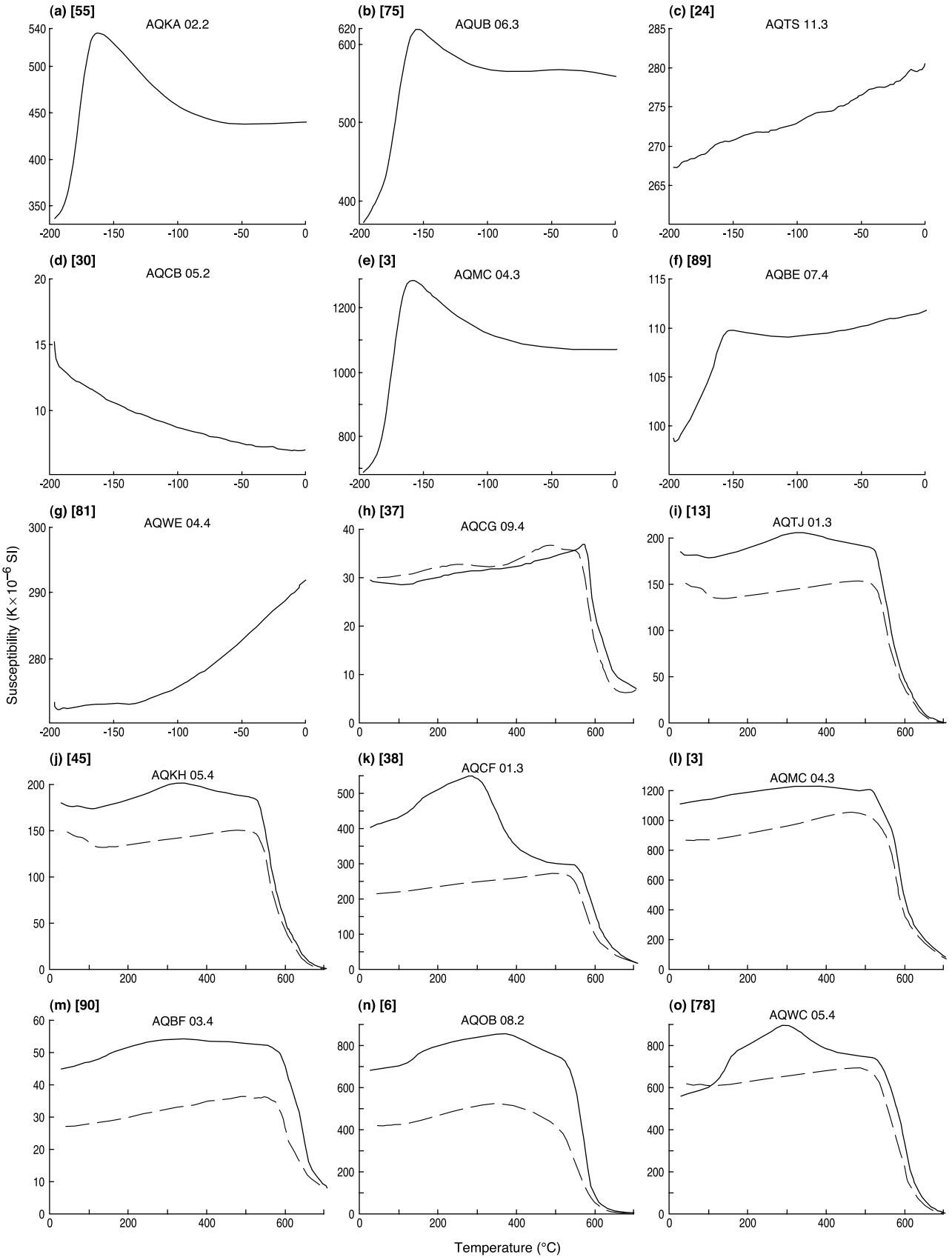


Figure 9 (a–g) Examples of thermomagnetic curves obtained with the AGICO CS-3 cryostat showing changes in susceptibility during warming up from liquid nitrogen temperature. (h–o) Examples of thermomagnetic curves obtained with the AGICO CS-3 furnace showing changes in susceptibility during heating (continuous curve) and cooling (dashed curve). The heating cycle has been carried out in argon gas. Numbers in parentheses following the headings a–o relate to site numbers in Table 1.

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Basalt and Boggabri Volcanics likewise show P values generally below 5% with exception of a flow-banded lava of the Boggabri Volcanics (87: ~9%) and two pyroclastics sites of the Werrie Basalt (83, 84: ~8%). Volcaniclastic rocks of the Currabubula Formation generally show slightly higher P values up to ~9% (e.g. 44, 45, 76) and, as expected, stronger predominance of oblate AMS ellipsoids than the volcanic rocks. There is a fair spread in ellipsoid shapes with sites showing well-defined triaxial ellipsoids, planar distributions, or ill-defined principal axes. The latter dominate, as expected, amongst the sites with lower anisotropy values.

Transport patterns

The *in situ* K_{max} axes for the Currabubula Formation sedimentary sites show gross north–south alignments (Figure 10a) that deviate only slightly from the predominant north-northwest–south-southeast tectonic grain (Figure 4) of the Werrie Syncline. Close agreement between magnetic fabric and tectonic grain has been observed also for the Rocky Creek Block (Klootwijk 2002), and it can again be demonstrated for the Werrie Block that the magnetic fabric pattern is unlikely to be of tectonic origin. The K_{min} axes concentrate better in stratigraphic

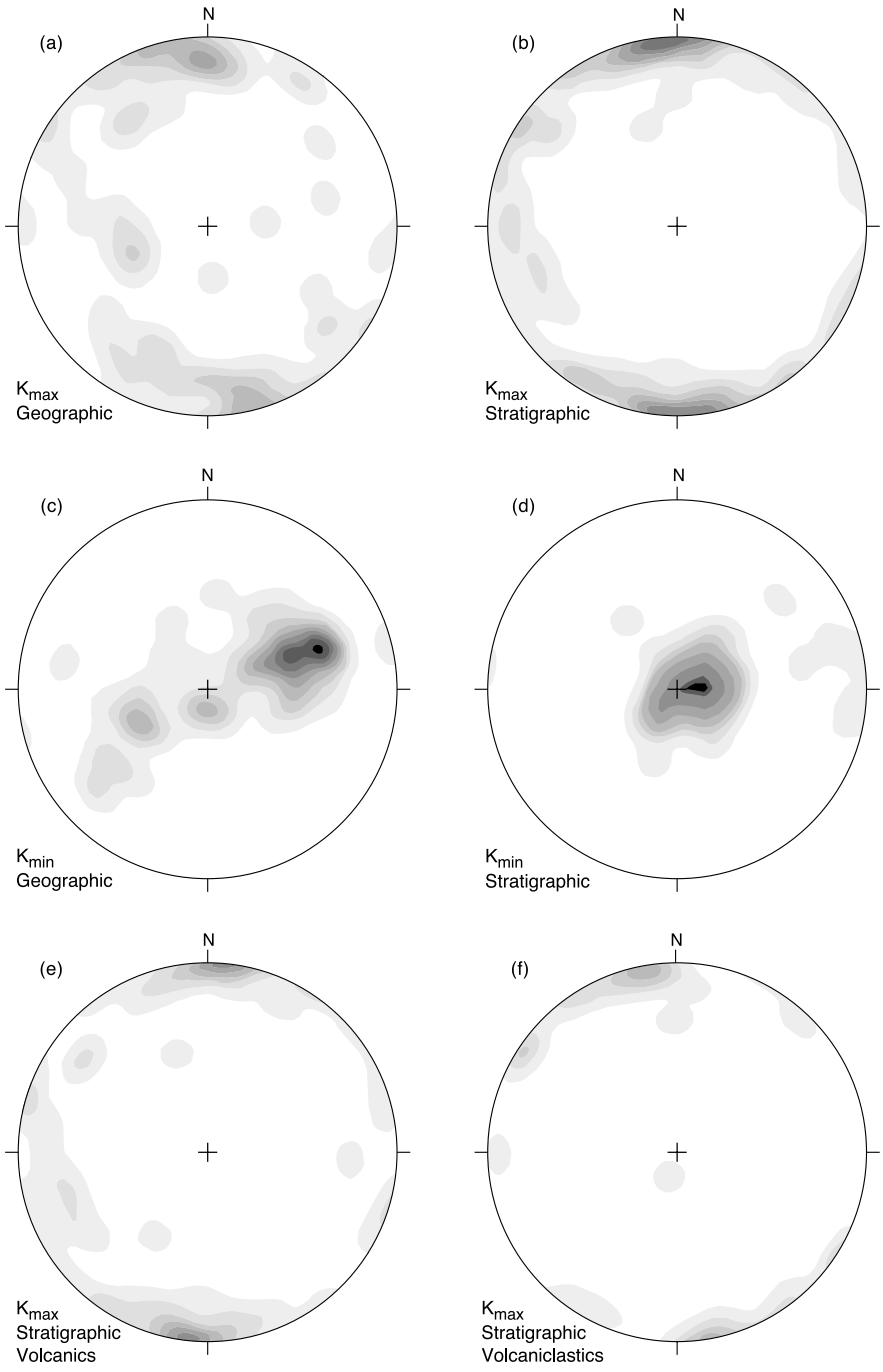


Figure 10 Contoured density distributions (equal-area projection) of AMS mean-tensorial axes for all measured Currabubula Formation sites. Lower hemisphere projection, contours for each integer value. (a) K_{max} axes, all sites in geographical coordinates (n = 67). (b) Same in stratigraphic coordinates. (c) K_{min} axes in geographical coordinates (overall mean: D = 76.3°, I = 73.2°, n = 67, k = 4.5, α_{95} = 9.3°). (d) Same in stratigraphic coordinates (D = 87.6°, I = 78.9°, n = 67, k = 11.9, α_{95} = 5.2°). (e, f) K_{max} axes of volcanic (e: n = 45) and volcaniclastic (f: n = 22) rock sites in stratigraphic coordinates, no clear evidence for imbrication.

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coordinates (unfolded: Figure 10d) than in geographical coordinates (*in situ*: Figure 10c), with the K_{min} axes showing a near-vertical mode and the K_{max} axes a near-horizontal mode. This is typical for a primary, depositional, magnetic fabric. The K_{max} axes for the volcanic rocks and for the volcaniclastic rocks show slightly different alignments in stratigraphic coordinates, with the former orientated slightly east of north (Figure 10e) and the latter slightly west of north and closer to the prevailing tectonic grain (Figure 10f). The near-horizontal K_{max} patterns do not show evidence for imbrication. The directional axis of transport can be established from these data, but not the sense of transport.

Palaeogeographical evidence indicates the presence of a (former) continental volcanic arc source to the west (McPhie 1983, 1984a, b; Roberts & Geeve 1999). Predominance of east–west rather than north–south aligned K_{max} axes therefore would be expected. The Rocky Creek Block study likewise has shown grossly north–south aligned K_{max} axes, that can be interpreted in terms of distal southern or northern sources for both blocks. Only a minority of sites in the Werrie Block show east–west aligned K_{max} axes (Figure 11b, c), but this probably reflects the odd occurrence of an oblate anisotropy ellipse with closely similar K_{max} and K_{int} values. East–west alignment of directionally significant K_{max} axes becomes prevalent, however, in the studied upper part of the Currabubula Formation, particularly so in the Piallaway Trig Ignimbrite Member (Figure 11b–e).

INTERPRETATION

Comparison with Australian pole path

It would be preferable to date Werrie Block overprints through comparison of their poles with the New England pole path. However, the latter is not yet available, leaving

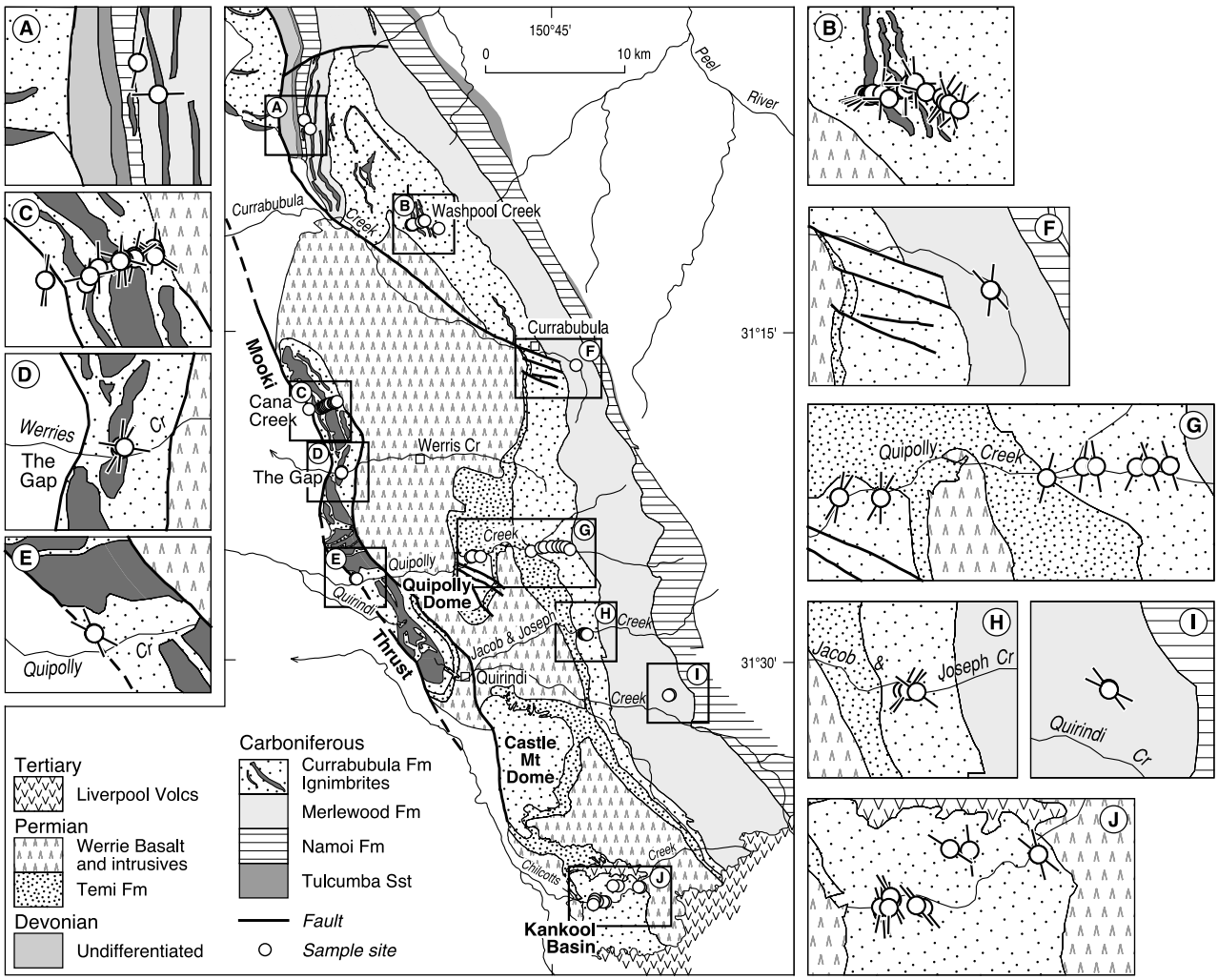


Figure 11 Orientation of AMS mean-tensorial K_{max} axes (bar through open circles in a–j) for individual sites (see Figure 4a) as an indication of the axes of transport. Note that the sense of transport has not been established in absence of magnetic evidence for imbrication (see Figure 10e, f). (a–j) Enlargements of sampled sections or sites. (a) Merlewood Formation, near Merlewood (Table 1: sites 5–7). (b) Washpool Creek (8–27). (c) Cana Creek (28–41). (d) The Gap (74–77). (e) Lower Quipolly (72, 73). (f) Merlewood Formation, at Belgamba Mountain (1, 2). (g) Quipolly Creek – Quipolly Dam (56–67). (h) Jacob and Joseph Creek (68–71). (i) Merlewood Formation, at Wallabadah (3, 4). (j) Chilcotts Creek (42–51).

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comparison with the Australian cratonic pole path as the practical substitute. Post-overprint deformation of the New England Orogen, internally and relative to the craton, remains to be resolved, but is not expected to notably affect a first-order comparison of overprint poles with the cratonic pole path.

Overprint magnetisations: post-folding

MIDDLE–LATE CENOZOIC OVERPRINTS

The LT component mean-site directions [Figure 12a–d: (LT1 + LT: D = 3.2°, I = –65.1°, n = 76, k = 134.0, α_{95} = 1.1°; mean pole — Plong = 141.8°E, Plat = 73.7°S, k = 71.3, A95 = 2.9°); (LT2 + LT: D = 3.3°, I = –64.6°, n = 76, k = 115.6, α_{95} = 1.5°; mean pole — Plong = 140.8°E, Plat = 74.0°S, k = 59.7, A95 = 2.1°)] are significantly different from the axial geocentric dipole direction (D = 0°, I = –51°), but are not far off the present local field direction (D = 11.5°, I = –62°). These mean-site directions for the Werrie Block are also comparable to the mean-site direction for the LT component of the Rocky Creek Block (Klootwijk 2002 figure 12b) and resemble an Early Miocene weathering overprint from the far-south coast of New South Wales (Nott *et al*. 1991 figure 9). The LT component may be indicative, therefore, of a mid-Tertiary weathering event of regional significance in southeastern Australia, if not beyond. Australia-wide observations of an Early to Middle Miocene weathering event have recently emerged (McQueen *et al*. 2002; Pillans 2002; Schmidt & Williams 2002). Alternatively, the component could be of recent viscous origin because of its exclusively normal polarity and resemblance to the present local field direction (Figure 12a, e).

Overprint magnetisations: pre-folding

PERMO-TRIASSIC OVERPRINTS

A minority of sites from the Currabubula Formation (Table 6*: 13 of 69) and a single site from the Merlewood Formation show HT components of pre-folding origin and reverse polarity with poles that are grouped or distributed in the southeast Pacific (Figure 12f). These poles grossly agree with the Australian Permo-Triassic pole path loop (Figure 12h), whose outline may be better indicated by the Indian pole path transferred to Australia (Figure 12i), and are interpreted as Permo-Triassic overprints acquired upon west-directed thrusting during mainly the Early Triassic phase of the Hunter–Bowen Orogeny. Comparable overprints that are prevalent in volcaniclastic rocks of the Rocky Creek Block (Klootwijk 2002 figure 13a) have been similarly interpreted.

PERMIAN OVERPRINTS

Five Cana Creek sites show HT components with very steep inclinations and with pole positions (Figure 12g) that are different from the bulk of the Currabubula HT poles, but are in gross agreement with the Permian segment of the Australian pole path (Figure 12h, i). These components occur in both volcanic and volcaniclastic rocks of the Currabubula Formation (Table 6*), predominantly near major structural zones such as the bounding thrust splays and an internal shear zone of the Quirindi Dome (Figures 4, 6) which may have facilitated fluid movements.

The vast majority of overprinted sites (16 of 20) appear fully remagnetised without evidence for a retained primary magnetisation component (Table 6*). Four sites only show evidence for partial, Permo-Triassic, overprinting. Two flow-banded silicic lava sites (54, 55) in the Kankool Basin clearly show partial overprinting (P0: Table 6*) of a primary component (P2: Table 2) and an incompletely cleaned pseudo-component (P1: Table 5*). One Merlewood Formation site (2) shows localised overprinting (O1, O2: Table 6*), with the remainder of the site apparently unaffected (P1, P2: Tables 2, 5*). One site of the Taggarts Mountain Creek Ignimbrite Member (22) shows an overprint component (P1: Table 5*) and an apparently incompletely cleaned primary component (P2?: Table 2).

Pre-folding Permian overprints have also been observed locally in the Rocky Creek Block (Klootwijk 2002 figure 12f) and more widely in the southern Tamworth Belt (Klootwijk 1996a, b; Geeve 2000a), the latter interpreted by Geeve (2000a) as latest Carboniferous – earliest Permian in age. Lackie and Schmidt (1993) also reported widespread occurrence of Permian overprints in the Kiah Limestone (Late Devonian – Early Carboniferous) and interpreted these as deformation-related, diachronous remagnetisations, of essentially pre-folding origin in the southwest of the Tamworth Belt and of syn-folding origin in the northwest.

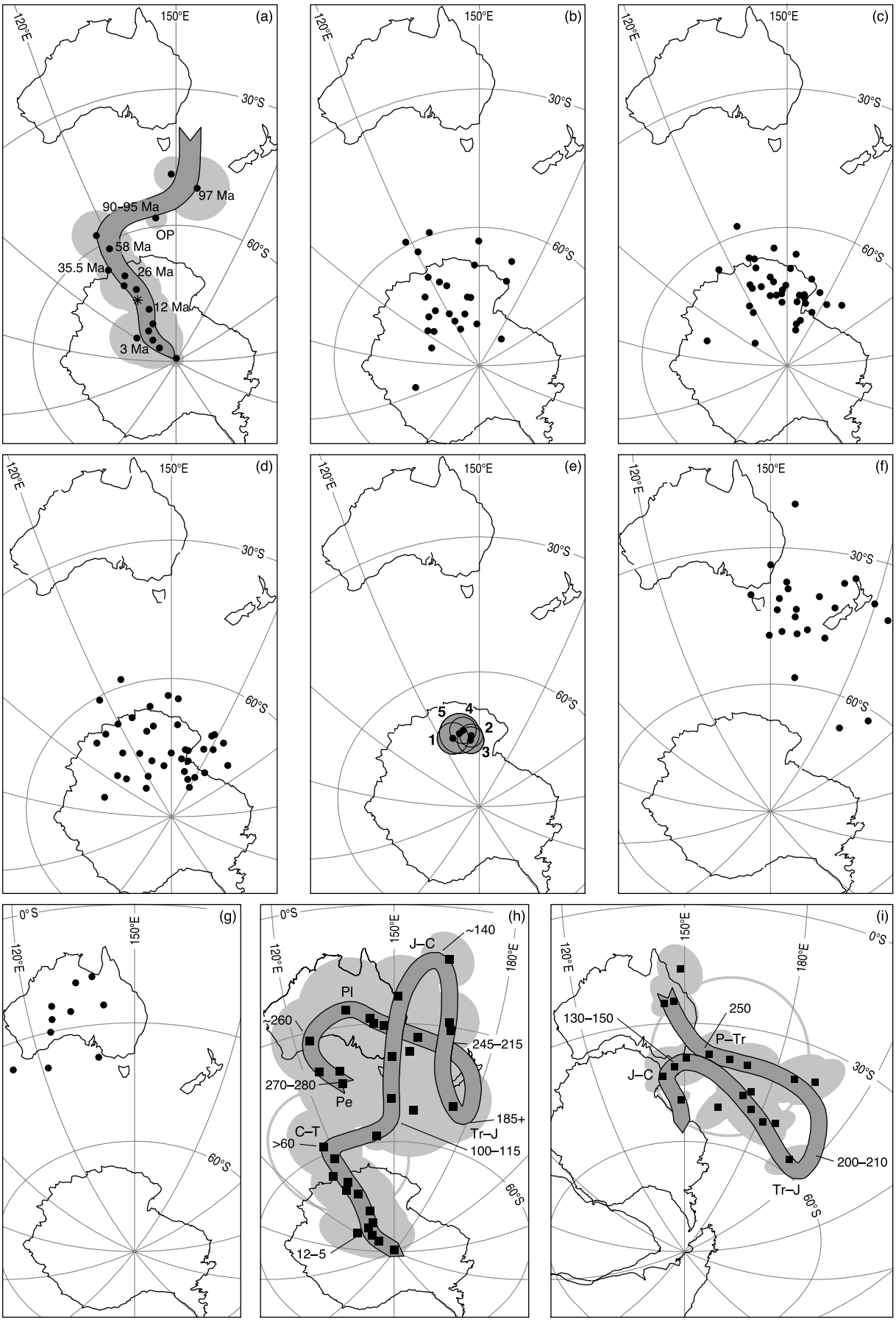
Conglomerate tests

Conglomerate tests have been carried out on the Currabubula Formation in the Quipolly Creek – Quipolly Dam section. Volcanic cobbles in the basal conglomerate site 56 [P1: k = 1.2, α_{95} = 134.1°, R = 1.92, n = 8; P2: k = 1.1, α_{95} = 148.7°, R = 1.79, n = 8, R_o (8, 95–99%) = 4.48–5.26], formerly known as Coeypolly Conglomerate, and volcanic cobbles and pebbles in the uppermost site 67 [Pr (P1, P2 not distinguished): k = 1.2, α_{95} = 92.8°, R = 2.74, n = 10, R_o (10, 95–99%) = 5.03–5.94] show positive conglomerate tests with widely distributed HT components, demonstrating the capability of volcanic rocks in the Currabubula Formation to retain their original magnetisation. In contrast, syn-formational pebbles and cobbles in a deeply red, silicified, diamictite (33) of the Currabubula Formation in the Cana Creek section show reasonably well-concentrating HT components with a steep mean direction [Tables 2, 6*: P1: k = 21.8, α_{95} = 9.1°, R = 12.45, n = 13, R_o (13, 95–99%) = 5.75–6.84; P2: k = 21.9, α_{95} = 10.0°, R = 10.54, n = 11, R_o (11, 95–99%) = 5.29–6.25] interpreted as a Permian overprint, prevalent in sites near major structural zones.

Primary magnetisation component structure

Considerable care has been taken to maximise directional information on the HT components, with thermal demagnetisation routinely continued in small steps till the magnetic intensity dropped below the resolution limit of the ScT magnetometer (~2×10^{-3} mAm^{-1}). A few sites had to be abandoned prematurely because development of viscous magnetisation at higher temperature steps impeded remanence measurements. Detailed analysis of thermal

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demagnetisation data has shown the general presence of one incompletely cleaned pseudo-component (P1) and one optimally cleaned component (P2), with occasional occurrence of two pseudo-components (P0, P1) or two more optimally cleaned components (P2, PH). These components have successively higher unblocking temperature ranges: up into the magnetite range for P0, if identified; into the upper magnetite range for P1; and into the uppermost magnetite–hematite range for P2 and, if present, PH. Comparison of mean-site directions shows the thermally hardest P2 (PH) component results to be better defined than the P1 pseudo-component results, with universally smaller α_{95} angles and often slightly lesser declinations and slightly higher inclinations (Figure 13; Tables 3, 7*).

Following the Rocky Creek study (Klootwijk 2002 tables 2, 7*) the highest temperature P2 component has been taken generally as the preferred, more precise, representation of the primary magnetisation for reason of its smaller α_{95} angle. The P1 component, and the occasional P0 component, have been taken, therefore, as incompletely cleaned pseudo-components. The Rocky Creek Block study (Klootwijk 2002) together with the subsequent discussion (Schmidt & McElhinny 2003) and reply (Klootwijk 2003) demonstrated the merit of, if not requirement for, distinguishing between the P1 and P2 components. These earlier results showed that where the P2 component passed the McFadden (1990) fold test at 95% or 99% confidence levels, the P1 component did so invariably as well, but with systematic deflection between the P1 and P2 mean-site directions (Klootwijk 2003 figure 3) and with superior precision for the P2 component (Klootwijk 2002 figure 14). The P1 component in its own right might have seemed well acceptable as a representative characteristic and primary component, but more extensive demagnetisation and more detailed analysis resulted in a more optimally cleaned and presumably better representative P2 result. The present analysis and interpretation of the Werrie Block succession follows and builds on this finding from the Rocky Creek Block study.

Components with intermediate unblocking temperature ranges were widely observed in the Rocky Creek Block and were interpreted as Permo-Triassic overprints associated with the Hunter–Bowen Orogeny. Such intermediate components have been less frequently observed in the present Werrie Block study.

Figure 12 Site virtual geomagnetic poles (b–d, f, g) and palaeomagnetic poles (e), all south poles, for: post-folding overprints of Cenozoic origin (b–e); pre-folding overprints of Permo-Triassic (f) and Permian (g) origin. (a) Cenozoic and Late Cretaceous pole path and pole positions (dots) (after Idnurm 1994 and Klootwijk in press a figure 8) for comparison with overprint poles shown in subfigures (b–e). The star indicates the present local virtual geomagnetic pole. (b–d): LT (b), or when distinguishable the magnetically softer LT1 (c) and the magnetically harder LT2 (d), overprints (dots) of normal polarity, shown for the Currabubula sites only; the overprints probably represent a Middle to Late Cenozoic weathering event or possibly a recent viscous component; (e) comparison of mean pole positions [1, LT component Werrie Block (WB); 2, LT1 component (WB); 3, LT2 component (WB); 4, LT component Rocky Creek Block; 5, overprint on the far south coast of New South Wales dated as Early Miocene (Nott *et al.* 1991)]. (f) HT component of probable Permo-Triassic origin (Table 6*). (g) HT component of probable Permian origin (Table 6*). (h) Australian Late Palaeozoic – Cenozoic pole path (compiled from data in Klootwijk 1996a tables 1.1, 1.2) and for the Late Cretaceous and Cenozoic (from Idnurm 1994 and Klootwijk in press a figure 8), approximate ages according to Klootwijk (1996b). (i) Indian latest Palaeozoic – Mesozoic pole path transferred to Australia, with India in a Gondwana reconstruction that is anchored to Australia in its present position. Legend for (h, i): Pe, Early Permian; Pl, Late Permian; P-Tr, Permo-Triassic; Tr-J, Triassic–Jurassic; J-C, Jurassic–Cretaceous; C-T, Cretaceous–Tertiary.

Primary magnetisation highlights

MERLEWOOD FORMATION

Primary magnetisations are predominantly of normal polarity [sites 1–5 (N); 6, 7 (R)]. Site-mean directions for the P2 components generally show low to equatorial inclinations, although moderate for site 1 (Table 2). Site-mean declinations show a considerable spread, although with good agreement for three sites of normal and reverse polarity [2, 3 (N), 6 (R)]. The McFadden (1990) fold test shows slightly better concentration of site-mean directions in stratigraphic coordinates (Table 3), but the improvement is not significant at the 95% confidence level. The dispersion remains considerable and the mean pole is poorly defined (Table 4). The McFadden and McElhinny (1990) reversal test is indeterminate. These observations on magnetic behaviour and mean directions are comparable to Schmidt's (1988) study of the Merlewood Formation, with reasonable agreement between both mean results despite their considerable imprecision.

CURRABUBULA FORMATION

Chilcotts Creek (42–55) P2 poles for the majority of sites (43–46, 48–53) track from Victoria Land, Antarctica, towards the southwest of Western Australia (Figure 14a). Four sites (42, 47, 54, 55) are located off the main track. Their rock-magnetic properties are not notably different from the other sites and are not indicative of lower magnetic stability. Poles for three 'off' sites (47, 54, 55) are probably affected by rheomorphic flow, but Permo-Triassic overprinting cannot be excluded for sites 54 and 55. The P2 pole for site 42, coarse-grained red beds overlying massive conglomerates, is close to the Permo-Triassic segment and also to the Early Cretaceous segment of the Australian pole path (Figure 12h, i). It is interpreted as an overprint related to the Hunter–Bowen Orogeny. The Chilcotts Creek pole positions outline a track comparable to the Namurian to Westphalian segment of the pole path for the Rocky Creek Block (Klootwijk 2002 figure 15), although individual poles do not show a clear younging trend (Figure 14a). McPhie's (1983, 1984a) mapping shows the Chilcotts Creek section to pre-date the main Currabubula Formation of the Werrie Basin (Figure 5), but the section's time-span is not constrained by isotopic dates. If the time-span is limited, the spread in pole positions could reflect secular variation rather than a younging trend. McFadden's (1990) fold test for the ten (43–46, 48–53) site-mean results is positive at the 99% confidence level for the P2 component (Table 3) as well as for the P1 pseudo-component (Table 5*). The positive

fold test results are well demonstrated from the good agreement between the P2 poles for sites 52 and 53 on the slightly east-northeast-dipping limb and the other eight sites on the moderately west-dipping western limb of the Kankool Basin anticlinal structure (Figures 4a, 14a). This mean pole position for the lowermost part of the Currabubula Formation locates slightly to the west of the mean pole for the Rocky Creek Conglomerate of the Rocky Creek Block (Figures 15a, 16a), in good agreement with stratigraphic correlation between the basal Currabubula Formation and the Rocky Creek Conglomerate (Figure 3).

Washpool Creek (8–27) Several sites from the lower part of the section have ill-defined (8, 9, 15, 16) or odd (14) mean directions and their pole positions (Figure 14b) are not considered further. Pole positions for the other Washpool Creek sites track from Tasmania through the Great Australian Bight towards the southwest of southwestern Australia (Figure 14b). The younging trend in primary poles is from east to west, but is irregular, with poles for the middle part of the section apparently deflected eastwards. Pole positions for the lower part of the Washpool Creek section (10–13, 17) are closer to Australia than for the Chilcotts Creek section and may reflect a younger part (3 in Figure 15a) of the Australian pole path, in agreement with the Rocky Creek Conglomerate-to-Lark Hill Formation trend observed for the Rocky Creek Block (Klootwijk 2002 figure 15). Poles for the Taggarts Mountain Ignimbrite Member sites (18–22) and for a single site from the base of the Piallaway Trig Ignimbrite Member (23) show a considerable spread in positions with several poles (18, 19, 23) apparently deflected towards the Permian and Permo-Triassic segments of the Australian pole path (Figure 12h, i), and with two poles (20, 22) in agreement with the Permo-Triassic segment. These apparently deflected poles are not used in defining the Werrie Block – Tamworth Belt pole path. Poles for the upper flow in the Washpool Creek section, the Piallaway Trig Ignimbrite Member (24–27), and a single pole for a crystal flow of the underlying Taggarts Mountain Ignimbrite Member (21) group closely to the west of southwestern Australia. These pole positions are similar to poles at comparable stratigraphic levels from other sections discussed hereafter and are taken as representative for the uppermost part of the Currabubula Formation. Rock magnetic characteristics for sites without an identified primary magnetisation component are not notably different from sites retaining such components and provide no constraints on selection. Bedding attitude hardly varies throughout the section and the fold test cannot be applied appropriately. Separate mean-site results have been determined for selected sites from the base (primary), the middle (overprint) and the top (primary) of the Washpool Creek section (Table 4).

Cana Creek (28–41) Several sites, close to the internal shear zone in the Cana Creek section (31–34: Figure 6) and close to the western base thrust (Mooki Thrust) of the Quipolly Dome (28: Figure 4), show P2 poles (Figure 14c) resembling the Permian pole-path trajectory (Figure 12g–i). They are interpreted as overprints facilitated by early fluid flow along fault zones which must have originated prior to Early Triassic westwards thrusting of the Tamworth Belt and are probably associated with initial development of the Bowen–Gunnedah–Sydney Basin system. P2 poles for the other sites form a meridional band extending from Antarctica to the Australian North West Shelf (Figure 14c). Two P2-poles for the Cana Creek Tuff Member (29, 30) at the base of the section show a mean pole which is in reasonable agreement with other mean poles for the lower part of the Currabubula Formation, that is for the base of the Washpool Creek section and for the Kankool section (Figure 14a–c; Tables 3, 4). P2 poles for the upper part of the Cana Creek section (35–41) show a considerable spread with no apparent stratigraphic trend and may represent secular variation. Their mean pole position (Table 4; 6 in Figure 15a) locates off southwestern Australia and may reflect the northward pole-path shift also observed in the correlated Rocky Creek Conglomerate and Lark Hill Formation of the Rocky Creek Block (Klootwijk 2002 figure 15). Their mean inclination (71.4°),

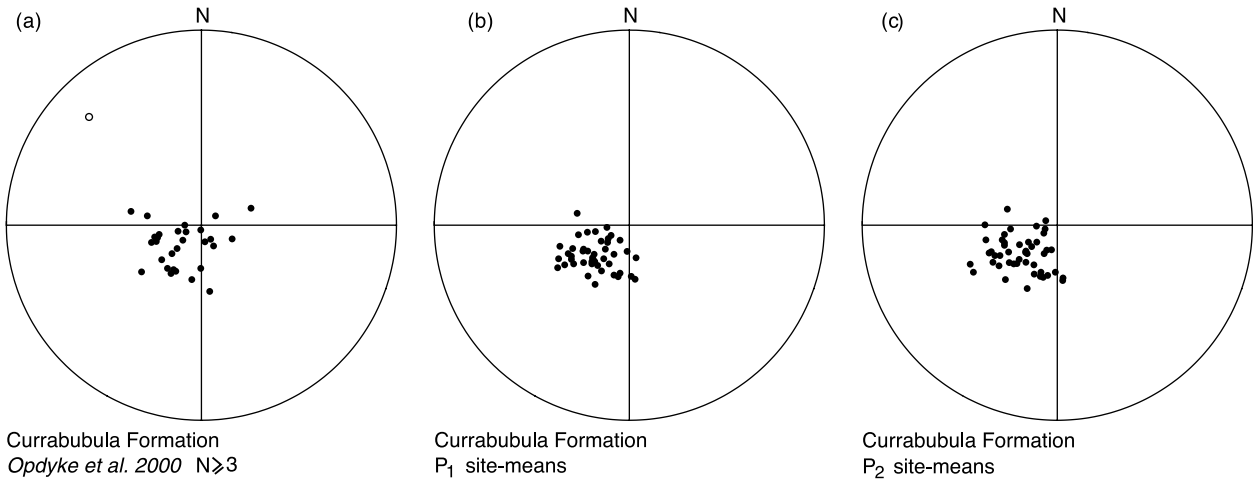


Figure 13 Comparison of the overall distribution of site-mean directions for the Currabubula Formation. (a) B-components (after Opdyke *et al.* 2000 table 1). (b) HT-P1 (occasionally P2, P0) (pseudo)-components (Table 5*): not shown are results listed in Table 5* under footnotes g and h. (c) HT-P2 (occasionally P, PH) primary components (Table 2): not shown are results listed in Table 2 under footnotes g, h and i. Full circles, lower hemisphere; open circles, upper hemisphere. Equal-area projection. Mean-site directions and statistics for relevant subsets of the data are given in Opdyke *et al.* (2000, table 3) for (a), Table 7* for (b), and Tables 3, 4 for (c).

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Table 3 Fold test (McFadden 1990), primary components (HT).

Result no[a]	Site Formation/member	*n*	Site nos (Comp)[b]	*In situ* DEC (°)	INC (°)	K[c]	α95[d] (°)	SCOS2[e]	Part unfold[f]	Unfolded DEC (°)	INC (°)	K[c]	α95[d] (°)	SCOS2[e] (SCOS1)	Confidence level (%)[g]
1	Merlewood Formation	7	1-7	172.6	21.1	7.5	23.6	2.977		181.7	19.8	13.0	17.4	1.392	
2	Currabubula Fm. base, Kankool Basin	10	43-46,48-53	134.6	71.0	26.3	9.6	7.398		209.1	69.4	118.6	4.5	1.112	99
4	Currabubula Fm. base, Werrie Basin	11	10-13,17,29,30, 58, 59,62,74	139.6	84.5	7.3	18.1	10.511		219.2	73.1	107.0	4.4	0.162	99
8	Currabubula Fm., top, Werrie Basin	18	24,25,26(P),27(PH), 35-41,64-66, 71-73,75(PH)	244.3	53.1	5.6	16.1	17.312	93%	243.5 243.6	65.5 64.7	99.3 130.6	3.5 3.0	6.738 (0.061)	99
i	Currabubula Fm., top selected, Werrie Basin	11	24,25,26(P), 27(PH),64-66, 71-73,75(PH)	243.4	76.3	10.8	14.6	10.871		240.6	61.6	233.9	3.0	1.536	99
9	Boggabri Volcanics	6	86(PH),87,88(P), 89-91(PH)	149.9	56.0	53.4	9.3	3.855	69%	176.1 161.1	78.0 71.7	147.7 236.9	5.5 4.4	3.813 (0.010)	
10	Werrie Basalt	9	77-85	183.0	71.3	17.1	12.8	6.837		199.8	72.2	29.6	9.6	1.395	99

The McFadden (1990) fold test has been applied routinely. The McFadden (1998) fold test requires at least two observations per cluster, a requirement not met by all datasets. [a]Pole number, see Table 4 and Figure 15. [b]Site numbers refer to Table 2 and relate to the P2 component unless indicated otherwise. [c]K represents the best estimate of the precision parameter κ (Fisher 1953). [d]α95 indicates the half angle of the circle of 95% confidence around the mean-site direction (Fisher 1953). [e]Critical values for different *n* of the test statistic SCOS2 for the correlation test (McFadden 1990) at the 95% (99%) confidence level are: 6 = 2.862 (3.919), 7 = 3.086 (4.253), 9 = 3.497 (4.849), 10 = 3.685 (5.120), 11 = 3.865(5.378), 18 = 4.940 (6.919). [f]The percentage of unfolding required to minimise SCOS1: directional statistics are detailed as an alternative to complete (100%) unfolding. [g]Fold tests that show at the 95% (99%) confidence level correlation between magnetisation and bedding in the *in situ* position and no correlation in the unfolded position.

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Table 4 Pole-path data.

Formation/site			Direction				South Pole position							
Pole no[a]	Formation / Member	Site no.[b]	n[b]	Decl. (°)	Incl. (°)	KD[c]	ED95[c] (°)	Lat. (°S)	Long. (°E)	Palaeolat. (°S)	Palaeolat.[d] (°S)	DP[c](°)	DM[c](°)	KP[c]	EP95[c] (°)
1	MERLEWOOD FORMATION	1–7	7	181.8[e]	20.8[e]			69.4	335.7	10.8[e]	10.0			17.1	15.0
2	BASE CURRABUBULA FM., KANKOOL BASIN	43–46, 48–53	10	209.1	69.4	118.6	4.5	60.0	115.2	53.2	51.8	6.6	7.7		
a f	*base The Gap*	*74*	*(11)*	*197.0*	*68.2*	*166.8*	*3.6*	*66.4*	*123.4*	*51.4*	*50.3*	*5.0*	*6.0*		
b	*base Quipolly Creek*	*58,59,62*	*3*	*205.4*	*73.4*	*49.7*	*17.7*	*57.4*	*126.7*	*59.2*	*58.1*	*28.4*	*31.7*		
c	*base Cana Creek*	*29,30*	*2*[g]	*221.0*	*70.0*	—	—	*53.1*	*110.6*	*54.0*	*52.8*	—	—		
3	base Washpool Creek	10–13,17	5	232.9	74.1	233.8	5.0	44.9	116.8	60.3	59.2	8.2	9.0		
4	BASE CURRABUBULA FM., WERRIE BASIN	10–13,17,29, 30,58,59,62, 74	11	219.8[e]	73.3[e]			51.6	118.6	59.0[e]	57.8			37.5	7.6
d	*middle Washpool Creek*[h]	*18,19,21–23*	*5*	*221.3*	*78.6*	*105.3*	*7.5*	*46.0*	*129.8*	*68.0*	*67.0*	*13.4*	*14.2*		
5	top Washpool Creek (Piallaway Trig Ignimbrite Member)	24–27	4	240.3	59.1	376.3	4.7	41.1	88.5	39.9	38.8	5.3	7.0		
6	top Cana Creek	35–41	7	250.3	71.4	105.2	5.9	36.3	109.9	56.1	54.9	9.0	10.3		
7	top Quipolly Creek	64–66	3	246.0	61.2	517.2	5.4	37.4	92.4	42.3	41.2	6.4	8.3		
e f	*top Tara Gap*	*71*	*(12)*	*247.5*	*64.4*	*136.9*	*3.7*	*37.1*	*97.5*	*46.2*	*45.1*	*4.8*	*6.0*		
f f	*L Quipolly, Piallaway Trig Ignimbrite Member*	*72*	*(11)*	*230.7*	*65.4*	*66.6*	*5.6*	*48.6*	*98.5*	*47.6*	*46.3*	*7.4*	*9.2*		
g f	*L Quipolly, Piallaway Trig Ignimbrite Member*	*73*	*(11)*	*226.5*	*69.3*	*544.2*	*2.0*	*50.4*	*107.2*	*52.9*	*51.6*	*2.9*	*3.3*		
h f	*top The Gap*	*75*	*(10)*	*238.5*	*58.1*	*111.8*	*4.6*	*42.3*	*86.5*	*38.8*	*37.6*	*5.0*	*6.8*		
8	TOP CURRABUBULA FM., WERRIE BASIN	24–27,35–41,64–66,71–73,75	18	243.7[e]	65.9[e]			39.8	99.6	48.2[e]	47.1			44.9	5.2
i	*Currabubula top selected*	*24–27,64–66, 71–73,75*	*11*	*240.5*[e]	*61.9*[e]			*41.5*	*92.6*	*43.1*[e]	*41.9*			*126.0*	*4.1*
9	BOGGABRI VOLCANICS	86–91	6	176.1	78.0	147.0	5.5	53.6	152.6	67.0	66.9	9.7	10.4		
10	WERRIE BASALT	77–85	9	199.8	72.2	29.5	9.6	61.2	128.7	57.3	55.8	15.0	17.0		

[a]See Figures 15–17 for pole path. Numbers indicate main results. Formation-mean results in capitals. Other characters (italics) indicate single- or multi-site results of lesser significance. See Table 3 for results with significant positive fold tests. Results with Fisher (1953) statistics listed under the KD and ED95 columns are derived from individual site-mean directions. Results with Fisher (1953) statistics listed under the KP and EP95 columns are derived from site VGPs. For the latter results an equivalent mean direction is calculated for the reference location of Werris Creek (footnote d). Note that these equivalent mean directions differ slightly from the mean directions detailed in Table 3 which are based on site-mean directions. [b]Number of site-mean results as detailed in Table 2 (*number of specimens for single site result*). [c]KD and ED95 indicate the best estimate for the precision parameter κ and the half-angle of the circle of 95% confidence after Fisher (1953) for the mean-site direction. KP and EP95 represent the same for the mean of individual site VGPs. DP and DM represent the half angles of the ellipse of 95% confidence derived from ED95. [d]Armidale (30.53°S, 151.67°E) reference location. [e]Werris Creek (31.33°S, 150.68°E) reference location. [f]Single site, site-mean result. [g]Mean of two site-mean results, no statistical parameters provided. [h]Unclear origin, possibly affected by Permian/Permo-Triassic overprinting?

however, is steeper than for the top of the Washpool Creek section (59.1°) and also for the top of the Quipolly Creek section (61.2°) discussed below. Rock magnetic results show the overprinted sites (28, 31–34) to be distinguishable from the other sites by prevalence of multidomain magnetite and hematite carriers.

Quipolly Creek Dam (56–67) Several sites with a steep to very steep site-mean inclination and a pole near the Permo-Triassic (57, 60, 63) or Late Permian (61) segments of the Australian pole path (Figures 12g–i, 14d) are interpreted as overprints (Table 6*). The remaining sites show poorly defined poles for the lower part of the section (58, 59, 62) and better defined, well-concentrating, poles for the top of the section at Quipolly Dam (64–66). The poles (Figure 14d) show a shift up-stratigraphy from an eastern Antarctic location, comparable to the Cana Creek Tuff sites

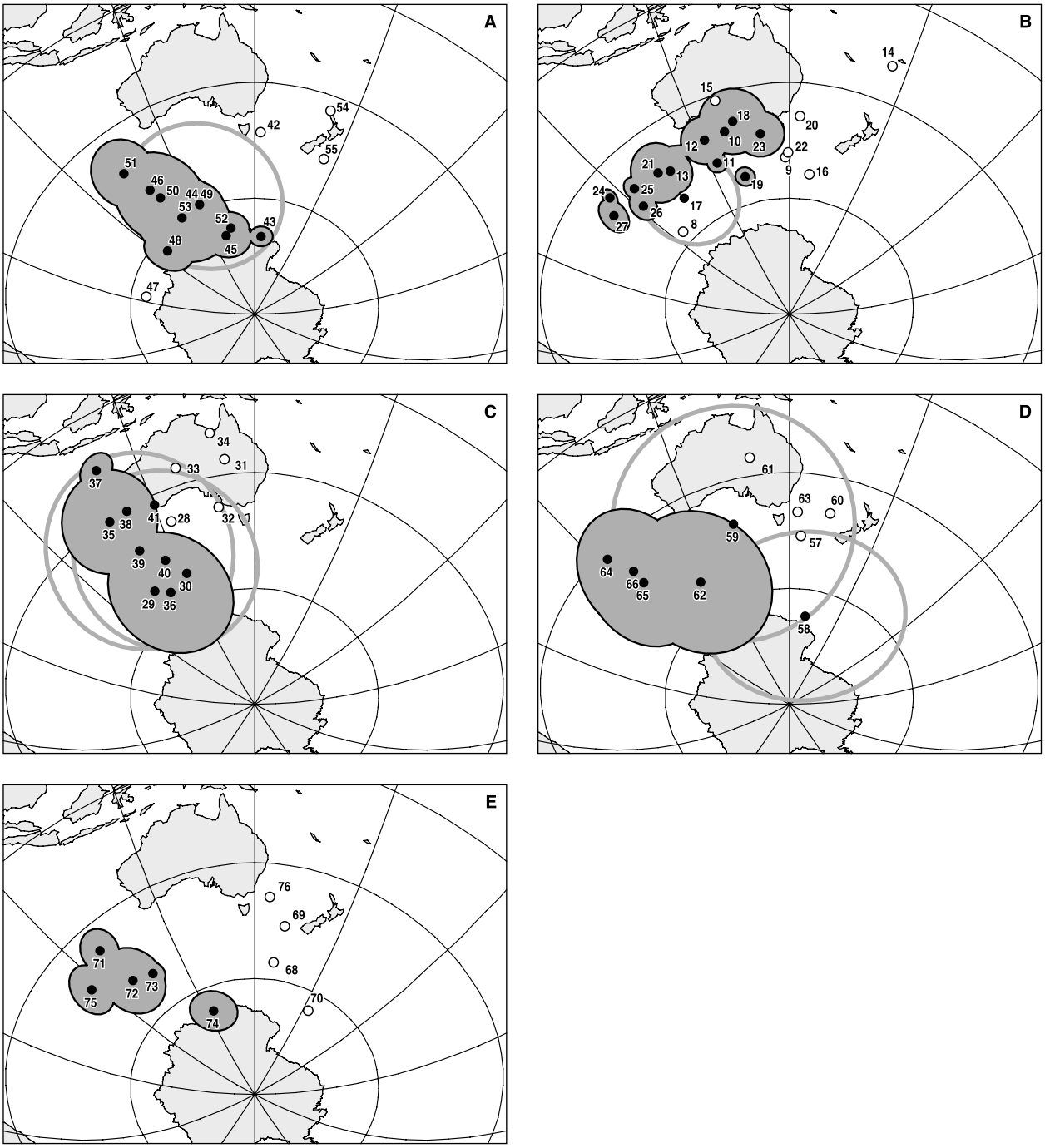


Figure 14 Pole positions for HT-P2 (occasionally P, PH) components from sites in the main sections of the Currabubula Formation. The numbers refer to site numbers in Table 1 (site detail), Table 2 (primary poles and poles of ambiguous or suspected secondary origin), and Table 6* (overprint and odd poles). (a) Chilcotts Creek. (b) Washpool Creek. (c) Cana Creek. (d) Quipolly Creek and Quipolly Dam. (e) Jacob and Joseph Creek, Lower Quipolly and The Gap. Primary magnetisations are indicated as full dots with α_{95}-equivalent error ellipses; secondary magnetisations as open circles without error ellipses. Very large error ellipses are shown as outlines only without shading. All poles are based on stratigraphic coordinates.

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(Figure 14a), to locations southwest of western Australia, in good agreement with results for the Kankool, Washpool and Cana Creek sections (Figure 14a–c). A distinguishing rock magnetic characteristic of the primary sites is the presence of hematite, whereas the overprinted sites are characterised by multidomain magnetite.

Other Currabubula Formation sites Most *Jacob and Joseph Creek* (*68–71)* sites show rather poorly defined P2 poles (68–70) that locate on the Permo-Triassic pole-path trajectory (Figures 12g–i, 14e) and are interpreted as overprints of that age. The uppermost site 71, in contrast, shows a well-defined pole whose position is in good agreement with results for the top of the Currabubula Formation and is interpreted as a primary magnetisation (Figure 14e). The two sites for the Piallaway Trig Ignimbrite Member from *Lower Quipolly (72, 73)* show P2 poles in good agreement with other results from the top member of the Currabubula Formation and are also interpreted as primary magnetisations (Figure 14e). Results from *The Gap (74–76)* vary in quality and interpretation. The uppermost silicic siltstone site (76) shows a reasonably well-defined P2 pole of apparent Permo-Triassic overprint origin. The lowermost unwelded tuff site (74) shows a well-defined P2 pole, seemingly in agreement with the early part of the Currabubula Formation track (Figure 14e). A primary origin is plausible, but cannot be corroborated for want of detailed stratigraphic and/or radiometric age control. The main crystal-flow site 75 shows a well-defined P2 pole which is interpreted as primary for reason of its good agreement with other P2 poles for the top of the Currabubula Formation (Figure 14e), in accordance with the site's stratigraphic position above the Piallaway Trig Ignimbrite Member (Figure 5). Rock-magnetic results show a MD-magnetite carrier in sites 68, 70 and 76 and a hematite carrier in site 69. These features are not, or not obviously, observed in the other primary sites and are therefore related to the Permo-Triassic overprint origin of P2 magnetisations from these four sites.

WERRIE BASALT (77–85)

Individual sites show well-defined P2 poles, apart from site 85. Most site poles are spread north of eastern Antarctica, but three poles (80, 82, 83) locate considerably off this grouping. Bedding control for the sites is generally poor, in particular for sites 82–85. Sites 82–84 show well-defined anisotropy ellipses, but application of bedding control based on the orientation of individual K_{min} axes increases, rather than decreases, the spread in poles. Two other palaeomagnetic studies on the Werrie Basalt (Schmidt 1988; Lackie & Schmidt 1993) also showed considerable spread in site-mean directions. The presently observed spread, therefore, could reflect inherent characteristics such as secular variation rather than erroneous bedding control. The mean-site result has a considerable α_{95} angle, but the McFadden (1990) fold test is positive at the 99% confidence level (Table 3). Rock-magnetic results show the prevalence of a multidomain magnetite carrier in all sites, apart from site 77. Sites 78 and 81, and less evidently 84, also show presence of a hematite carrier.

BOGGABRI VOLCANICS (86–91)

The highest temperature components for the Lower Permian Boggabri Volcanics (P2 and PH) show generally well-defined site-mean directions, although with a larger α_{95} value for site 91. The site poles concentrate quite well with a mean-site pole to the south of Tasmania, about intermediate between the above described Permian Werrie Basalt pole and Schmidt and Lackie's (1993) pole for the same. Rock-magnetic results show the general presence of both a magnetite and a hematite carrier with

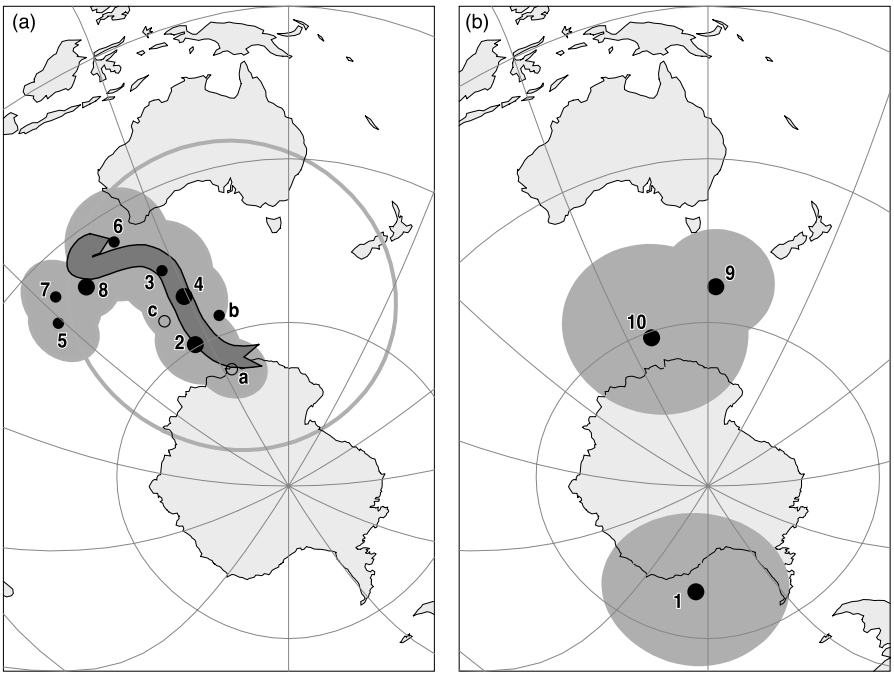


Figure 15 (a) Mean poles and pole path for the Currabubula Formation. (b) Mean poles for the Merlewood Formation, Werrie Basalt and Boggabri Volcanics. The numbers and characters refer to pole positions listed in Table 4, with fold-test results listed in Table 3. Large full circles, formation-mean poles; small circles, section-mean poles (full symbol, $n \geq 3$; open symbol $n < 3$) and single-site poles (open symbol). See caption to Figure 14.

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indications for multidomain magnetite (88–91) from low-temperature susceptibility cycling, and for single-domain magnetite (86, Hopkinson peak) from high-temperature cycling.

Fold tests and mean results

The McFadden (1998) fold test could not be universally applied in the Rocky Creek Block study (Klootwijk 2002) because of limited cluster sizes, and the McFadden (1990) fold test was used instead. For consistency, the latter test has been applied also in the present study.

Three mean results have been determined for different parts of the Currabubula Formation (Tables 3, 4, 7*), that is for the Kankool Basin section and for the lower and upper parts of the Currabubula Formation of the Werrie Basin. The Kankool Basin section pre-dates the Werrie Basin sections according to McPhie's (1983, 1984b) mapping and this is borne out by their mean directions and pole positions (Tables 3, 4; Figure 15). The Werrie Basin sections show a considerable and rather consistent shift in site virtual geomagnetic poles (Figure 14). Mean-site directions and pole positions for the two lower members (Cana Creek, Iventure Ignimbrite) and the two upper members (Taggarts Mountain Ignimbrite, Piallaway Trig Ignimbrite) of the Currabubula Formation and their equivalents are significantly different indeed (Tables 3, 4; Figure 15).

The fold test for the upper part of the Currabubula Formation shows optimal concentration at slightly less than full unfolding (93%: Table 3 result 8). The reason for this is unclear. The fold test includes results from both limbs of the Werrie Syncline: the eastern limb and the tectonically more complex western limb. The optimal partial, rather than full, unfolding could reflect unnoticed local rotations within sections from the western limb, and/or, perhaps, incomplete separation of primary and overprint components. An alternative mean result (Table 3 result i), excluding the site-mean results from the tectonically affected Cana Creek section, shows optimal concentration at full unfolding. The Boggabri Volcanics also show more optimal partial unfolding (69%: Table 3 result 9), possibly reflecting unnoticed rheomorphic effects in some of the lavas.

The seven mean-site directions for the four studied formations all show positive fold tests for both the P2 primary components and the P1 pseudo-components, with confidence levels of 99% for five of the seven results in each set, and with a slightly superior precision for all P2 components (Tables 3, 7*). The P2 components for the Curra-

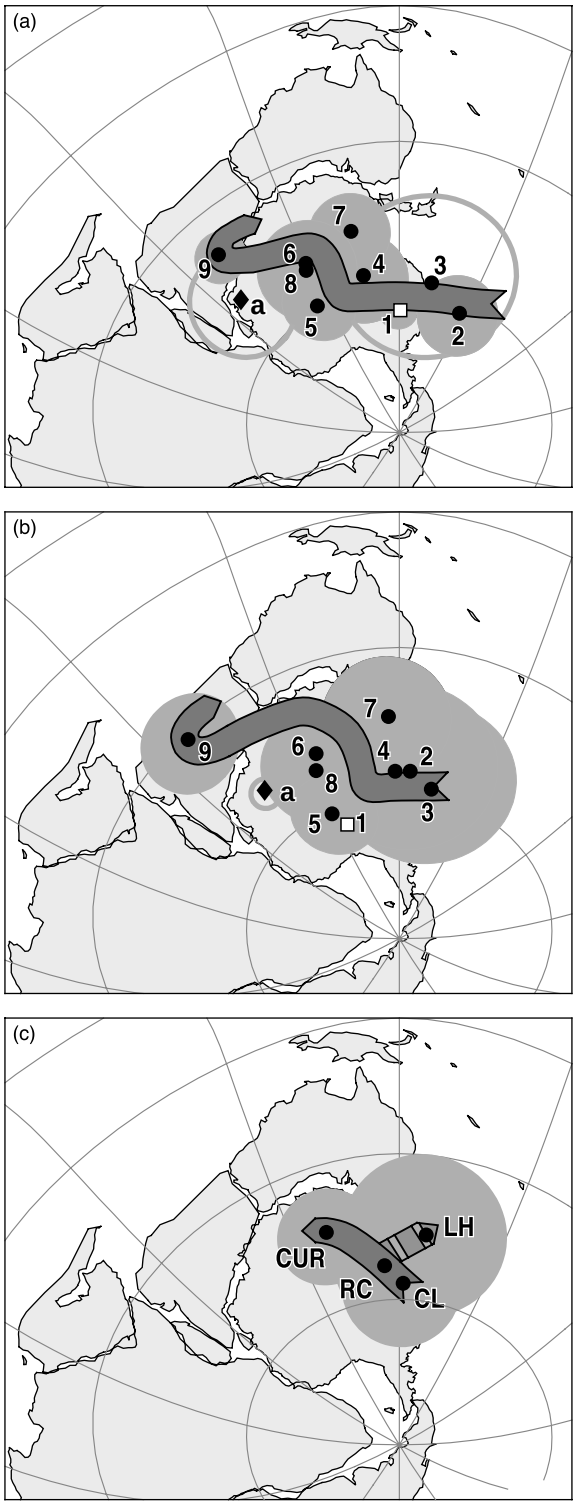


Figure 16 Comparison of pole path trajectories based on the combined Late Carboniferous results for the Clifden – Lark Hill Formations of the Rocky Creek Block and the Currabubula Formation of the Werrie Block. (a) Pole path based on primary component (P2) results (see legend below and listings in Table 4 and Klootwijk 2002 table 4). (b) Pole path based on equivalent pseudo-component (P1) results (see Table 7* and Klootwijk 2002 table 7*). (c) Pole path based on Opdyke *et al.*'s B-component results (Opdyke *et al.* 2000 table 3). 1, Kooringal Dacite (Klootwijk 2002 table 4: b); 2, Peri Rhyolite (Klootwijk 2002 table 4: 2); 3, tuffaceous beds above the Boomi Rhyolite (Klootwijk 2002 table 4: d); 4, Rocky Creek Conglomerate (Klootwijk 2002 table 4: 4); 5, base Currabubula Formation, Kankool Basin (Table 4: 2); 6, base Currabubula Formation, Werrie Basin (Table 4: 4); 7, Lark Hill Formation, Rocky Creek (Klootwijk 2002 table 4: 6); 8, Lark Hill Formation, Tops Road (Klootwijk 2002 table 4: 5); 9, top Currabubula Formation, Werrie Basin (Table 4: 8); a, Upper Clifden Formation – lower Rocky Creek Conglomerate (Klootwijk 2002 table 4: 3, probable overprint); CL, Clifden Formation (Opdyke *et al.* 2000 table 3); RC, Rocky Creek Conglomerate (Klootwijk 2002); LH, Lark Hill Formation (Klootwijk 2002); CUR, Currabubula Formation (Klootwijk 2002). See caption to Figure 14.

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bubula Formation consistently show slightly higher inclinations and mostly slightly lower declinations than the P1 results. This minor, but systematic, difference in the main Werrie Block results manifests itself as a slight offset to the southwest of the P2 versus the P1 pole path (see below). This supports and extends the observation of a comparable offset for the Rocky Creek Block results (Klootwijk 2003 figure 3).

Dating control, magnetic polarity and secular variation

Isotopic age control on the studied formations is limited to two U–Pb SHRIMP zircon dates for the Currabubula Formation (Opdyke *et al.* 2000). They define an age of 313.6 ± 3.6 Ma for an unnamed ignimbrite near the base of the formation and an age of 306.0 ± 4.2 Ma for the Piallaway Trig Ignimbrite Member (Washpool Creek section) near the top of the formation. The Currabubula Formation thus covers a time-span of ~8 million years within the Permo-Carboniferous Reverse Superchron, which lasted from *ca* 318 Ma (Opdyke *et al.* 2000) to *ca* 261–265 Ma (Haag & Heller 1991; Menning 1995, 1997; Opdyke & Channel 1996). All palaeomagnetic results for the Currabubula Formation obtained here and by Opdyke *et al.* (2000) show reverse polarity, as expected.

The time-span of the Currabubula Formation is relatively short, but the site virtual geomagnetic poles show a considerable spread (Figure 14). This may reflect apparent polar wander during Currabubula Formation time, tracking from near eastern Antarctica to near southwestern Australia, and also secular variation, with a considerable spread expected for high latitude sites during a superchron (Merrill *et al.* 1996) (Figure 3).

The mean results for the base of the Currabubula Formation of the Kankool Basin and for the base and top of the Currabubula Formation of the Werrie Basin are all based on substantial sets of sites (10–18) of both volcanic and volcaniclastic rocks. These results are therefore not expected to reflect bias due to non-averaged secular variation effects. This cannot be excluded though for the mean results of the Boggabri Volcanics and the Werrie Basalt which are based on limited sets, 9 and 6 respectively, of volcanic units. The mean result for the Merlewood Formation represents two flows. It has not been established whether the seven studied sites may represent more than three cooling units. This mean result, therefore, may be affected by secular variation.

Late Palaeozoic pole path for the northwestern Tamworth Belt

The P2 pole-path trajectory for the Currabubula Formation, from north of eastern Antarctica to west of southwestern Australia, is in agreement with the P2 poles for the correlated Rocky Creek Conglomerate and Lark Hill Formation of the Rocky Creek Block (cf. Figure 15a with Klootwijk 2002 figure 15). The Currabubula trajectory trends slightly to the west of the Rocky Creek Block trajectory. This offset is in the opposite sense to that expected from the slight difference in regional strike between the Werrie and Rocky Creek Blocks. Pending further pole-path detail this minor offset is taken to be of lesser significance and a joint pole path is constructed for the upper successions of both blocks, that is the Clifden Formation, Rocky Creek Conglomerate and Lark Hill Formation of the Rocky Creek Block and the Currabubula Formation of the Werrie Block (Figure 16a). This combined P2 pole path is offset slightly, but systematically, to the southwest of the combined P1 pole path (Figure 16b). Both the P2 and the P1 pole path show general agreement with the poles obtained by Opdyke *et al.* (2000) (Figure 16c) for the same formations. However, Opdyke *et al.* (2000) did not distinguish between P2 primary components and P1 pseudo-components and the trajectory indicated by their poles (Figure 16c) locates, as expected, between the P2 (Figure 16a) and the P1 (Figure 16b) pole-path trajectories for the combined results of the Rocky Creek and Werrie Blocks.

The pole determined for the Lower Permian Werrie Basalt (10 on Figure 15b) is not well defined and locates considerably off Lackie and Schmidt's (1993 figure 1) pole, with the pole for the Lower Permian Boggabri Volcanics (9 on Figure 15b) located in between. The Early Permian pole path remains essentially undefined between the latest Carboniferous pole for the top of the Currabubula Formation (Figure 15a, 16a) and these three Early Permian poles. Further pole-path detail may be obtained by inclusion of latest Carboniferous and Early Permian poles from northeastern Queensland (Klootwijk *et al.* 1993; Lackie 2000; Anderson *et al.* 2003; Clark & Lackie 2003), but only after correction for displacement, if any, relative to the southern New England Orogen.

The pole for the Merlewood Formation (Figure 15b) is poorly defined, but is close to an earlier, also poorly defined, pole by Schmidt (1988) for the same formation. This agreement and the presence of both normal and reverse polarities in the present results lends some credibility to its representation as a primary pole for the formation, provided that the sampled beds are flows and not intrusions (see stratigraphic overview). The age range of the Merlewood Formation is not clear, and may reach from earliest Late Visean to as far back as earliest Early Visean, possibly Tournaisian–Visean (Figure 3). In absence of SHRIMP dating control, it is possible, if not probable, that the here-defined pole for the lower part of the Merlewood Formation pre-dates the pole for the High Valley andesite tuff of the Rocky Creek Block. This is indicated on a tentative extension of the joint pole path for the Rocky Creek and Werrie Blocks from the Late Carboniferous (Figure 16a) into the Early Carboniferous (Figure 17a).

DISCUSSION

Regional overprint patterns

A post-folding overprint of probable Middle to Late Cenozoic origin has been observed widely throughout the Werrie and Rocky Creek Blocks and may be indicative for a regional weathering event throughout southeastern Australia, if not Australia-wide. In contrast to the Rocky Creek Block study, Late Cretaceous overprinting associated with rifting of the Tasman Sea has not been observed.

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Pre-folding overprints of Permian and Permo-Triassic origin have been observed throughout the Werrie Block, with Permian overprints more prevalent and Permo-Triassic overprints less prevalent than in the Rocky Creek Block. Several other studies of the Tamworth Belt reported widespread Late Palaeozoic – Early Mesozoic overprinting (Schmidt & Lackie 1993; Lackie & Schmidt 1993; Geeve 2000a), but not Opdyke *et al.*'s (2000) extensive study of the Werrie and Rocky Creek Blocks. This is understandable because the latter study concentrated, partly for SHRIMP dating control, on ignimbrites and these are less prone to overprinting than other volcanic and volcaniclastic lithologies as demonstrated here and previously (Klootwijk 2002). The present observation of more prevalent Permian overprinting in the Werrie Block than in the Rocky Creek Block, is in line with Geeve (2000a) observation of pervasive pre-folding overprinting in the Rouchel, Gresford and Myall Blocks of the southern Tamworth Belt, interpreted by him as latest Carboniferous to earliest Permian in age. Syn-folding overprints have not been established in the present study, in contrast to Schmidt and Lackie's (1993) observations of syn-folding and pre-folding overprints throughout the Kiah Limestone, interpreted by them to reflect northward migration of remagnetisation across the Tamworth Belt during the Permian. The more widespread recording of overprints in carbonates by Schmidt and Lackie (1993) is again not surprising because the ignimbritic volcanic and volcaniclastic rocks studied here were selected for their capability to retain a primary magnetisation and to resist overprinting.

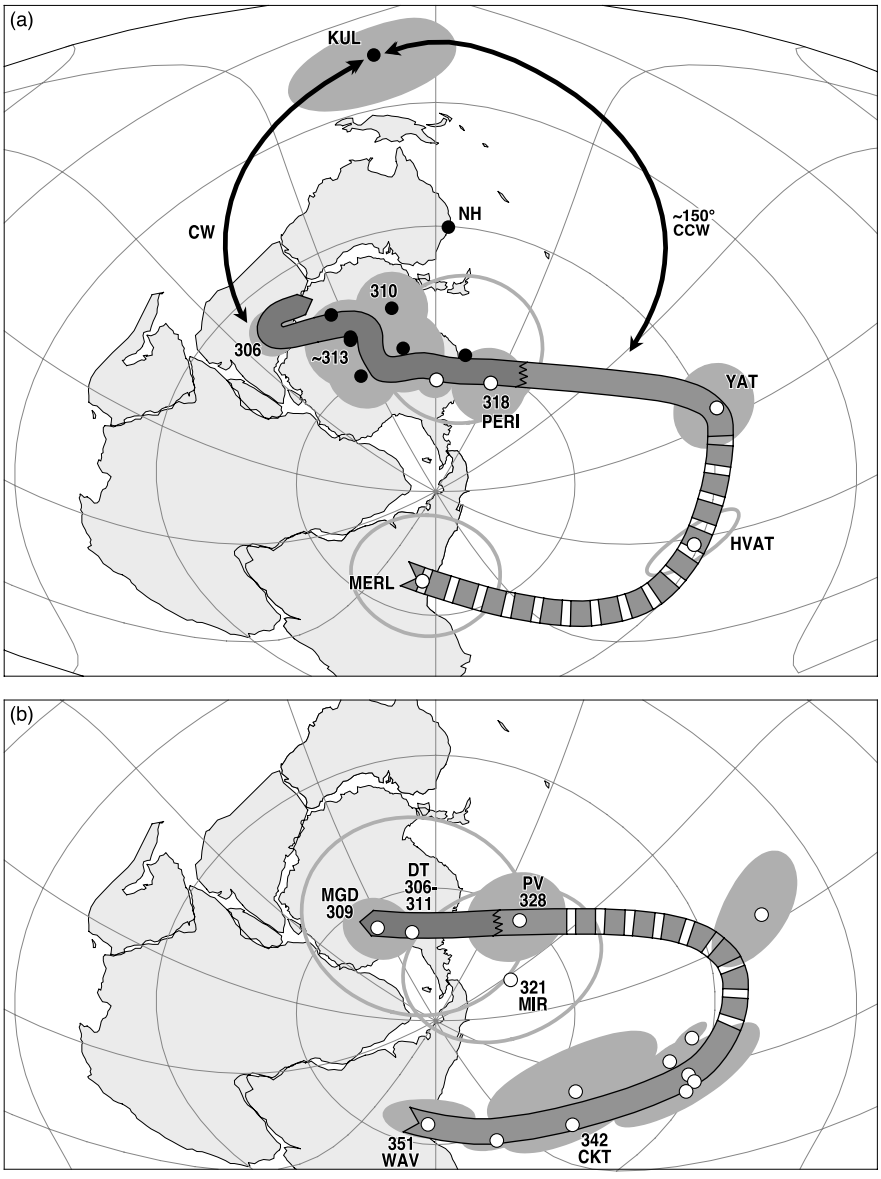


Figure 17 Comparison of the here-detailed Carboniferous pole path for the northwestern Tamworth Belt: (a) Rocky Creek and Werrie Blocks with a preliminary pole path for the southwestern Tamworth Belt; (b) Rouchel and western Gresford Blocks. Poles without acronym in (a) are the poles shown in Figure 16a and listed in Table 4 or Klootwijk (2002 table 4). Palaeomagnetic poles are shown as full circles. Open circles represent single site or multiple site poles, possibly affected by secular variation; MERL, Merlewood Formation (Table 4: 1); HVAT, High Valley andesite tuff (Klootwijk 2002 table 4: a); YAT, Yuendoo Rhyolite (Klootwijk 2002 table 4: 1); PERI, Peri Rhyolite (Klootwijk 2002 table 4: 2). Data for (b) are listed in Klootwijk (1996b table 1.3: all represent preliminary results): WAV, Waverley Formation [Rouchel Block (RB)]; CKT, Curra Keith Tongue (RB); PV, Paterson Volcanics [Gresford Block (GB)]; MIR, Mirannie Ignimbrite Member (GB); DT, Mt Durham Tuff Member (GB); MGD, Matthews Gap Dacite Tuff Member (Pokolbin Hills). Comparison of the northwestern Tamworth Belt pole path (a) with the pole for the Namurian Kullatine Formation (KUL: Schmidt *et al.* 1994) from the Northern Hastings Block (NH) shows the likelihood of an about 150° counterclockwise rotation of the Northern Hastings Block relative to the northwestern Tamworth Belt. Schmidt *et al.* (1994) figure 9), in contrast, interpreted their Kullatine Formation pole in terms of a 130° clockwise (or 230° counterclockwise) rotation of the Northern Hastings Block relative to the Australian craton. See caption to Figure 14.

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Pole-path control on rotational deformation of the Tamworth Belt

One of the main aims of the Tamworth Belt study is to define a Carboniferous pole path for the New England Orogen by integrating mainly Late Carboniferous results for the Rocky Creek and Werrie Blocks with mainly Early Carboniferous results for the Rouchel and Gresford Blocks. This requires control on rotational deformation, if any, between the northwestern and southwestern parts of the Tamworth Belt. Pending forthcoming results from detailed palaeomagnetic studies on the Rouchel and Gresford Blocks, such control can be established tentatively through comparison of the detailed results for the Werrie and Rocky Creek Blocks (Figure 17a) with preliminary results for the Rouchel and Gresford Blocks (Klootwijk 1996a, b). A tentative Carboniferous pole path trajectory for the southwestern Tamworth Belt is shown in Figure 17b, based on selected preliminary results for the Rouchel and Gresford Blocks (Klootwijk 1996b table 1.3). The selection includes results for the Waverley, Isismurra, Mt Johnstone and Seaham Formations of the Rouchel and western Gresford Blocks and for the Matthews Gap Dacite Tuff Member of the Pokolbin Hills, but does not include results for the Gilmore Volcanic Group of the eastern Gresford Block. Results for the latter have not been included because of their complexity and because the rotational deformation of the eastern Gresford Block may differ from that of the western Gresford and Rouchel Blocks by reflecting the evolution of the Manning Orocline instead.

The tentative pole path for the southwestern Tamworth Belt shows an 'eastwards' younging trend through the Waverley (350.9 ± 3.8 Ma: Roberts *et al.* 1995a) and Isismurra Formations (Curra Keith Tonque: 342 ± 3.2 Ma: Roberts *et al.* 1995a) (Figure 17b). This is followed by a 'westwards' younging trend through the Mt Johnstone Formation (Paterson Volcanics: 328.5 ± 1.4 Ma: Claoué-Long *et al.* 1995), Seaham Formation (Mirannie Ignimbrite Member: 321.3 ± 4.4 Ma: Roberts *et al.* 1995b; Mt Durham Tuff Member) and the Matthews Gap Dacite Tuff Member (309 ± 3 Ma: Gulson *et al.* 1990). Two of the four preliminary poles on this younger part of the path are poorly determined, but taken as a group they outline a pole path trajectory that shows good agreement with the pole path trajectory for the upper successions of the Rocky Creek and Werrie Blocks (Figure 17a). This agreement indicates no significant, if any, rotational deformation between the northwestern and southwestern parts of the Tamworth Belt. Quantification and further qualification of this preliminary conclusion awaits the detailed outcomes of the forthcoming Rouchel and Gresford Block studies. Good agreement between poles from regions on opposite sides of the Hunter Thrust (to the south, Matthews Gap Dacite Tuff Member pole from the Pokolbin Hills; to the north, other poles) supports this preliminary conclusion.

There is also coincidence between the pole for the High Valley andesite tuff of the Rocky Creek Block (HVAT in Figure 17a) and poles for ignimbrite members of the upper Isismurra Formation (Figure 17b). Correlation of these sampled units is stratigraphically permissible, but fortuity cannot be excluded in absence of SHRIMP dating control. General agreement between the ill-defined pole for the Merlewood Formation of the Werrie Block and the better defined pole for the Waverley Formation of the Rouchel Block is somewhat surprising. Absence of rotational deformation would impose a Tournaisian age on the studied part of the Merlewood Formation, whereas available stratigraphic control indicates an earliest Early Visean, possibly Tournaisian–Visean, lower age limit.

Reference pole path(s): choice and implications

The present tentative observation of no significant rotational deformation between the northwestern and southwestern Tamworth Belt contrasts with the conclusion of Geeve (1999; 2000b), Roberts and Geeve (1999), and Geeve *et al.* (2002) of a counterclockwise rotation by 70–80° of the combined Rouchel and Gresford Blocks versus the Australian craton. Their conclusion depends on preference for the SLP-pole path (Klootwijk 1996a figure 4) as the representative Late Palaeozoic pole path for the Australian craton and the New England Orogen. In contrast, comparison with the alternative KG-pole path for cratonic Australia (Klootwijk 1996a figure 5) indicates no significant rotational deformation of the Rouchel and Gresford Blocks versus the Australian craton. This is in agreement with the present observation of no significant rotation between the southwestern and northwestern parts of the Tamworth Belt.

Different choices for the reference pole path also lead to contrasting conclusions on rotation of the Northern Hastings Block and on evolution of the Manning Orocline. Schmidt *et al.* (1994 figure 9) compared their well-defined pole for the Namurian Kullatine Formation with the SLP-pole path for the Australian craton and concluded that there was an 130° clockwise, or an 230° counterclockwise, rotation of the Northern Hastings Block relative to cratonic Australia. This was also taken as the rotation of the Northern Hastings Block relative to the western Tamworth Belt. Direct comparison of the pole for the Kullatine Formation with the here defined pole path for the northwestern Tamworth Belt leads to a different conclusion (Figure 17a). Clockwise rotation does not appear an option because the Kullatine pole locus intersects the pole path for the northwestern Tamworth Belt in far too young a location for this option to be sustainable. In contrast, counterclockwise rotation over about 150° appears a viable option, under the assumption that the studied redbeds of the Kullatine Formation pre-date the Peri Rhyolite from the Clifden Formation of the Rocky Creek Block. This assumption is permissible within available stratigraphic constraints. Such an 150° counterclockwise rotation is in good agreement with predictions based on analysis of the region's structural evolution (Korsch & Harrington 1987; Schmidt *et al.* 1994; Leitch & Lara 2000).

CONCLUSIONS

Palaeomagnetic, rock magnetic and magnetic fabric studies of volcanic and volcaniclastic rocks of the Werrie Block, that is mainly the Currabubula Formation and also

the Merlewood Formation, Werrie Basalt and Boggabri Volcanics, have shown the following.

(1) A pervasive post-folding, probably Middle to Late Cenozoic, overprint with low unblocking temperature ranges (LT). This overprint mirrors a pervasive overprint in the northward adjacent Rocky Creek Block and may represent a regional weathering event in southeastern Australia, if not Australia wide.

(2) Pre-folding overprints with high unblocking temperature ranges (HT) and Permo-Triassic and Permian acquisition ages. The Permian overprint is more dominant and the Permo-Triassic less so compared with corresponding overprints in the Rocky Creek Block.

(3) A primary magnetisation component (HT-P2) accompanied by an incompletely cleaned pseudo-component (HT-P1) with unblocking temperatures generally in the uppermost magnetite–hematite range (P2) and in the upper magnetite range (P1) respectively. The pole path trajectory defined by the P2 component is in good agreement with the trajectory defined by the correlated Rocky Creek Conglomerate and Lark Hill Formation of the Rocky Creek Block. The combined results for the Werrie and Rocky Creek Blocks further detail the Namurian–Westphalian segment of the pole path for the northwestern Tamworth Belt and extend it into the Stephanian.

(4) A primary magnetisation result with rather poor precision for several andesite levels in the Merlewood Formation of the Werrie Syncline that are of presumed volcanic rather than intrusive origin, and of Visean, to perhaps, Tournaisian–Visean age. Primary magnetisation results with rather poor precision have also been determined for the Lower Permian Werrie Basalt and Boggabri Volcanics.

(5) Integration of the Currabubula and Merlewood Formation results for the Werrie Block with results for the Rocky Creek Block (Klootwijk 2002) into a Carboniferous pole path for the northwestern Tamworth Belt and comparison with a preliminary pole path for the Rouchel and western Gresford Blocks of the southwestern Tamworth Belt (Klootwijk 1996a) indicates no significant rotational deformation between the northwestern and southwestern parts of the belt.

(6) Comparison of the pole for the Namurian Kullatine Formation of the Northern Hastings Block with the pole path for the northwestern Tamworth Belt indicates counterclockwise rotation over ~150° of the Northern Hastings Block versus the northwestern part of the belt, in agreement with structural estimates.

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REFERENCES

ANDERSON K. L., LACKIE M. A., CLARK D. A. & SCHMIDT P. W. 2003. Paleomagnetism of the Newcastle Range, northern Queensland: eastern Gondwana in the Late Paleozoic. *Journal of Geophysical Research* **108**, 2282–2297.

AUBOURG C., KLOOTWIJK C. T. & KORSCH R. J. 1994. Magnetic fabric and palaeomagnetic studies in the Texas and Coffs Harbour Blocks, New England Orogen. *Australian Geological Survey Organisation Record* **1994/58**.

BECKETT J. & WATSON N. 1999. A strato-tectonic interpretation of the Gunnedah Basin aeromagnetic data. *In*: Flood P. G. ed. *Regional Geology Tectonics and Metallogeny New England Orogen NEO'99 Conference*, pp. 189–202. University of New England, Armidale.

BENSON W. N. 1920.The geology and petrology of the Great Serpentine Belt of New South Wales. Part IX. The geology of the Currabubula district. *Proceedings of the Linnean Society of New South Wales* **45**, 285–317.

BROWNLOW J. W. 1999. Geological setting and the origin of the Early Permian Boggabri Volcanics, Gunnedah Basin, N.S.W. *In*: Flood P. G. ed. *Regional Geology Tectonics and Metallogeny New England Orogen NEO'99 Conference*, pp. 325–333. University of New England, Armidale.

CAPRARELLI G. 1998. Werrie Basalt and Alum Mountain Volcanics: geochemical signatures and geodynamic importance. *Geological Society of Australia Abstracts* **49**, 67.

CAREY S. W. 1934. The geological structure of the Werrie Basin. *Proceedings of the Linnean Society of New South Wales* **59**, 351–374.

CAREY S. W. 1937. The Carboniferous sequence in the Werrie Basin. *Proceedings of the Linnean Society of New South Wales* **62**, 341–376.

CLAOUÉ-LONG J. C., COMPSTON W., ROBERTS J. & FANNING C. M. 1995. Two Carboniferous ages: a comparison of SHRIMP zircon dating with conventional zircon ages and $^{40}Ar/^{39}Ar$ analysis. *In:* Berggren W. A., Kent D. V., Aubry M.-P. & Hardenberg J. eds. *Geochronology Time Scales and Global Stratigraphic Correlation*, pp. 3–21. SEPM Special Publication **54**.

CLARK D. A. & LACKIE M. A. 2003. Palaeomagnetism of the Early Permian Mount Leyshon Intrusive Complex and Tuckers Igneous Complex, North Queensland, Australia. *Geophysical Journal International* **153**, 523–547.

COLLINS W. J. 1991. A reassessment of the 'Hunter–Bowen Orogeny': tectonic implications for the southern New England Fold Belt. *Australian Journal of Earth Sciences* **38**, 409–423.

COLLINS W. J., OFFLER R., FARRELL T. R. & LANDENBERGER B. 1993. Recent advances in understanding the geological development of the New England Province of the New England Orogen. *In*: Flood P. G. & Aitchison J. C. eds. *New England Orogen Eastern Australia NEO'93 Conference*, pp. 69–84. University of New England, Armidale.

FERGUSSON C. L. & LEITCH E. C. 1993. Late Carboniferous to Early Triassic tectonics of the New England Fold Belt, eastern Australia. *In*: Flood P. G. & Aitchison J. C. eds. *New England Orogen Eastern Australia NEO'93 Conference*, pp. 53–59. University of New England, Armidale.

FISHER R. A. 1953. Dispersion on a sphere. *Proceedings of the Royal Society of London* **A217**, 295–305.

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FLOOD P. G. & AITCHISON J. C. 1988. Tectonostratigraphic terranes of the southern part of the New England Orogen. *In*: Kleeman J. D. ed. *New England Orogen Tectonics and Metallogenesis*, pp. 7–10. University of New England, Armidale.

FLOOD R. H., CRAVEN S. J., ELMES D. C., PRESTON R. J. & SHAW S. E. 1988. The Warrigundi Igneous Complex: volcanic centres for the Werrie Basalt NSW. *In*: Kleeman J. D. ed. *New England Orogen Tectonics and Metallogenesis*, pp. 166–171. University of New England, Armidale.

GEEVE R. J. 1999. The palaeomagnetism of the latest Visean Paterson Volcanics: a reassessment. *In*: Flood P. G. ed. *Regional Geology Tectonics and Metallogeny New England Orogen NEO'99 Conference*, pp. 81–89. University of New England, Armidale.

GEEVE R. J. 2000a. Pre-folding overprints in the southern Tamworth Belt. *In*: Klootwijk C. ed. *Australian Palaeomagnetism, Rockmagnetism and Environmental Magnetism 2000 Abstracts*, pp. 12–13. Australian Geological Survey Organisation Record **2000/36**.

GEEVE R. J. 2000b. Palaeomagnetic results indicate anticlockwise rotation of the southern Tamworth Belt during the Hunter–Bowen Orogeny. *In*: Klootwijk C. ed. *Australian Palaeomagnetism, Rockmagnetism and Environmental Magnetism 2000 Abstracts*, pp. 10–11. Australian Geological Survey Organisation Record **2000/36**.

GEEVE R. J., SCHMIDT P. W. & ROBERTS J. 2002. Paleomagnetic results indicate pre-Permian counter-clockwise rotation of the southern Tamworth Belt, southern New England Orogen, Australia. *Journal of Geophysical Research* **107**, 2196–2210.

GLEN R. A., KORSCH R. J. & WAKE-DYSTER K. D. 1993. A deep seismic cross-section through the Tamworth Belt: preliminary interpretation of 4 seconds two-way time data. *In*: Flood P. G. & AITCHISON J. C. eds. *New England Orogen Eastern Australia NEO'93 Conference*, pp. 101–104. University of New England, Armidale.

GREENTREE M. R. & FLOOD R. H. 1999. Constraining the location of the Kelvin Thrust. *In*: Flood P. G. ed. *Regional Geology Tectonics and Metallogeny New England Orogen NEO'99 Conference*, pp. 95–104. University of New England, Armidale.

GULSON B., DIESSEL C. F. K., MASON D. R. & KROGH T. E. 1990. High precision radiometric ages from the northern Sydney Basin and their implication for the Permian time interval and sedimentation rates. *Australian Journal of Earth Sciences* **37**, 459–469.

HAAG M. & HELLER F. 1991. Late Permian to Early Triassic magnetostratigraphy. *Earth and Planetary Science Letters* **107**, 42–54.

HARRINGTON H. J., BRAKEL A. T., HUNT J. W., WELLS A. T., MIDDLETON M. F., O'BRIEN P. E., HAMILTON D. S., BECKETT J., WEBER C. R., RADKE S., TOTTERDELL J. M., SWAINE D. J. & SCHMIDT P. W. 1989. Permian coals of Eastern Australia. *Bureau of Mineral Resources Bulletin* **231**.

HARRINGTON H. J. & KORSCH R. J. 1985. Tectonic model for the Devonian to mid Permian of the New England Orogen. *Australian Journal of Earth Sciences* **32**, 163–179.

HENLEY J. E., BROWNLOW J. W., MCDONALD K. M. & BARNES R. G. 1999. Geological compilation of the southernmost New England Orogen. *In*: Flood P. G. ed. *Regional Geology Tectonics and Metallogeny New England Orogen NEO'99 Conference*, pp. 159–161. University of New England, Armidale.

HOLCOMBE R. J., STEPHENS C. J., FIELDING C. R., GUST D., LITTLE T. A., SLIWA R., MCPHIE J. & EWART A. 1997. Tectonic evolution of the northern New England Fold Belt: Carboniferous to Early Permian transition from active accretion to extension. *In*: Ashley P. M. & Flood P. G. eds. *Tectonics and Metallogenesis of the New England Orogen*, pp. 66–79. Geological Society of Australia Special Publication **19**.

IDNURM M. 1994. New Late Eocene pole for Australia, time-averaging of remanence directions, and palaeogeographic reference systems. *Geophysical Journal International* **117**, 827–833.

JENKINS R. B. & COLLINS W. J. 1998. Contrasting character of Carboniferous and Early Permian volcanism in the southern New England Fold Belt: tectonic implications. *Geological Society of Australia Abstracts* **49**, 226.

JENKINS T. B. H., CRANE D. T. & MORY A. J. 1993. Conodont biostratigraphy of the Visean Series in eastern Australia. *Alcheringa* **17**, 211–283.

JONES P. J. 1996. Carboniferous (Chart 5). *In*: Young G. C. & Laurie J. R. eds. *An Australian Phanerozoic Timescale*, pp. 110–126. Oxford University Press, Melbourne.

KIRSCHVINK J. L. 1980. The least-square line and plane and the analysis of palaeomagnetic data. *Geophysical Journal of the Royal Astronomical Society* **62**, 699–718.

KLOOTWIJK C. 1985. Palaeomagnetism of the Tasman Fold Belt: indication for mid-Carboniferous large-scale southward displacement of the New England region. *Geological Society of Australia Abstracts* **14**, 124–127.

KLOOTWIJK C. T. 1996a. Phanerozoic configurations of Greater Australia: evolution of the North West Shelf. Part Two, palaeomagnetic and geologic constraints on reconstructions. *Australian Geological Survey Organisation Record* **1996/52**.

KLOOTWIJK C. T. 1996b. Phanerozoic configurations of Greater Australia: evolution of the North West Shelf. Part Three, palaeomagnetic data bases. *Australian Geological Survey Organisation Record* **1996/53**.

KLOOTWIJK C. T. 2002. Carboniferous palaeomagnetism of the Rocky Creek Block, northern Tamworth Belt, and the New England pole path. *Australian Journal of Earth Sciences* **49**, 375–405.

KLOOTWIJK C. T. 2003. Reply. Carboniferous palaeomagnetism of the Rocky Creek Block, northern Tamworth Belt, and the New England pole path. *Australian Journal of Earth Sciences* **50**, 131–135.

KLOOTWIJK C. T. Sedimentary basins of eastern Australia: palaeomagnetic constraints on geodynamic evolution in a global context. *In*: Korsch R. J. & Totterdell J. M. eds. *Evolution and Petroleum Potential of the Bowen, Gunnedah and Surat Basins, Eastern Australia*. Australian Geological Survey Organisation Bulletin (in press a).

KLOOTWIJK C. T. DM Tangorin DDH1 Drillcore, Cranky Corner Basin, magnetostratigraphy, palaeomagnetism and rockmagnetism. In: Facer R. & Stevenson D. K. eds. *The Cranky Corner Basin*. Department of Mineral Resources, Sydney (in press b).

KLOOTWIJK C., GIDDINGS J. W. & PERCIVAL P. 1993. Palaeomagnetic reconnaissance of Upper Palaeozoic volcanics, northeastern Queensland. *Australian Geological Survey Organisation Record* **1993/36**.

KORSCH R. J. 1977. A framework for the Palaeozoic geology of the southern part of the New England Geosyncline. *Journal of the Geological Society of Australia* **25**, 339–355.

KORSCH R. J. & HARRINGTON H. J. 1981. Stratigraphic and structural synthesis of the New England Orogen. *Journal of the Geological Society of Australia* **28**, 205–226.

KORSCH R. J. & HARRINGTON H. J. 1987. Oroclinal bending, fragmentation and deformation of terranes in the New England Orogen, Eastern Australia. *In*: Leitch E. C. & Scheibner E. eds. *Terrane Accretion and Orogenic Belts*, pp. 129–139. American Geophysical Union Geodynamics Series **19**.

KORSCH R. J., JOHNSTONE D. W. & WAKE-DYSTER K. D. 1997. Crustal architecture of the New England Orogen based on deep seismic reflection profiling. *In*: Ashley P. M. & Flood P. G. eds. *Tectonics and Metallogenesis of the New England Orogen*, pp. 29–51. Geological Society of Australia Special Publication **19**.

KORSCH R. J., WAKE-DYSTER K. D. & JOHNSTONE D. W. 1993. The Gunnedah Basin—New England Orogen deep seismic reflection profile: implications for New England tectonics. *In*: Flood P. G. & Aitchison J. C. eds. *New England Orogen Eastern Australia NEO'93 Conference*, pp. 85–100. University of New England, Armidale.

LACKIE M. A. 2000. Palaeomagnetic investigation of Late Palaeozoic volcanic units from the Northern New England Orogen, Queensland. *In*: Klootwijk C. ed. *Australian Palaeomagnetism, Rockmagnetism and Environmental Magnetism. 2000 Abstracts*, p. 22. Australian Geological Survey Organisation Record **2000/36**.

LACKIE M. A. & SCHMIDT P. W. 1993. Remagnetisation of strata during the Hunter–Bowen Orogeny. *Exploration Geophysics* **24**, 269–274.

LACKIE M. A. & SCHMIDT P. W. 1996. Thermal events in the Sydney-Bowen Basin as defined by palaeomagnetism. *Geological Society of Australia Abstracts* **43**, 319–328.

LEITCH E. C. 1993. The floor of the Gunnedah Basin north of the Liverpool Range. *In*: Tadros N. Z. ed. *The Gunnedah Basin, New South Wales*, pp. 335–348. Geological Survey of New South Wales Memoir Geology **12**.

LEITCH E. C. & LARA P. 2000. The structural geology of the Kindee District and the age and timing of displacement of the Hastings Block, Southern New England Fold Belt. *Geological Society of Australia Abstracts* **59**, 297.

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LENNOX P. G. & FLOOD P. G. 1997. Age and structural characterisation of the Texas megafold, southern New England Orogen, eastern Australia. *In*: Ashley P. M. & Flood P. G. ed. *Tectonics and Metallogenesis of the New England Orogen*, pp. 161–177. Geological Society of Australia Special Publication **19**.

LENNOX P. G. & ROBERTS J. 1988. The Hastings Block—a key to the tectonic development of the New England Orogen. *In*: Kleeman J. D. ed. *New England Orogen Tectonics and Metallogenesis*, pp. 68–77. University of New England, Armidale.

LOWRIE W. 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. *Geophysical Research Letters* **17**, 159–162.

MANSER W. 1968. *Geological Map of the New England 1:100 000 Wingen Sheet (no 359), with parts of nos 350, 351, 260, with marginal text and map of the Timor Anticline.* University of New England, Armidale.

MCELHINNY M. W., POWELL C. MCA. & PISAREVSKY S. A. 2003. Palaeozoic terranes of eastern Australia and the drift history of Gondwana. *Tectonophysics* **362**, 41–65.

MCFADDEN P. L. 1990. A new fold test for palaeomagnetic studies. *Geophysical Journal International* **103**, 163–169.

MCFADDEN P. L. 1998. The fold test as an analytical tool. *Geophysical Journal International* **135**, 329–338.

MCFADDEN P. L. & MCELHINNY M. W. 1990. Classification of the reversals test in palaeomagnetism. *Geophysical Journal International* **103**, 725–729.

MCKELVEY B. C. & MCPHIE J. 1985. Tamworth Belt. *In*: Wagner R. H., Winkler Prins C. F. & Granados L. F. *The Carboniferous of the World 2. Australia, Indian Subcontinent, South Africa, South America and North Africa*, pp. 15–23. Instituto Geologico y Minero de Espana and Empressa Nacional ADARO de Investigaciones Mineras, S.A., Madrid.

MCKERROW W. S., MAC NIOCAILL C., AHLBERG P. E., CLAYTON G., CLEAL C. J & EAGAR R. M. C. 2000. The Late Palaeozoic relations between Gondwana and Laurussia. *In*: Franke W., Haak V., Oncken O. & Tanner D. eds. *Orogenic Processes: Quantification and Modelling in the Variscan Belt*, pp. 9–20. Geological Society of London Special Publication **179**.

MCPHIE J. 1983. Outflow ignimbrite sheets from Late Carboniferous calderas, Currabubula Formation, New South Wales, Australia. *Geological Magazine* **120**, 487–503.

MCPHIE J. 1984a. Permo-Carboniferous silicic volcanism and palaeogeography on the western edge of the New England Orogen, north-eastern New South Wales. *Australian Journal of Earth Sciences* **31**, 133–146.

MCPHIE J. 1984b. Eruption styles and setting of Late Palaeozoic silicic volcanic centres, New England Orogen, northeastern New South Wales. PhD thesis, University of New England, Armidale (unpubl.).

MCPHIE J. 1987. Andean analogue for Late Carboniferous volcanic arc and arc flank environments of the western New England Orogen, New South Wales, Australia. *Tectonophysics* **138**, 269–288.

MCQUEEN K. G., PILLANS B. J. & SMITH M. L. 2002. Constraining the weathering history of the Cobar region, western NSW. *Geological Society of Australia Abstracts* **67**, 426.

MENNING M. 1995. A numerical time scale for the Permian and Triassic periods: an integrated time analysis. *In*: Scholle P. A., Peryt T. M. & Ulmer-Scholle D. S. eds. *The Permian of Northern Pangea, Volume 1: Paleogeography, Paleoclimates, Stratigraphy*, pp. 77–97. Springer Verlag, Berlin.

MENNING M. 1997. Permian magnetostratigraphy of South China, global correlation, and time scale calibration. *In*: Shi G. R. ed. *The Permian of Eastern Tethys: Biostratigraphy, Palaeogeography and Resources*, pp. 98–101. Deakin University Technical Paper **1997/3**.

MERRILL R. T., MCELHINNY M. W. & MCFADDEN P. L. 1996. *The Magnetic Field of the Earth: Paleomagnetism, the Core, and the Deep Mantle.* Academic Press, San Diego.

MOORE D. J. 1974. The stratigraphy and sedimentation of the Late Devonian – Carboniferous sequence in the northern end of the Werrie Syncline NSW. BSc (Hons) thesis, University of New England, Armidale (unpubl.).

MOORE M. E., GLEADOW J. W. & LOVERING J. F. 1986. Thermal evolution of rifted continental margins: new evidence from fission tracks in basement apatites from southeastern Australia. *Earth and Planetary Science Letters* **78**, 255–270.

MOORE D. & ROBERTS J. 1976. The Early Carboniferous marine transgression in the Merlewood Formation, Werrie Syncline, New South Wales. *Journal and Proceedings of the Royal Society of New South Wales* **109**, 49–57.

MURRAY C. G., FERGUSSON C. L., FLOOD P. G., WHITAKER W. G. & KORSCH R. J. 1987. Plate tectonic model for the Carboniferous evolution of the New England Fold Belt. *Australian Journal of Earth Sciences* **34**, 213–236.

NOTT J. F., IDNURM M. & YOUNG R. W. 1991. Sedimentology, weathering, age and geomorphological significance of Tertiary sediments on the far south coast of New South Wales. *Australian Journal of Earth Sciences* **38**, 357–373.

O'SULLIVAN P. B., COYLE D. A., GLEADOW A. J. W. & KOHN B. P. 1996. Late Mesozoic to Early Cenozoic thermotectonic history of the Sydney Basin and the Eastern Lachlan Fold Belt. *Geological Society of Australia Abstracts* **43**, 424–432.

OPDYKE N. D. & CHANNEL J. E. T. 1996. *Magnetic Stratigraphy*. Academic Press, San Diego.

OPDYKE N. D., ROBERTS J., CLAOUÉ-LONG J., IRVING E. & JONES P. J. 2000. Base of the Kiaman: its definition and global stratigraphic significance. *Geological Society of America Bulletin* **112**, 1315–1341.

PILLANS B. J. 2002. Climate-driven weathering episodes during the last 200 Ma in southern Australia. *Geological Society of Australia Abstracts* **67**, 428.

POGSON D. J. 1972. *Geological Map of New South Wales, Scale 1:1 000 000.* Geological Survey of New South Wales, Sydney.

POGSON D. J. & HITCHINS B. L. 1973. *New England 1:500 000 Geological Sheet*. Geological Survey of New South Wales, Sydney.

PRATT W. 1996a. *Gunnedah Coalfield South Regional Geology, 1:100 000 Map*. New South Wales Department of Mineral Resources, Sydney.

PRATT W. 1996b. A geological interpretation of the Maules Creek aeromagnetic survey. *Coal and Petroleum Geology Report, New South Wales Department of Mineral Resources* (unpubl.).

PRATT W. 1998. *Gunnedah Coalfield North Regional Geology, 1:100 000 Map*. New South Wales Department of Mineral Resources, Sydney.

RAZA A., HILL K., KORSCH R. & BROWN R. 1998. Geological evolution of the central eastern margin of Australia: fission track thermochronology. *In*: Finlayson D. M. & Jones L. E. A. eds. *Mineral Systems and the Crust – Upper Mantle of Southeast Australia*, pp. 159–161. Australian Geological Survey Organisation Record **1998/2**.

ROBERTS J., CLAOUÉ-LONG J. C. & JONES P. J. 1995a. Australian Early Carboniferous Time. *In*: Berggren W. A., Kent D. V., Aubry M. P. & Hardenbol J. eds. *Geochronology Time Scales and Global Stratigraphic Correlation*, pp. 23–40. SEPM Special Publication **54**.

ROBERTS J., CLAOUÉ-LONG J. C., JONES P. J. & FOSTER C. B. 1995b. SHRIMP zircon age control of Gondwanan sequences in Late Carboniferous and Early Permian Australia. *In:* Dunay R. E. & Hailwood E. A. eds. *Non-Biostratigraphic Methods of Dating and Correlation*, pp. 145–174. Geological Society of London Special Publication **89**.

ROBERTS J. & ENGEL B. A. 1987. Depositional and tectonic history of the southern New England Orogen. *Australian Journal of Earth Sciences* **34**, 1–20.

ROBERTS J., ENGEL B. & CHAPMAN J. 1991. *Geology of the Camberwell, Dungog, and Bulahdelah 1:100 000 sheets 9133, 9233, 9333*, Geological Survey of New South Wales, Sydney.

ROBERTS J. & GEEVE R. 1999. Allochthonous forearc blocks and their influence on an orogenic timetable for the Southern New England Orogen. *In*: Flood P. G. ed. *Regional Geology Tectonics and Metallogeny New England Orogen NEO'99 Conference*, pp. 105–114. University of New England, Armidale.

ROBERTS J., JONES P. J. & JENKINS T. B. H. 1993. Revised correlations for the Carboniferous marine invertebrate zones of eastern Australia. *Alcheringa* **17**, 353–376.

SCHEIBNER E. & BASDEN H. (Editors) 1998. Geology of New South Wales—Synthesis. Volume 2 Geological Evolution. *Geological Survey of New South Wales Memoir Geology* **13 (2)**.

SCHMIDT P. W. 1988. A rapid Carboniferous polar shift of New England from palaeomagnetism. *In*: Kleeman J. D. ed. *New England Orogen Tectonics and Metallogenesis*, pp. 193–198. University of New England, Armidale.

SCHMIDT P. W., AUBOURG C., LENNOX P. G. & ROBERTS J. 1994. Palaeomagnetism and tectonic rotation of the Hastings Terrane, eastern Australia. *Australian Journal of Earth Sciences* **41**, 547–560.

SCHMIDT P. W. & CLARK D. A. 2000. Paleomagnetism, apparent polar-wander path & paleolatitude. *In*: Veevers J. J. ed. *Billion-year earth history of Australia and neighbours in Gondwanaland*, pp. 12–17. GEMOC Press, Sydney.

SCHMIDT P. W., CLARK D. A. & RAJAGOPALAN S. 1993. An historical perspective of the Early Palaeozoic APWP of Gondwana: new results from the Early Ordovician Black Hill Norite, South Australia. *Exploration Geophysics* **24**, 257–262.

SCHMIDT P. W. & LACKIE M. A. 1993. Permian remagnetisations and the deformation of the New England Fold Belt. In: Flood P. G. & Aitchison J. C. eds. *New England Orogen eastern Australia NEO'93 Conference*, pp. 299–307. University of New England, Armidale.

SCHMIDT P. W. & MCELHINNY M. W. 2003. Discussion. Carboniferous palaeomagnetism of the Rocky Creek Block, northern Tamworth Belt, and the New England pole path. *Australian Journal of Earth Sciences* **50**, 129–131.

SCHMIDT P. W. & WILLIAMS G. E. 2002. Palaeomagnetic dating of the Hamersley surface and deep weathering in the Pilbara – northern Yilgarn region, WA. *Geological Society of Australia Abstracts* **67**, 431.

SHAW R. D., WELLMAN P., GUNN P., WHITAKER A. J., TARLOWSKI C. & MORSE M. 1995. 'Australian crustal elements' map: a geophysical model for the tectonic framework of the continent. *Australian Geological Survey Organisation Research Newsletter* **23**, 1–3.

SHAW R. D., WELLMAN P., GUNN P., WHITAKER A. J., TARLOWSKI C. & MORSE M. 1996. Guide to using the Australian crustal elements map. *Australian Geological Survey Organisation Record* **1996/30**.

SHAW S. E. & FLOOD R. H. 1993. Carboniferous magmatic activity in the Lachlan and New England Fold Belts. *In*: Flood P. G. & Aitchison J. C. eds. *New England Orogen eastern Australia NEO'93 Conference*, pp. 113–121. University of New England, Armidale.

SYMONDS P. A., CALWELL J. B., STRUCKMEYER H. I. M., WILLCOX J. B. & HILL P. J. 1996. Mesozoic rift basin development off eastern Australia. *Geological Society of Australia Abstracts* **43**, 528–542.

TADROS N. Z. 1993. The Gunnedah Basin New South Wales. *Geological Survey of New South Wales Memoir Geology* **12**.

TAIT J., SCHÄTZ M., BACHTADSE V. & SOFFEL H. 2000. Palaeomagnetism and Palaeozoic palaeogeography of Gondwana and European terranes. *In*: Franke W., Haak V., Oncken O. & Tanner D. eds. *Orogenic Processes: Quantification and Modelling in the Variscan Belt*, pp. 21–34. Geological Society of London Special Publication **179**.

VAN DER VOO R. 1993. *Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans*. Cambridge University Press, Cambridge.

VEEVERS J. J., CONAGHAN P. J. & POWELL C. MCA. 1994. Eastern Australia. *In*: Veevers J. J. & Powell C. McA. eds. *Permian–Triassic Pangean Basins and Foldbelts Along the Panthalassan Margin of Gondwanaland*, pp. 11–171. Geological Society of America Memoir **184**.

VEEVERS J. J., POWELL C. MCA. & ROOTS S. R. 1991. Review of seafloor spreading around Australia. I. Synthesis of the patterns of spreading. *Australian Journal of Earth Sciences* **38**, 373–389.

VICKERS M. D. 1993. The Lower Permian Werrie Volcanics: facies and tectonic setting. *In*: Flood P. G. & Aitchison J. C. eds. *New England Orogen eastern Australia NEO'93 Conference*, pp. 323–330. University of New England, Armidale.

WOODWARD N. B. 1995. Thrust systems in the Tamworth Zone, southern New England Orogen, New South Wales. *Australian Journal of Earth Sciences* **42**, 107–117.

ZIJDERVELD J. D. A. 1967. AC-demagnetization of rocks: analysis of results. *In*: Collinson D. W., Creer K. M. & Runcorn S. K. eds. *Methods in Palaeomagnetism*, pp. 254–286. Elsevier, Amsterdam.

ZIJDERVELD J. D. A. 1975. Paleomagnetism of the Esterel rocks. PhD Thesis, University of Utrecht, Utrecht (unpubl.).

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SUPPLEMENTARY TABLES

Table 5 High temperature (HT) pseudo-components in stratigraphic coordinates.

Table 6 High temperature (HT) overprints in stratigraphic coordinates.

Table 7 Fold test (McFadden 1990), pseudo-components (HT).